

NHS Pathways Team

How to Build NHS Pathways

Version	Summary of changes	Date
1.0	First draft for consultation	05/04/2006
2.0	Revisions following consultation: 2.1.1.1 – “JumpID” changed to “OrderNo” 2.1.1.2 - additional negative jump code added. “-6” jumps to 1 st party safety pathway 2.2 – field lengths added to data tables 2.4.1 note added on display of Crown copyright notice 2.4.1.1 note added on alternative screen fonts 3.2.1 rules for data transfer added 3.2.3 – A note on future requirements re message routing 6.0 section added on customizing disposition alternative. 7.0 section added for non-triage calls to ambulance service	26/05/2006
3.0	Revisions to tables to reflect inclusion of CMS Directory of services functionality Revisions of tables and interfaces to reflect modifications to Body maps Revisions of tables and interfaces to reflect modifications to care advice screens Instructions regarding evaluating expressions in the Transfer fields of triagelogic. Addition of skillset 4 and T4 field in triagelogic	12/01/2008
4.0	Revision of care advice instructions to reflect exclude function and disposition transfer code changes. Inclusion of rules section to describe Control Type 3 and 4 questions Removal of some redundant fields from tables	1/10/2009
4.1	Section 8 added re supporting documentation	14/3/2010
4.2	2.3.3.1 Restart Triage clarified	20/5/2010
4.3	2.3.2.6 Clarification re include in report status of control type 3 and 4 questions. Clarification of operators used in integer type 3 and 4 questions	30/5/2010
4.4	2.2.8.3 System flags table updated with extra REAP values	16/5/2010
5.0	“-5” explicitly allowed in triagelogic Dx rank added to dispomap to improve ambulance stand down/triage cut off. Ambulance priority type field added to dispomap to map Dx codes to R1, R2 etc Cxkeyword character limit increased Update on DOS screen to include reject all and list of reject reasons Continue from validation screen update to handle early exit transfers to nurses Keypoint title word change Section on Attend Incident heavily modified re introduction of new pathways Section on interoperability edited to reflect change in control over interoperability governance General review to tidy and provide clarification	24/05/2012
6.0	Specify value of 2 allowed in cases where timeframe in hours can be set for deferred ambulance disposition Introduction of new field in the dxmap table for sd substitution when disposition is overridden by clinician. To ensure legal sg/sd/disposition value is used to interrogate the dos.	

	Amended button in inline care advice to show “save and close” rather than “next” when jump value is -9 see 2.4.3.1 Introduction of new field in CareAdvice table to enable filtering of Cx sets by disposition (not enabled until Pathways 6.7.0. Clarification of validation from early exit added – 2.4.7.5 Guidance re the Call Summary Report added – 10.3 Clarification re. transferring calls from one Pathways instance to another where the release number is different.	
6.1	Field length for cxspecific increased to 250	19/6/13
Excerpt	Addition of SubstituteGroup field onto dispomap export table for use when clinician overrides disposition 9.1.2 Addition of specify box upon selecting override disposition	Via email 06/12/13
7.0	Clarification on wording for clinician validation boxes in order to reflect different styles used Clarification of the specify value of 2 in the txanswers table this is used in all configurations Clarification of the format of the export data for reporting purposes (section 3) Increase of character limit for Dispomap::DxItemRationale Inclusion of Reception point values in disposition items table, dispomap table, care advice items table (FOHINT) 2.2 Character limits throughout document reviewed and amended if necessary Addition of DxCode to DispositionItems table and correction of description of actions for this field in Dispositions Interface skill set 1 and 3 Order of the tables amended to reflect the order of the fields in the CSV files provided by NHS Pathways Addition to collect separate care advice text, CxID and answer number in triagerecord Ambulance stand down/rank cut off is now part of standard NHS Pathways implementation and not an optional extra. The administration screen data for the rank cut off is now provided in a table DxRank as part of csv data set Replacement of the original case record and triage record tables with new versions which reflect output needed for CQI Webapi. New table::field names replaced throughout this document. Old tables have been moved to the end of the document and and new column added onto them to ease transference onto new format. Clarification of the type of reports and what fields should be included in these reports from the pathways system data	18/12/2013
7.1	Addition of mers virus system variable – see 2.2.6 Correction to the triage start point for ambulance clinicians – Skill Set 4 – see 2.1.2. Start point should be “0” not “5” Section 3.3.1 has been removed as transfer of data between systems is now governed by 111 interoperability specification.	24/06/2014
7.2	Addition of viral haemorrhagic fevers system variable – see 2.2.6	04/08/2014
7.3	Update to table 3.1.4.1.1 to include postcode in case record for reporting purposes – note added at end of table	20/08/2014
10.0	Change of Tech spec versioning convention to align with Pathway releases (R10). Addition of another radio button option in the disposition override interface (2.5.5.2) : Disposition considered inappropriate – Repeat caller (3 calls within 4 days) Updated CQI WebApi URLs and updated JSON fields.	06/01/2015

10.1	Updated CQI output data dosRequestFailId lookup value list	18/02/2015
11.0	Clarification of Body Map ordering (2.5.4) On leaving the disposition interface, also set the TriageRecord::QuID to the selected disposition code (2.5.5.1) Interim Care Advice screen should check for Cx_Dx applicability before rendering the radio button keyword. If no match is found, the keyword should not be displayed (2.4.2.12) Added note clarifying Pathway CxKeywords should be picked up on pathway jumps (2.1.1.2) Removed references to Solo <u>Updated data column maxlengths:</u> DispositionItems:: DispoAdviceCode CareAdvice:: KwSpecific, KWGroup, CX_DX CxExclude:: CxKeyword, ExcludeKw Pathways:: CxKW TxQuestions::Qu_Rationale TxAnswers:: SnoMed_description <u>Column names renamed (in release CSV files):</u> Pathways::PathwayTitle.htm (full-stop removed) DispositionItems::Active SPA (space removed) DxRank::DxRank (column header added to export file for consistency) SystemFlag::System Flag (space removed) SystemFlag::Term (renamed to SystemFlagDescription) TxAnswers::Dispo_key_Procedure (underscore instead of hyphen) <u>New columns (in release CSV files):</u> BodyMap::Ordering TxQuestions::QuestionTextDigital, Qu_RationaleDigital TxAnswers::AnswerTextDigital, AnswerRationaleDigital (new digital fields used in Online Digital setting only, and should be ignored for Telephone triage) CareAdviceItems::CxTextDigital	09/11/2015
11.0.1	Addition of sepsis system variable – see 2.2.6 2.3.1.1 skillset typos corrected Standardise Maxlength for CxKW fields to 255 (Pathways:: CxKW, TxAnswers::CxSpecific) to match TriageRecord::CxKeywords Clarification of clinician alternative disposition list ordering – see 2.5.5.2	07/04/2016
12.0	No changes. Version number updated in line with release.	04/07/2016
13.0	Clinician disposition override screen (Dx2) – list of override options updated. See 2.5.5.2 DxMap::AmbPriorityType column maxlength increased	20/09/2016
13.0_CAS	New content added to support Clinical Assessment Service (or Clinical Hubs) functionality. Includes 2 new tables called Skillsets and BodyMapOverrides. Chapter on Clinical Assessment Service provides more details, along with the relevant table definitions. Changes also made to 2.4.2.6 and 2.4.2.9 to replace hard coded logic with lookups from the skillset table	22/03/2017
14.0	No changes. Version number updated in line with release.	01/06/2017
15.0	Added journeyId into the IDT spec as an additional field to submit to Pathways. The Call Validation section (2.5.7) was amended to exclude SCM users. Contact Information updated.	28/11/2017

	NHS Number as a required IDT field added. Request for it to be PDS-traced only. Sepsis system variable now removed from 2.2.6.2 as this is incorporated into Pathways without a switch. Updated triage cut-off section to highlight difference between Dx000 and Dx001	
15.0 revision 1	Updated the Call Summary Report to specify that UserComment text should be included where entered.	03/01/2018
16.0	Improved the definition of the NHS Pathways Age Groups. Amended formatting across the document.	14/05/2018
16.0 revision 1	Add detail on "Dental", and remove Dental "Light" references	03/08/2018
17.0	2.5.5 updated to clarify that Inline Dispositions are displayed using the Single Selection Triage (Tx1) interface. 2.2.1.2, 2.5.5, 2.5.5.1, 2.5.5.2, 2.5.6, 8.1.1: Removed references to Skillset IDs used to determine if the skillset is a CallHandler or Clinician type. This is determined by Skillset::SkillsetUserType .	21/09/2018
17.0.1	Section 12 added to describe the implementation of the HCP and IFT frameworks. 2.4.2.1 and 2.4.3.1 have been updated to correctly reflect the process for selecting the Entry Point when starting and restarting a Triage. Also added note to refer to section 12 for HCP and IFT process.	30/01/2019
18.5.0	Added Symptom Group (2.2.11) and Symptom Discriminator (2.2.12) NHS Pathways Data Table definitions. Updated Directory of Services rejection reasons as requested by NHS Digital DoS team.	21/05/2019
18.5.0_revision1	Added clarification in section 3.1.4 and 3.1.5 regarding IDT data item mandation within the json packet. Some items are only mandatory if certain conditions are met. E.g. call handler fields being mandatory only if call handler was involved. Added nhsEnglandAreaCode as a new mandatory IDT data field into section 3.1.4.	20/06/2019
18.5.0_revision2	Added clarification regarding required data fields in section 3.1.4.1.	25/09/2019
19.3.0	Removed section 9.1.1 regarding DoS rejection reasons as DoS team now host those separately.	23/10/2019
20.3.0	Moved instructions regarding interfaces for the DoS from Section 9 (Connecting to the DoS) to Section 2.5.6 (DoS Interfaces) Altered instructions regarding interfaces for the DoS (Section 2.5.6) to explain difference in program flow between Call Handlers and Clinicians where a DoS call is required Removed reference to DoS Exception IDs Added instructions to Section 2.5.5.1 regarding Dx1 interface, to include a DoS return Altered instructions on No 5 in the table within 2.5.5.1 regarding the text on the Next button Updated screenshots of Dx1 and Dx2 (Sections 2.5.5.1 and 2.5.5.2) for consistency Removal of image showing message while DoS call is being undertaken, replaced by written instructions so this can apply to both Dx1 and the full DoS screen Change status of returned DoS services not recorded as Rejected or Accepted from 'Displayed' to 'Returned' (Sections 2.3.2.3 and 3.1.4.1.3) Updated section 9.1.1.1 to include the URL for the current list of Reject Reasons.	17/01/2019
21.1.0	Up-versioned for R21.1.0.	02/06/2020
22.1.0	Added Alpha/Beta/Production field into the version.xml file of the release files so that the intended state of each release can be identified.	16/06/2020

	Add requirement to include the Worsening Interim Care Advice keyword. Sections 2.4.2.12 and 2.5.7 updated. Added to 11.2.3 section on Senior Clinician Module: “The pathway selected by the user should be recorded in the TriageRecord so that it is displayed on the Pathways Selected line of the Call Summary Report.” Updated section 2.2.6 System Variables to highlight that we would expect host system suppliers to automatically import the SystemVariables.csv file so that if NHS Pathways add new variables, there is no manual work needed on behalf of suppliers or providers before this release content can be used. Added to 2.5.5.2 (Disposition Interface for CL) instructions for the labelling of the Next/Save and Close button. Updated numbered image to include button.	
23.2.0	Section 2.2.4.5: New column (in release CSV files): CxCareAdvice::AlternativeTitleDigital (new digital field used in Online Digital setting only, and should be ignored for Telephone triage) Sections 2.2.2.2, 2.2.2.3, 2.2.4.6: Updated Online Digital fields to be longer and HTML based. Section 10.2: Updated example Call Report to include User Comments added on Care Advice screens. Section 2.2.9.1, 11.1.1: Added wording to prevent ensure only skillsets approved for use by NHS Pathways should be enabled (to cover skillsets used for specific purposes – e.g. ED Streaming. Also updated the example table of skillsets. Section 2.5.8: Added the ability to Restart the Triage from the Call Validation screen. Section 11.4: Injury Module details added.	01/09/2020
24.1.0 and 24.2.0	Upversioned to R24.1.0 Section 3.1.4.1.1 Added a request for the collection of host system version into the CQI (IDT) API (supplierRelease) Added reference to the new ED Streaming and Redirection skillset that can be found in the skillsets table. Purely for awareness as no host system other than 111 Online uses this skillset. Merged the standalone PaCCS spec into this document (Sections 2.2.13, 2.5.5.2, 2.5.9, 9, 13) Section 2.5.7 updated to default Worsening Care Advice on the Home Care Screen. Clarify that only the latest SD encountered should be stored in section 2.3.2.1. Also clarified that only the latest is to be sent to DoS (section 9) Section 11.2.3 updated to clarify that Care Advice Keywords should be taken from the overridden pathway in SCM.	06/10/2020
25.1.0 and 25.2.0	Upversioned to R25.2.0 Updated contact information.	02/03/2021
26.2	Clarification that CxKeywords should be picked up when jumping into Pathways from PaCCS – (Sections 13.2 and 13.2.3)	05/03/2021
27.2	PaCCS update to make the “Considered” column optional (Section 2.5.9.2) Added extra field into CCS Jumps csv file to indicate the type of button it represents. Also removed the space in the “Pathway Jump” column name to be consistent with other columns (Section 2.2.13.6).	04/05/2021
28.2	Alternative Text field added to skillset.csv definition (Section 2.2.9.2) New Alternative Wording CSV file added (Section 2.2.14)	15/06/2021

	Skillset ID in use at the time a Triage Record is created to be included in the IDT data for each Triage Record (Sections 3.1.4.1.2, 3.1.6.1) Removed Contact Information (Section 1.5) Wording added to clarify that the clTriageDispoCode should be set when a Disposition is reached for a PaCCS call (Sections 2.5.5.2 and 13.1.3)	
29.2	Add early notification that the Digital columns have been deprecated and will be removed in a future release (Sections 2.2.2.2, 2.2.2.3, 2.2.4.5, 2.2.4.6).	19/08/2021
30.2	Minor text change to reflect that the DoS Programme requested the addition of a new "servicestatus" for the IDT data packed. This new servicestatus (ID = 4) supports a status of "bulkReject" which only one supplier is using. Update to state that the Question Rationale field can contain newline indicators ("n") to enhance readability for users (Section 2.2.2.2) Upversioning to R30.2.	13/10/2021

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1 Introduction

1.1 About NHS Pathways

NHS Pathways is a decision support system for users to sort patients to a safe level of care in a safe timeframe.

The NHS Pathways suite of products comprises of three modules:

-  Module 0 – Emergency care assessment – filters those who require ambulance or other urgent care support due to immediate threat to life from reported symptoms, mechanism of injury or current location/environment
-  Module 1 – Continued symptom-based assessment and prioritisation for an initial contact by a patient or 3rd party, once immediate threat to life has been excluded but imminent threat to life has not been ruled out.
-  Module 2 – Clinician assessment - clinical decision support for clinical assessment for those callers with more complex problems that have not already been directed to another provider (999, A&E, GP urgent, etc) or to self-care by Module 1 users.

Modules 0 and 1 are deliberately designed to enable use by trained call handlers including processing of patients through to disposition where appropriate. Dispositions in Module 0 will, by definition, involve ambulance or similar service. Dispositions in Module 1 can include the obvious high-end alternatives (999, A&E, etc.) but also include options through to self-care, where appropriate.

For patients not disposed to another provider, Module 2 picks up the assessment with a ‘validation exercise’ by the clinician, and continues along the appropriate path (or begins again/enables change where worsening or new symptoms have arisen in the meantime).

Key to the system is the setting of ‘gates’ to determine by organisation, location or even individual the point at which Module 1 ends and transfer to a module two operator is required. Clearly this can happen against level of acceptable risk, organization role (e.g. ambulance service may only wish to identify 999 calls and hand rest over to another organization for additional assessment).

1.2 Purpose of this document

This document sets out the data structures, interfaces and controls of the NHS Pathways clinical triage system. It is provided to enable coherent, comprehensive and effective embedding of NHS Pathways within other functional systems used in the management of healthcare delivery.

NHS Pathways and the Department of Health reserve the right to make additions, alterations and deletions to this document at any time in writing to any parties in receipt of an official copy.

Supply of this document does not in any way imply or convey rights to market, sell, license or otherwise offer NHS Pathways to any other individual or organisation.

System developers wishing to embed NHS Pathways into their own products should use this document in conjunction with:

-  The supplier licence agreement
-  Sample data tables
-  Sample images for use in body map screens

1.3 Restrictions on use

This document is for the express use of only relevant technical personnel within named organisations to enable NHS Pathways to be effectively embedded within the wider system functionality provided by the named organisation. Only the organisation named on the front of this document may use it for the express purpose of achieving effective embedding and no other purpose.

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A separate licence agreement provides full details regarding the use of this document and associated materials.

1.4 Conventions, Acronyms and Abbreviations

1.4.1 Conventions used in this document

-  Table names such as **Pathways** are shown in bold
-  Field names such as *BlockOrder* are shown in italics
-  Where field names are used not adjacent to table information, the table and field name is given together as **TriageLogic::QuID** to show that the field: *QuID* belongs to the table: **TriageLogic**

1.4.2 Acronyms and Abbreviations

-  DoS – Directory of Services
-  OOH – Out of Hours Service
-  SPA – Single Point of Access Service (111)
-  AMB – Ambulance
-  FOH – Front of House (reception point)
-  RP – Reception Point
-  CAS – Clinical Assessment Service
-  CSV – Comma Separated Values
-  Intelligent Data Tool (IDT)
-  Continuous Quality Improvement (CQI)

1.5 Contact Information

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2 Technical Specifications

2.1 Overview

NHS Pathways is a system of interlinked individual pathways. A single pathway consists of a series of triage questions (Tx) that link to other questions depending on the answer chosen. Therefore, the route through a pathway is determined by the answers selected by the user until eventually an end point or disposition (Dx) is reached. This disposition will identify the level of care required within a timeframe deemed safe based on the symptoms provided and history described. Following the disposition, care advice (Cx) may also be given in order to support the patient with their symptoms while they await care. This is the normal route through a Pathway as follows:

Start Triage Tx-----> Tx-----> Tx----->Tx -----> Dx-----> Cx End Triage

However, for pathways in an ambulance control setting, it is possible that care advice and dispositions will appear more than once and be interspersed with triage questions. For instance, an answer may lead to an instant “999” disposition in order to trigger the dispatch of an ambulance but then be followed by more questions and care advice to provide pre-arrival advice to the caller.

Start Triage Tx----> Dx---->Tx---->Cx--->Cx--->Tx----->Tx-----> Cx-----> End Triage

2.1.1 Triagelogic

As stated above, the route through a Pathway is determined by the answers that a user selects. The logic behind this is contained within the fields *JumpID*, *T1*, *T2*, *T3* and *T4* of the **Triagelogic** table.

2.1.1.1 JumpID's with a positive integer

If answer 1 of a question has a *JumpID* of 40 then the next question to be presented to the user is the question identified in the **Triagelogic** table with an *OrderNo* of 40 within the same pathway. This is true of any positive integer.

2.1.1.2 Jump ID's with a negative one

If, however, the **Triagelogic:JumpID** is -1 then this indicates a jump to a different Pathway or a disposition given by the transfer codes provided in the fields **Triagelogic:T1**, **T2**, **T3** and **T4**. Which of these codes is used depends on the skill set of the current system user (see 2.1.2 – Skill Sets below). When jumping from one pathway to another, any CxKeywords attached to the target pathway should be picked up and recorded in **TriageRecord::CxKeywords**.

2.1.1.3 JumpID's with other negative numbers

JumpID's with other negative numbers have special meanings and add extra functionality to the triage logic. The following codes are currently in use but developers should note that this is a likely means by which further triage logic will be added in the future.

JumpID	Action
-2	Jump to appropriate pathway selection screen (body map) depending on age/gender of patient - no transfer codes.
-3	Loop to “3 rd party safety pathway” SP3.0 – do safety pathway and return to point in current pathway given by transfer codes in <i>T1</i> , <i>T2</i> , <i>T3</i> , <i>T4</i> .
-4	Return to pathway from loop. This will only appear in pathways that have been looped to. Return point given by transfer codes in <i>T1</i> , <i>T2</i> , <i>T3</i> , <i>T4</i> .
-5	Jump to Care Advice interface – this will follow a disposition but can also be in triagelogic.
-6	Loop to “1 st party safety pathway” SP1.0 – do safety pathway and return to point in current pathway given by transfer codes in <i>T1</i> , <i>T2</i> , <i>T3</i> , <i>T4</i> .
-9	End pathway – code sent to host system that triage is complete or has been terminated early. No transfer codes.

2.1.2 Skill Sets

Which transfer code is used when the **TriageLogic:JumpID** is -1 is dependent on the skill set of the user. In the past, there were predominately six skill sets identified (see table below) but more are being added to extend NHS Pathways functionality, in particular for CAS. To achieve this, a skillset table was introduced, which enabled new skillsets to be dynamically added in the release data (see section 2.2.9).

Skillset	Description
1	Call handlers in an OOH urgent care setting.
2	Nurses or other clinicians with a comparable skill set/level of training in an OOH urgent care setting.
3	Call handlers in an ambulance control setting.
4	Nurses or other clinicians with a comparable skill set/level of training in an ambulance setting.
5	Call handlers in a SPA/111 urgent care setting.
6	Nurses or other clinicians with a comparable skill set/level of training in a SPA/111 urgent care setting.
	more defined in skillsets table

As an example of how different skillsets may use the system, a call handler (skill set 1) may triage a call through all the jump codes until a point is reached when a -1 in *T1* indicates a nurse disposition, e.g. “speak to nurse within 1 hour”. At this point the call handler may give interim care instructions and the call is then queued for a nurse to continue the triage. For a nurse starting the same call, a transfer to nurse disposition would be inappropriate and the transfer code in *T2* would transfer the nurse to more triage questions, until he/she reaches a disposition. On the other hand if the nurse picked the part-triaged call from the queue then he/she would validate the call handler’s work then proceed from the point where the call handler left off, using the transfer code in *T2* to continue the triage.

In an ambulance control setting a call taker’s skill set is different to that of a call handler in an out of hours setting. If an OOH call handler reaches a “999” disposition this call will be immediately transferred to the ambulance service. However, in an ambulance control setting, the call handler may need to give pre-arrival advice and ask further questions of the caller. In this case, transfer codes in *T3* are used to direct the triage.

Purely for awareness, a new skillset was added for R23 called “ED Streaming and Redirection” but only 111 Online will be using this skillset.

2.1.3 Case insensitive matching

Case-insensitive matching should be used in both the application and within database queries. Therefore, **Tx12345** should be treated the same as **TX12345**, as they are essentially the same item. The same applies for care advice keywords i.e. **Bruising** should be treated the same as **bruising**.

2.2 NHS Pathways Data tables

NHS Pathways provides a set of CSV data tables used both in the triage system and as lookup data tables.

2.2.1 Pathways

2.2.1.1 Pathways: Definition

A pathway consists of triage questions, dispositions, care advice and the triage logic that connects these items together. The **Pathways** table itself contains the headline data about the pathway; its title, target group, key points and care advice to be associated with the Pathway.

At present some fields still have text in the CSV files whereas the CaseRecord refers to ID numbers e.g. agegroup, gender. The pathways system will be moving over to using the ID numbers within the CSV files and will inform suppliers when this will be introduced. In the meantime, the text will be used for matching fields however the ID number value needs to be sent in the data package to NHS Pathways.

2.2.1.2 Pathway Modules

Pathways are grouped into modules. Currently there are three pathway modules.

-  Module 0 – Pathways in this group are used to start the triage. The selection for the user is automatic and depends on the patient's age group and gender and whether the caller is calling about his/herself or someone else (1st or 3rd party).
-  Module 1 – Pathways in this group are selected by the user from a list on the pathways selection screen (see pathway selection by user (BodyMap)). The pathways presented for selection depend on the patient's age group, gender, whether the call is about a trauma or non-trauma. An additional filter, clinical area, is selected by the user on the Pathways selection screen. Not all the Pathways are listed here as some are only accessible via another 'menu' pathway.
-  Module 2 – clinician modules are continuations of Module 1 Pathways only available to users with a **Skillset::SkillsetUserType** of CL. They do not appear in the pathways selection screen but are accessed by a transfer code in **TriageLogic:T2 or T4** which is operational when a user of **Skillset::SkillsetUserType** CL is using the Pathway system.

2.2.1.3 Pathways Table - Field Definitions

Field	Type/properties	Size	Example	Notes
<i>PW_ID</i>	Required primary key	10	PW1	Primary key pathway identifier
<i>Version</i>	Text required	10	1.1.1	The version number of the given pathway – significant in data transfer between systems – see 3.2
<i>PathwayTitle</i>	Text required	100	Stopped Breathing, Trauma, Choking	Informs advisor of nature of pathway. Appears at top of Pathways interface
<i>Module</i>	Integer	1	0 or 1 or 2 etc	Which pathway module a pathway belongs to. Information only
<i>Party</i>	Numeric	1	1, 3, 4 or null	Identifies the pathways to be used at start of triage based on 1 st or 3 rd person caller 4 is used to identify the GP Urgent pathway. Other <i>Party</i> values may be used in the future to identify specialist pathways
<i>PW_Age</i>	Text required	50	Adult Child Toddler Infant	List of the target age group(s) of the pathway. Some pathways are suitable for more than one age group and will therefore

				have more than one value in a list. Values are separated by “ ”
<i>PW_Gender</i>	Text Required	12	Female Male	List of the target gender(s) of the pathway. Some pathways are suitable for both genders
<i>PW_ClinicalArea</i>	Text	30	Head and neck throat	Value used for grouping pathways in pathway selection screen. A pathway may have more than one value from list. Separated by pipeline
<i>PW_Trauma</i>	Text	20	Trauma Non trauma	Pathways are either trauma, non trauma or both or neither (null allowed)
<i>KP_Use</i>	text	500	An individual has injured their hand and/or wrist and broken the skin within the last 7 days	Used in pathway selection screen when a pathway is selected. This and <i>KP_NotUse</i> are concatenated with other text to provide the pathway's keypoints text. Line breaks are shown as “/n” in text.
<i>KP_NotUse</i>		1000	A hand or wrist injury in the last 7 days has not broken the skin (Hand or Wrist Injury, Blunt)./nAny hand or wrist injury also includes an injury to the arm above the wrist (Arm Injury, Blunt or Penetrating)	Used in pathway selection screen when a pathway is selected (see 2.3.1). This and <i>KP_Use</i> are concatenated with other text to provide the pathway's keypoints text
<i>CxKW</i>	text	255	Medication-Paracetamol	A keyword attached to some pathways which provides a value for the value list on the care instructions screen. There may be more than one value separated by pipeline
<i>PathwayTitleHtm</i>	Text	105	Ov1	Identifies the overview document linked to the pathway (NOT currently in use)
<i>SymptomGroup</i>	integer	4	200	The group of symptoms to which the pathway belongs. Forms part of the criteria to query DoS

2.2.2 Triage Questions (Tx)

2.2.2.1 Triage Questions and Answers: Definition

Data for triage questions is held in two tables, the first containing the question data, the second holding related answer data. One question can have up to seven related answers. Triage questions consist of a question and its rationale, which explains why the question is being asked. The related answers contain the answer number – used to sort the answers for display, the answer text - which the user selects, and the answer rationale, which provides further information about the answer. In addition, various other pieces of information, such as snomed codes and *Cxkeywords*, are stored with the answer selected. Nearly all questions are single select but it is possible that future development may require multi-select answers.

2.2.2.2 TxQuestions - Field Definitions

Field	Type	Example	Size	Notes
<i>QuID</i>	Required primary key	Tx1066	Max 20	Primary key: Tx codes refer to triage questions
<i>QuestionText</i>	Text required	Are features of severe respiratory distress described?	200	The question text for display
<i>Qu_Rationale</i>	Text	This question seeks to establish whether features associated with severe respiratory distress can be excluded	300	The reason for asking the question – displayed with the question text. “\n” is used to identify a line break for presentational purposes.
<i>ControlType</i>	Integer required	1,2,3,4	1	<i>ControlType</i> = 1 indicates answers are single select. <i>ControlType</i> = 2 indicates answers are multi-select. <i>ControlType</i> = 3 indicates a Tx to set a variable, not displayed to user. <i>ControlType</i> = 4 indicates a Tx to evaluate a variable. The variables can be system set, patient dependent or set by a <i>ControlType</i> 3 Tx, not displayed to user
<i>QuestionTextDigital*</i>	Text(HTML)	Have you noticed any of the following problems since the injury?	5000	The question text for display in online setting Used in Online Digital setting only, and should be ignored for Telephone triage

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<i>Qu_RationaleDigital</i> *	Text(HTML)	To find out if you have had a serious injury to the spine.	5000	The reason for asking the question in online setting Used in Online Digital setting only, and should be ignored for Telephone triage
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* Please note that the *QuestionTextDigital* and the *Qu_RationaleDigital* fields are now deprecated and will be removed in a future release of Pathways.

2.2.2.3 TxAnswers - Field Definitions

Field	Type	Example	Size	Notes
<i>QuID</i>	required	Tx1066	20	Relates to parent question
<i>AnswerNo</i>	Integer Required	1 to 7	1	The answer number. Determine sort order in display <i>QuID + AnswerNo</i> is primary key
<i>AnswerText</i>	Text required	The individual is collapsed, cold and clammy	150	The answer text for display either as a single or multi-select item list
<i>AnswerReportText</i>	text	The individual was described as collapsed, cold and clammy.	250	The report text for the chosen answer. This is displayed in the full call report.
<i>SystemFlag</i>	integer	1,2,3 etc	20	Sets system variables needed by either pathways or the host system. There may be more than one value separated by pipeline (See below)
<i>DispoDisplayText</i>	text	Not breathing	250	Provides the rationale for why a particular disposition has been reached. Displayed on the triage interfaces and in the summary report.
<i>SnoMed_description</i>	VarChar		200	The snomed code corresponding to the <i>Dispo_key_description</i>
<i>SnoMed_procedure</i>	VarChar		200	The snomed code corresponding to the <i>Dispo_key_procedure</i>
<i>Dispo_key_description</i>	VarChar		200	The snomed preferred term for the <i>DispoDisplayText</i>
<i>Dispo_key_Procedure</i>	VarChar		200	Snomed preferred term which indicates that a particular procedure may be required if this answer is selected
<i>AnswerRationale</i>	Text	This means the individual is unable to stand. They will not be alert and usually be sweaty and deathly pale or blue.	700	Supporting information for a given answer. “\n” is used to identify a line break for presentational purposes. Max 4 lines of rationale
<i>CxSpecific</i>	Text	Medication-Paracetamol	255	A keyword attached to some answers which provide a value for the value list on the care instructions screen.

				Answers may have more than one value, pipeline separated
<i>Specify</i>	Integer	0 or 1 or 2	1	If specify is set to 1 then user input is required to complete the answer. If specify is set to 2 then user input is required to enter a time
<i>Positive</i>	Integer	0 or 1	1	This field is a flag for the Validation Screen. Answers with a value of “1” should be highlighted in this screen
<i>SymptomDiscriminator</i>	integer	2000	4	A token which is picked up during triage. Forms part of the criteria to query DoS
<i>AnswerTextDigital*</i>	Text(HTML)	You've tenderness of the bones of your spine (help)	5000	The Answer Text for display in online setting. Used in Online Digital setting only, and should be ignored for Telephone triage
<i>AnswerRationaleDigital*</i>	Text(HTML)	This means your neck and anywhere along the backbone will be painful, when touched. This does not mean pain in the back from lifting a heavy weight.	5000	The Answer Rationale for display in online setting. Used in Online Digital setting only, and should be ignored for Telephone triage
<i>SNOMEDCTIDs</i>	Text	39104002 24484000	200	Pipe separated list of SNOMED CT IDs applicable to the Answer stem.

* Please note that the *AnswerTextDigital* and the *AnswerRationaleDigital* fields are now deprecated and will be removed in a future release of Pathways.

2.2.3 Dispositions (Dx)

2.2.3.1 Dispositions: Definition

Dispositions are the recommended outcomes of triage and range from ambulance to self care. As explained in 2.1, dispositions can occur inline with other triage items, or at the end of the triage. Inline dispositions will have an order number and a *jumpId* in the same way as other items in the **Triagelogic** table. This currently only happens in an ambulance control and 111/SPA settings. The disposition tables provide the data for both inline and endpoint dispositions.

2.2.3.2 Dispositions field definitions

Field	Type	Example	Size	Notes
<i>DispoCode</i>	VarChar	Dx01	10	The code used to identify the level of disposition e.g. Dx011 = Ambulance 8 minutes.
<i>DxID</i>	VarChar Required primary key	Dx12345	Max 20 characters	The disposition ID to be used for this code. All dispositions have a <i>DxID</i> .
<i>DispoText</i>	Text required	The disposition is 999 ambulance required	200	The disposition text for the disposition, this is not displayed to user.

2.2.3.3 DispositionItems field definitions

Field	Type	Example	Size	Notes
<i>DxID</i>	VarChar Required	Dx12345	Max 20 characters	Identifies the parent disposition the item belongs to.
<i>DxItemNo</i>	integer required	1, 2 etc	1	The item number – determines sort order for display. There can be more than one example of each item but which is to be displayed is determined by the active fields below.
<i>DxItemText</i>	Text required	The disposition is 999	200	The item for user selection.
<i>DxReportText</i>	Text required	Instructions given were an emergency ambulance is needed now.	300	The selected item text used in the call report.
<i>DxItemRationale</i>	Text	THIS OPTION IS FOR USE WHEN THE PERSON ACCEPTS THE DISPOSITION	220	Explains when to select item.
<i>DxCode</i>	VarChar Required	Dx04	10	Identifies the disposition code of the answer stem selected.

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<i>DispoAdviceCode</i>	VarChar	Dx01	10	Links the <i>DxitemNo</i> to the appropriate Disposition Advice in the Care Advice table via matched text in the <i>KWSpecific</i> field.
<i>TransferCode</i>		-5, -7 or -9 or specific pathway transfer e.g. PT3.0	10	This is the Disposition items own triagelogic. -5 indicates go to Interim Care Advice Screen, -7 the home care advice screen -9 initiates run the save and close procedure.
<i>Active</i>	integer	0 or 1	1	Legacy field only used internally by NHS Pathways. Not used by suppliers.
<i>ActiveOOH</i>	integer	0 or 1	1	Denotes item is to be displayed in OOH setting provided “ <i>ActiveOOH</i> ” is also set in <i>DxMap</i> table.
<i>ActiveAmbulance</i>	integer	0 or 1	1	Denotes item is to be displayed in Ambulance setting provided “ <i>ActiveAmbulance</i> ” is also set in <i>DxMap</i> table.
<i>ActiveSPA</i>	Integer	0 or 1	1	Denotes item is to be displayed in SPA setting provided “ <i>ActiveSPA</i> ” is also set in <i>DxMap</i> table.
<i>ActiveFOHINT</i>	Integer	0 or 1	1	Denotes item is to be displayed in Reception Point Co-located ED and UCC setting provided. “ <i>ActiveFOHINT</i> ” is also set in <i>DxMap</i> table.

2.2.3.4 DispoMap Table

Not all dispositions are available at every site or there may be a local disposition that provides the same service as one indicated by NHS Pathways. This table allows the organisation using NHS Pathways to map their own disposition words to the Disposition codes and to make a disposition available/unavailable to service and or skill set.

Field	Type	Example	Size	Notes
<i>DispoCode</i>	VarChar Required primary key	Dx011	10	The code used to identify the level of disposition e.g. Dx011 = ambulance 8 minutes.
<i>PDTDisposition</i>	text	The disposition is “999” ambulance required	200	The standard PDT disposition text.
<i>LocalDisposition</i>	Text required	The disposition is “999” ambulance required	200	Local text for display in interfaces.
<i>DxRank</i>	integer	1 through to 1000	4	Used to set triage cutoff/ambulance stand-down. 10 is currently the highest-ranking disposition. The list of cut off choices are provided in DxRank CSV.
<i>IncludeOOH</i>	Integer required	0 or 1	1	If set to 1 then disposition will be available to clinicians in OOH for override.
<i>IncludeAmbulance</i>	Integer required	0 or 1	1	If set to 1 then disposition will be available to clinicians in ambulance setting for override.
<i>IncludeSPA</i>	Integer required	0 or 1	1	If set to 1 then disposition will be available to clinicians in SPA for override.
<i>ActiveOOH</i>	integer	0 or 1	1	Identifies dispositions which are active in OOH.
<i>ActiveAmbulance</i>	integer	0 or 1	1	Identifies dispositions which are active in ambulance setting.
<i>ActiveSPA</i>	integer	0 or 1	1	Identifies dispositions which are active in SPA setting.
<i>CMSGroup</i>	integer	23	5	The DoS disposition code according to urgency. This value forms part of the request sent to DoS. If the value is NULL, then the DoS will not be called.

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<i>AmbPriorityType</i>	VarChar	R1, R2, AR, AT, P2T etc	5	Clarifies which Dx codes lead to ambulance dispatch and at what level. Local settings possible but refer to Pathways.
<i>SubstituteCode</i>	integer	4003 etc	4	Indicates which SD code to use when clinician overrides system disposition and new disposition will go to DoS.
<i>ActiveFOHINT</i>	Integer	0 or 1	1	Identifies dispositions which are active in the Reception Point Integrated ED and UCC setting.
<i>SubstituteGroup</i>	Integer	1207	4	Indicates which SG code to use when clinician overrides system disposition. The new disposition should be the one in the request that goes to DoS.

2.2.4 Care Advice (Cx)

2.2.4.1 Care Advice: Definition

Care advice is given to the caller as instructions to be followed in the treatment of their symptoms either as home care advice or while awaiting treatment. The advice may also include details of how to obtain/prepare for treatment. A care advice topic, e.g. Wound Management, will have several advice items for the user to give. The topic + items is referred to as a care advice set. The totality of homecare advice sets is referred to as the care advice bank.

Care advice sets presented to the user are matched to the patient age and gender. Care advice items are always multi-selectable. There can be up to 15 care advice items in a set. Home care advice bank is only presented to a clinician on reaching a home management type disposition.

2.2.4.2 In-line and post-disposition care advice

Normally care advice is presented to the user following the disposition. The triage answers selected and the disposition reached determines what care advice sets are presented to the user. If a triage answer with a *Cxkeyword* is selected during triage then that topic is presented following disposition. For example, if a user selects 2 “a raised temperature was described” this answer has *Cxkeywords* “Fever” and “Medication, pain and or fever” attached so both these care advice sets will be presented to the user following disposition this will have type set as ‘Interim’ advice.

In some circumstances it is necessary for care advice to be given during triage questioning. For instance, a patient who has a nosebleed may be given symptom relief advice before continuing the triage. Similarly, instructions such as CPR may be given in an ambulance setting before more triage questions determine other symptoms. In these cases the Cx set is referred to explicitly in the triage logic and will have a type set as ‘inline’.

2.2.4.3 Dispositions and Care Advice

As stated above, care advice is pre-selected for users depending on their position in a triage or the *Cxkeywords* collected from Pathway and answer selection during triage. However, if the disposition is home care which is currently only reached by clinicians then the clinician is able to view the whole of the care advice bank for that age and gender of patient. The **DispositionItems::TransferCode** field determines whether the user is directed to the Cx1 screen (predetermined advice) or Cx2 screen (care advice bank). Cx1 screen should only display sets of the “interim” type whereas the Cx2 screen displays sets of the “home care” type.

2.2.4.4 Disposition Advice

Some care advice sets are designed specifically to be attached to dispositions and provide instructions on accessing the service identified e.g. “I will arrange for a doctor to speak to the individual”. Such advice has a *KwSpecific* value matching the **DispositionItem::DispoAdviceCode** e.g. Dx106 and is displayed with the disposition rather than in the Care Advice interface. As such it does not have a *Cxkeyword* value and cannot be accessed from the Care Advice selection lists, this has a type set as ‘dispo’.

2.2.4.5 CareAdvice Table: Field Definitions

Field	Type	Example	Size	Notes
<i>CxID</i>	Required Primary Key	Cx12345	10	The care advice ID. All care advice has an ID beginning with Cx.
<i>CxAge</i>	Text required	Adult Child	50	List of target age groups. Matches the age group of the patient so that only care advice suitable for that age group is presented. Pipe separated where more than one age group is target.

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<i>CxGender</i>	Text required	Male Female	20	List of target genders for the care advice. Can be either or both, pipe separated.
<i>CxKeyword</i>	Text	Symptom-Bruising	200	Matches the keywords collected during triage from Pathway selection and answer.
<i>KwSpecific</i>	text	Dx01	200	Identifies care advice sets for display as specific disposition instructions.
<i>KWGroup</i>	text	Abdomen fever	500	One or more groups that a care advice set belongs to. Required to provide group value list for Care Advice Cx2 screen.
<i>Type</i>	text	Home care, interim, in line, dispo	20	Type “home care” is displayed in Cx2 screen, “interim” in Cx1 screen. Others are displayed when specifically called.
<i>Cx_Dx</i>	Text	Dx11 Dx12 Dx14	1000	List of target dispositions of the care advice. Only care advice identified by the disposition reached is presented. Pipe separated where a care advice set is used by more than one disposition.
<i>AlternativeTitleDigital*</i>	Text(HTML)	Cold exposure	5000	An alternative title (if provided) for display instead of KWSpecific in the online digital setting. Used in Online Digital setting only, and should be ignored for Telephone triage

* Please note that the *AlternativeTitleDigital* field is now deprecated and will be removed in a future release of Pathways.

2.2.4.6 CareAdviceItems: Field Definitions

Field	Type	Example	Size	Notes
<i>CxID</i>	Required	Cx12345	10	The care advice ID. Relates to the CareAdvice table.
<i>CxNo</i>	integer required	1,2,3, etc	2	The item number. Determines the sort order in the display. Max 15.
<i>CxText</i>	Text required	Antibiotics are not usually needed to treat a sore throat	200	The item text.
<i>Include</i>	integer	0 or 1	1	Some items are descriptive separators and do not need a checkbox in the display. 1 = display checkbox, 0 = do not display checkbox.
<i>ActiveOOH</i>	integer required	0 or 1	1	Flag that this item be displayed in OOH setting.
<i>ActiveAmbulance</i>	integer required	0 or 1	1	Flag that this item be displayed in ambulance setting.
<i>ActiveSPA</i>	integer required	0 or 1	1	Flag that this item be displayed in SPA setting.
<i>ActiveFOHINT</i>	integer required	0 or 1	1	Flag that this item be displayed in Reception Point integrated ED and UCC setting.
<i>CxTextDigital*</i>	Text(HTML)	Antibiotics are not usually	5000	The item text for display in online setting.

		needed to treat a sore throat		Used in Online Digital setting only, and should be ignored for Telephone triage
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* Please note that the *CxTextDigital* field is now deprecated and will be removed in a future release of Pathways.

2.2.4.7 CxExclude Field Definitions

Field	Type	Example	Size	Notes
<i>CxKeyword</i>	Text	toothache	200	The overriding keyword to be included.
<i>ExcludeKw</i>	Text	Medication, pain	200	The keyword to be excluded if the value in <i>CxKeyword</i> is collected.

2.2.4.8 BodyMap: Field Definitions

The body maps are used for pathway selection. The BMGroup determines which value list an item is displayed in, the BMItem is one of the match fields that determine which pathways are available to the user for selection and corresponds to the **Pathways::PW_ClinicalArea** field.

Field	Type	Example	Size	Notes
<i>BMGroup</i>	Text Required	head	50	The group name.
<i>BMItem</i>	Text required	nose	50	The item text for the body map list.
<i>Ordering</i>	Integer Required	1	1	The ordering of the BMItem within each BMGroup (display on BodyMap 2.5.4).

2.2.5 TriageLogic

2.2.5.1 TriageLogic: Definition

An explanation of how triagelogic works has been given in the introduction. The TriageLogic table contains the jump logic for navigating through a Pathway. A single record in the **TriageLogic** table has match fields to the tables above so that the system knows what data to display in a given interface and what item (Tx, Cx, Dx) to display next for a given user input. As such it is the key table for driving the Pathways system

2.2.5.2 TriageLogic: Field Definitions

Note there is no key field in this table as such. A given Pathway may have several answers at a particular order number. However, the combination of PW_ID with Order No and AnswerNo is unique to the triagelogic table.

Field	Type	Example	Size	Notes
<i>PW_ID</i>	required	PW123	10	The Pathway that the record belongs to. Relates to the Pathways table.
<i>OrderNo</i>	integer	0, 20000	10	All Pathways start with an order number 0. Because of insertions and deletions during authoring, the progression of numbers is not always even.
<i>Qu_ID</i>	required	Tx1066 or Cx1112 or Dx1071	20	The question that is to be used at this point in the Pathway. The prefix (Tx, Cx, Dx) tells the system which tables to pull data from. E.g. a Tx prefix tells the system to pull related question data from TxQuestions .

<i>AnswerNo</i>	integer	1 or 2 etc	2	The answers to be used at this point in the Pathway. Combined with the <i>Qu_ID</i> and <i>AnswerNo</i> will find a primary key match from the relevant table.
<i>JumpID</i>	Integer required	-1 or 1000 or 200 etc	10	Identifies the next record to jump to. Negative numbers have specific meanings while positive numbers indicate the next order number in the current Pathway to jump to.
<i>T1</i>		Dx01	20	If the jump code is “-1” then this field gives the block_order_calc, a disposition or an expression to evaluate to decide where to go next.
<i>T2</i>		PW124.0	20	If the jump code is “-1” then this field gives the block_order_calc, a disposition or an expression to evaluate to decide where to go next.
<i>T3</i>		PW125.10	20	If the jump code is “-1” then this field gives the block_order_calc, a disposition or an expression to evaluate to decide where to go next.
<i>T4</i>		Dx02, PW1234.50		If the jump code is “-1” then this field gives the block_order_calc, a disposition or an expression to evaluate to decide where to go next.
<i>Rank</i>	integer	0 to 1000		Rank indicates the highest disposition a given answer selected could lead to further down the triage. “0” indicates unranked and therefore no action should be taken. The actual dispositions are given by the DxRank column in the dispomap table. See section on Look Ahead and Stand down below for explanation.

2.2.6 System Variables

2.2.6.1 System Variables Definition

These are values that are read from the SystemVariables.csv file in the release files that should be automatically imported as part of the release upgrade. New system variables can be added in any release and so should be expected and handled automatically by the host system. These variables can be amended in the host system by either the suppliers or via an interface for high level administrators at site according to local policy if locally there is a need to differ from the national expectation. Some of these values are evaluated by rules within the pathways triage system, others are for information only such as NHS Pathways Release Version.

2.2.6.2 System Variable - Field Definitions

Field	Type/properties	Size	Example	Notes
<i>Variable Name</i>	Text required	100	SYSTEM_FLUPANDEMIC	No spaces. Should start with 'SYSTEM_'. This would be the value in the TxQuestions::QuestionText of the evaluate variable question.
<i>VarType</i>	Text required	20	STRING	Describes the type of variable.

<i>Explanation</i>	Text required	200	Flu Epidemic Level	Description of the system variable.
<i>Values</i>	Text Required	100	pandemic	This is the positive value which would be in the TxAnswers::AnswerText field of the evaluate variable question.

A non-exhaustive list of variables to offer examples:

VariableName	VarType	Explanation	Values
SYSTEM_FLUEPIDEMIC	STRING	Flu Epidemic Level	epidemic
SYSTEM_FLUPANDEMIC	STRING	Flu Pandemic Level	pandemic
SYSTEM_VERSION	STRING	Pathways Release ID	12.0.0
SYSTEM_RPContentED	STRING	Reception point with ED	RPED
SYSTEM_RPContentUCC	STRING	Reception point with UCC	RPUCC
SYSTEM_MERS	STRING	MERS virus	mers
SYSTEM_VHF	STRING	Viral Haemorrhagic Fevers	Vhf

2.2.7 System Flags

2.2.7.1 System Flag Definition

These are values that are set by the Pathways system and may be used either by the host system or by the Pathways system. For example, the trauma value is required to determine which pathways are displayed at the Pathway selection screen, whereas the scene unsafe flag may trigger an activity in the host dispatch system.

2.2.7.2 System Flag Table

Field	Type/properties	Size	Example	Notes
<i>SystemFlagCode</i>	Number	3	1	The number which would be found in the TxAnswers::SystemFlag field.
<i>SystemFlagDescription</i>	Text required	200	scene safe	Describes the meaning of the system flag.

2.2.7.3 System Flags set by answer selection

As mentioned above the **TxAnswers::SystemFlag** field holds values that can alter the state of system flags which may be needed by Pathways or the host system. Flags defined to date are:

TxAnswers::SystemFlag Value	Action
0	Set Pathways trauma flag to "non trauma"
1	Set Pathways trauma flag to "trauma"
2	Trapped at scene
3	scene safe
4	Scene unsafe
5	Police informed
6	Fire informed
7	Crew arrived
9	AED on scene

2.2.8 DxRank

2.2.8.1 DxRank Definition

These are described below in ambulance stand down/triage cut off section. The values in the DxRank table are to be used as the options presented to the administrator setting the cut off levels. These should not be accessible to call handlers and clinicians.

2.2.8.2 DxRankTable

Simple look up table for DxRank and what each value refers to. For more information on where DxRank is used in Triage cut off, please refer to Chapter 8.

Field	Type/properties	Size	Example	Notes
<i>DxRank</i>	Number	4	170	The DxRank cut off level.
<i>DxText</i>	Text required	255	Cut after ambulance response	Describes the meaning of the cut off level.

2.2.9 Skillsets

2.2.9.1 Skillsets Definition

Look up table for Skillsets used in the Pathways system. For more information on how Skillsets are used to drive CAS please refer to Chapter 11. Skillsets should only be enabled where approved by the Pathways team.

2.2.9.2 Skillsets Table

Field	Type/properties	Size	Example	Notes
<i>SkillsetId</i>	Number required	3	5	Unique numeric identifier for the skillset. Once created for a skillset, this value is not expected to change.
<i>SkillsetDescription</i>	Text required	200	111 Call Handler	Text description for the skillset i.e. 111 Call Handler
<i>SkillsetUserType</i>	Text required	2	CH or CL	Used to describe the user type i.e. whether the skillset is classed under a Call Handler (CH) skillset or a Clinician (CL) skillset. CQI reporting contain numerous columns to record when the call started, when disposition was reached, when the call ended etc. These exist for both Call Handler (i.e. chCallStartTime, chDispositionTime, chCallEndTime) and Clinician (i.e. clCallStartTime, clDispositionTime, clCallEndTime). Therefore, the SkillsetUserType will denote which column type the data should be stored against.
<i>Service</i>	Text required	3	OOH or AMB or	Used to specify the service type. This relates to the ActiveOOH, ActiveAMB, ActiveSPA, ActiveFOH

			SPA or FOH	columns used within the main NHS Pathways system. Therefore, a skillset with a value of “SPA” should use the ActiveSPA column logic within the appropriate Care Advice and Disposition tables.
<i>EntryPathwayId</i>	Text required	50	PW1808 or AgeGenderParty	For skillsets 1 to 6, there is no change to the mechanism of finding the entry pathway ID – it is still based on a look up of Age, Gender and Party against the Pathways table. For newer skillsets such as Pharmacy and HCP Callback, the EntryPathwayID and EntryOrderNumber fields in the skillsets table should be used to determine the starting pathway and order number for that skillset.
<i>EntryOrderNumber</i>	Number required		0	As above
<i>Tbranch</i>	Text required	2	T3	Rather than creating new Tbranch columns for each of these skillsets, one of the existing T pathway triage logic branches will be used (i.e. T1 to T4) and this is specified in the “Tbranch” column.
<i>EarlyExitPathwayID</i>	Text required	50	PEE4	The early exit pathway and Order number to be used for this skillset.
<i>EarlyExitOrderNumber</i>	Number required		500	As above
<i>Enabled</i>	Boolean		True	Used to describe if the skillset is enabled in the current system. This value can be changed to suit the local system setup/requirements i.e. in a 999 only site, only the 999 skillsets would be enabled (True), with the rest set to disabled (False).
<i>AlternativeDisplayText</i>	Text	50	EDS Online	Used to indicate if alternative text is to be used for this skillset. Used in Online Digital setting only, and should be ignored for Telephone triage A pipe separated list of tokens indicating the order alternative text will be used. These tokens are used to identify relevant alternative text rows from the AlternativeText table, so need to match the content of the ItemType field. In the example given, the skillset will use “EDS” specific text if available, otherwise it will use “Online” specific text if available, and finally will use standard Pathways Text if neither of the previous alternatives are available per item.

2.2.10 BodyMapOverrides

2.2.10.1 BodyMapOverrides Definition

Look up table for Overriding the Body Map pathway selected. Currently only used for Senior Clinician Module as part of the Clinical Assessment Service - please refer to Chapter 11.

2.2.10.2 BodyMapOverrides Table

Field	Type/properties	Size	Example	Notes
<i>SkillsetId</i>	Number required	3	12	SkillsetId which this override refers to
<i>BodyMapPathway</i>	Text required	10	PW1345	Pathway selected on the BodyMap, or an asterisk wildcard
<i>BodyMapPathwayJumpsTo</i>	Text required	10	PW1821	Pathway to jump to (effectively overriding the pathway selected by the user).

2.2.11 Symptom Groups

2.2.11.1 SymptomGroups Definition

Look up table for Symptom Groups. Each release will now contain a data table containing all Symptom Groups used by the release. The table contains the ID and description associated with each Symptom Group.

Should the host system display the Symptom Group at any point, this table can be used to extract the appropriate description for it.

2.2.11.2 SymptomGroups Table

Field	Type/properties	Size	Example	Notes
<i>SgId</i>	integer required		1231	Id of the Symptom Group
<i>SgDesc</i>	Text required	100	Drowning Incident	Description of the Symptom Group

2.2.12 Symptom Discriminators

2.2.12.1 SymptomDiscriminators Definition

Look up table for Symptom Discriminators. Each release will now contain a data table containing all Symptom Discriminators used by the release. The table contains the ID and description associated with each Symptom Discriminator.

Should the host system display the Symptom Discriminator at any point, this table can be used to extract the appropriate description for it.

2.2.12.2 SymptomDiscriminators Table

Field	Type/properties	Size	Example	Notes
<i>SdId</i>	integer required		4496	Id of the Symptom Discriminator
<i>SdDesc</i>	Text required	100	AMB ectopic pregnancy with TLOC	Description of the Symptom Discriminator

2.2.13 Pathways Clinical Consultation Support (PaCCS)

2.2.13.1 PaCCS Data Definition

The PaCCS data set consists of 5 different tables:

- ☐ CCSTemplates
- ☐ CCSConditions
- ☐ CCSTemplateConditions
- ☐ CCSConditionSupportingInformation
- ☐ CCSJumps

Details on how to use this data can be found in the related PaCCS sections of this specification.

2.2.13.2 CCSTemplates Table

This table contains the definitions of the Templates used within PaCCS.

Field	Type	Example	Size	Notes
TemplateId	Text required	CS0078	20	Primary key identifier
Version	Text required	1.0.1	10	The version number of the template
TemplateName	Text required	Headache	50	The name of the template
Synonyms	Text	Migraine Pain	250	Pipe separated list of synonyms for searching
SymptomGroup	Integer required	1112	4	The default Symptom Group associated with the template
ServicesJump	Text required	PW752.0	20	The entry point into standard Pathways used when the user clicks on the Service Search button.
ETCJump	Text required	PW1812.5	20	The entry point used when the user clicks on the Emergency Treatment Centres button.

2.2.13.3 CCSConditions Table

This table contains the definitions of the Conditions which appear within a Template.

Field	Type	Example	Size	Notes
ConditionId	Text required	Cn1234	20	Primary key identifier
ConditionName	Text required		250	The name of the condition
SymptomGroup	Integer	1112		If non-Null, this SG will override that of the template.
Importance	Integer required	1		1 = High (red) 2 = Normal (amber) 3 = General Marker (green)
AmbJump	Text	PW123.4	20	Null if disposition not appropriate
HomeMgmtJump	Text	PW234.5	20	Null if disposition not appropriate
ServicesJump	Text	PW752.0	20	Null if disposition not appropriate
ETCJump	Text	PW1812.5	20	Null if disposition not appropriate
Gender	Text required	male female	12	Pipe separated list of Genders the condition is appropriate for.

Field	Type	Example	Size	Notes
Age	Text	adult child toddler infant	50	Pipe separated list of Age groups the condition is appropriate for.

2.2.13.4 CCSTemplateConditions Table

This table contains the link between Templates and Conditions, defining which Conditions appear in which Templates, and the order they appear within each Template. It does not contain the link between the Green Marker Conditions (Importance = 3) and Templates, as all Green Markers must be displayed on every Template ordered alphabetically by **CCSCondition::ConditionName**.

Field	Type	Example	Size	Notes
TemplateID	Text required	CS0078	20	The template identifier
ConditionID	Text required	Cn1234	20	The condition identifier
OrderNo	Integer required	1		Condition order within the given template

2.2.13.5 CCSConditionSupportingInformation Table

This table contains the Supporting Information applicable to the defined Conditions. Each Condition may have 0 or more SupportingInformation records. When displayed, these should be ordered as per the defined OrderNo field for each Condition.

The Supporting Information HTML column can contain encoded links to external websites. Should these links need updating, we will either inform you of the update needed to your database, or produce an updated release file if there are multiple updates needed. This process may be further refined post-beta.

Field	Type	Example	Size	Notes
ConditionId	Text required	Cn1234	20	References ConditionId in CCSConditions.csv. Part 1 of primary key identifier
CategoryTitle	Text required	Overview	100	Title of the category of the supporting information that is to be displayed to the user.
CategoryId	Text required	ov	100	A consistent id for a category of supporting information. Part 2 of the primary key
OrderNo	Integer required	1		The order in which to display the various categories of supporting information for a condition to the user. E.g 1) overview 2) symptoms, 3) risk factors
Text	Text (html)	An epileptic seizure is the transient occurrence of signs or symptoms due to abnormal electrical activity in the brain. This manifests itself as a disturbance of consciousness, behaviour, emotion, motor function, or sensation.	5000	The html content for a particular category of advice for a condition.

2.2.13.6 CCSJumps Table

This table contains the definition of the buttons that appear at the top of the PaCCS User Interface.

Field	Type	Size	Example	Notes
Label	Text required	100	Ambulance /Immediate Threat to Life	The navigation text
OrderNo	Number required		1	The order of the navigation
PathwayJump	Text required	20	PW1812.0	The destination of the navigation
ButtonType	Text required	2	AD	The two character code indicating the type of button to be rendered. Options are: AD –Ambulance HC –Home Care SS –Service Search ET –Emergency Treatment Centres Buttons should be rendered in the same style as the Disposition Buttons against a Condition (e.g. AD type buttons should be Red). This code should also be used when populating the quID field of the PaCCS TriageRecord as detailed in section 13.2.3

2.2.14 Alternative Text

2.2.14.1 Alternative Text Definition

Used in Online Digital setting only, and should be ignored for Telephone triage

Table used to contain alternative text used for certain implementations of Pathways. Whether to use Alternative Text is determined by the AlternateDisplayText field in the Skillsets table. Where this is set, the wording to display to the user should be sourced from this Alternative Text release file where available, before reverting to the standard Pathways text if an alternative is not available.

For example, if the Skillset being used has the AlternateDisplayText set to “EDS”, when displaying the Answer Text for the second answer of Tx123456, the system should look in the Alternative Text file for a row with the DisplayType set to “EDS”, the ItemType set to “TxText”, the ItemId set to “Tx123456”, and the ItemNumber set to 2. The text to display will be in the ItemText field. Similarly, a Skillset can have multiple AlternateDisplayText set, giving a priority list of alternative texts to use. For example, having AlternateDisplayText set to “EDS|Online” would mean that the Skillset should use alternative “EDS” text if available, or else, alternative “Online” text if available, or else use the standard Pathways text.

2.2.14.2 AlternativeText Table

Field	Type/properties	Size	Example	Notes
<i>DisplayType</i>	Text required	20	EDS	AlternateDisplayText Identifier
<i>ItemType</i>	Text required	20	TxText	The type of text defined. Options are: TxText – Question Text

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				TxRationale – Question Rationale TxAnsText – Answer Text TxAnsRationale – Answer Rationale DxDispositionText – Disposition Text CxTitle – Care Advice Title CxItemText – Care Advice Item Text
<i>ItemId</i>	Text required	20	Tx123456	The unique identifier for the item whose alternative text is defined. This could be a Question, Care Advice or Disposition code.
<i>ItemNumber</i>	Integer		1	Where an Item has multiple instances (e.g. Answers), this will contain the number of the item. For example, where the second Answer of a Question is being defined, this column will contain the number 2. Where an item can only have one definition (e.g. a Care Advice Title), this column should not be used when finding the relevant text.
<i>ItemText</i>	Text required	5000	Are you confused?	The alternative text to be displayed for this type/item/ID/Number.

2.3 Host System Data Tables

2.3.1 Host System Data

When a user interacts with the Pathways system, their actions are recorded. The records for one triage episode all belong to the same *SupplierCallRef*.

The following tables show the data that the Pathways system will need to obtain from the host system in order to construct triage records of the user interactions with the system. This data is user and call dependent. The field names will depend on the host system but have been labelled as follows in order to show relationships with Pathways tables later. There are some field names that need to remain consistent with the NHS Pathways system in order to ensure the data is imported correctly into the IDT webservice. **The fields that need the naming convention to be maintained are double underlined**

2.3.1.1 SystemUser: Field Definitions

If a call is taken solely by a call handler then only the *chusercode* and *chskillset* will be needed. Where the call is passed from a call handler to a clinician then both will be required. The values should be unique to the site.

Field	Type	Example	Notes
<u>chUsercode</u>	Text	Host defined	Call Handler UserID. All triage records need a <i>UserID</i> as more than one user may triage a call if collected from a queue.
<u>chSkillSet</u>	Integer	1,3,5,7	Determines which transfer codes in triagelogic to use.
<u>clUsercode</u>	Text	Host defined	Clinician UserID. All triage records need a <i>UserID</i> as more than one user may triage a call if collected from a queue.
<u>clSkillSet</u>	Integer	2,4,6	Determines which transfer codes in triagelogic to use.

2.3.1.2 Patient: Field Definitions

Field	Type	Example	Notes
<u>ageGroupId</u>	required	Adult, Child, Toddler or Infant	Adult: 16 years and over Child: 5 to under 16 years Toddler: 1 to under 5 years Infant and Neonate: Under 1 year. Should be evaluated to one of these groups by the host system.
<u>genderId</u>	required	Male or female	Needed to access care advice sets and dispositions and to determine which transfer codes to use.
<u>partyId</u>	Integer required	1 or 3 or 4	Needed to select correct Module 0 Pathway.

2.3.2 Pathways Case Data

Data from the triage process along with associated patient and user information needs to be recorded in order to:

-  Provide an accurate historical record of the triage
-  Data analysis for the operating site and the NHS Pathways team
-  Transfer to other systems, co-located or elsewhere for further triage – this is now the payload of the national interoperability message.

The three data tables shown below indicate the data required for by the NHS Pathways team for reporting purposes. For historical record and transfer purposes it is assumed that further patient/caller demographics would be included. The data is transferred to NHS Pathways via methods described in section 3 below. Section 3 gives full details of the format and url to be used. Other fields are included in these tables and these are used for the generation of written/electronic reports to be sent to other interested parties such as GP OOH, patient's own GP, etc, or utilised in house for call audit.

A Pathways case is made up of a single **case** record, multiple **triage** records and if a call to the DoS was made – multiple **service return** records.

2.3.2.1 CaseRecord

Most, if not all of this data will already be collected by the host system but is included here to indicate the NHS Pathways team requirements and where the data is to be used.

NB There are cases where the call is fully completed from end to end by the clinician. In these instances, we would expect all the 'ch' call handler fields to be empty but the 'cl' clinician fields to be completed. Similarly, there will be instances where the call handler completes the call and only the 'ch' fields would be returned.

Field	Type	Example	Size	Notes
<u>siteId</u>	Integer <i>Required</i>	2		Unique ID provided by the NHS Pathways team to identify the site.
<u>supplierCallRef</u>	Text <i>Required</i>	30ac7afe-1e34-4848-a06a-ff605a299f62	36	System suppliers case number. This is the value that NHS Pathways can use to request call reports and call recordings
<u>ageGroupId</u>	Integer <i>Required</i>	1, 2, 3 or 4		1 (Adult), 2 (Child), 3 (Toddler), 4 (Infant) NB when finding matches in the Pathways tables the text is used NOT the integer
<u>ageInYears</u>	Integer	13		Patient age in years (rounded down, to nearest whole year). This should be an age of '0' if under 1 year of age. If infant is reported as 15months/18months, the age is '1'. Null value if age in years is not known.
<u>genderId</u>	Integer <i>Required</i>	1, 2 or 3		1 (Male), 2 (Female), 3 (Unknown) NB when finding matches in the Pathways tables the text is used NOT the integer
<u>partyId</u>	Integer <i>Required</i>	1 or 3		1 (1 st party), 3 (3 rd Party)
<u>Trauma</u>	Boolean <i>Required</i>	true or false	5	Trauma flag

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<u>chUserCode</u>	Text	NFT	Max 50 characters	Call Handler user code. This should be unique to each site
<u>chSkillsetId</u>	Integer	3		Call Handler skillset 1 (OOH Call Handler), 3 (999 Call Handler), 5 (111 Call Handler), 7 (RP Call Handler) 9 (Repeat Prescription) 10 (Dental) 11 (HCP Callback)
<u>chCallStartTime</u>	DatetimeOffset	2013-10-25T15:54:00+0100	24	Call Handler call start time
<u>chTriageStartTime</u>	DatetimeOffset	2013-10-25T15:55:13+0100	24	Call Handler triage start time
<u>chDispoTime</u>	DatetimeOffset	2013-10-25T16:02:06+0100	24	Call Handler dispo time
<u>chTriageEndTime</u>	DatetimeOffset	2013-10-25T16:04:33+0100	24	Call Handler triage end time
<u>chCallEndtime</u>	DatetimeOffset	2013-10-25T16:05:01+0100	24	Call Handler call end time
<u>chTriageDispoCode</u>	Text	Dx38	Max 10 characters	Call Handler triage dispo code
<u>chFinalDispoCode</u>	Text	Dx38	Max 10 characters	Call Handler final dispo code
<u>chOverrideFlag</u>	Boolean	true or false	5	Call Handler Disposition override flag
<u>clUserCode</u>	Text	PSF	Max 50 characters	Clinician user code. Unique to the site
<u>clSkillsetId</u>	Integer	4		Clinician skillset 2 (OOH Clinician), 4 (999 Clinician), 6 (111 Clinician), 8 (RP Clinician) 12 (Senior Clinician Module)
<u>clCallStartTime</u>	DatetimeOffset	2013-10-25T16:05:20+0100	24	Clinician call start time
<u>clTriageStartTime</u>	DatetimeOffset	2013-10-25T16:06:00+0100	24	Clinician triage start time
<u>clDispoTime</u>	DatetimeOffset	2013-10-25T16:09:56+0100	24	Clinician dispo time
<u>clTriageEndTime</u>	DatetimeOffset	2013-10-25T16:11:25+0100	24	Clinician triage end time
<u>clCallEndtime</u>	DatetimeOffset	2013-10-25T16:12:03+0100	24	Clinician call end time
<u>clTriageDispoCode</u>	Text	Dx05	Max 10 characters	Clinician triage dispo code

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<u>clFinalDispoCode</u>	Text	Dx06	Max 10 characters	Clinician final dispo code
<u>clOverrideFlag</u>	Boolean	true or false	5	Clinician Disposition override flag
<u>symptomGroup</u>	Integer	1112		The group of symptoms to which the pathway belongs. This should be the last value captured by the system.
<u>symptomDiscriminator</u>	Integer	4003		The discriminator that is produced by the completion of the Pathways triage. This should be the last value captured by the system – later SDs replace earlier ones.
<u>releaseVersion</u>	Text <i>Required</i>	6.2.2	Max 10 characters	Pathways version in use for this case/triage
<u>clinicalAreaId</u>	Integer	1 to 22		ClinicalAreaId from the pathway selected 1 (eye) 2 (head) 3 (breast) 4 (urinary) 5 (arm) 6 (back) 7 (hip & leg) 8 (skin and hair) 9 (not area specific) 10 (flank) 11 (rectum/anus) 12 (shoulder) 13 (groin) 14 (genital) 15 (abdomen) 16 (chest) 17 (face) 18 (neck) 19 (ear) 20 (nose) 21 (mouth) 22 (throat)
<u>callerType</u>	Text	Some caller type text here	Max 50 characters	Text field to capture caller type(if used). Mainly applicable to 999 sites.
<u>callReason</u>	Text <i>Required if CallerType is populated</i>	Some reason text here	Max 50 characters	Text field to capture caller reason (if used). Mainly applicable to 999 sites.
<u>doSTransactionId</u>	String	30ac7afe-1e34-4848-a06a-ff605a299f62	36	This is the value sent back from the DoS when a search is completed and results sent back. This needs to be captured in order to link the whole patient episode together.

2.3.2.2 TriageRecord

The table below indicates the data to be collected as the user triages a caller. There will be multiple records in this reporting table for each patient triage episode.

Field	Type	Example	Size	Notes
TriageRecordID	String	1	50	Unique ID or serial number generated by the host system which identifies this individual triage record

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<u>supplierCallRef</u>	String	30ac7afe-1e34-4848-a06a-ff605a299f62	36	This field links the triage record to the case record. Only the caserecord::supplierCallRef is sent back to NHS Pathways
<u>userCode</u>	Text	NFT	Max 50 characters	All triage records need a UserCode as more than one user may triage a call if collected from a queue. This will be the <u>clUserCode</u> or <u>chUserCode</u> already collected for caserecord. Each triagerecord needs to identify which cl or ch was the current user
<u>timeIn</u>	DatetimeOffset	2013-10-25T15:55:13+0100	24	Creation date time of TriageRecord.
<u>timeOut</u>	DatetimeOffset	2013-10-25T15:55:46+0100	24	Date time record is exited – e.g. by activating next procedure.
<u>pwld</u>	Text	PW123	Max 20 characters	PathwayID.
<u>orderNo</u>	Integer	100		OrderNo of the triage record.
<u>qulD</u>	Text	Tx220074	Max 20 characters	Question (Tx130011), Care Advice (Cx150010), dispo code (Dx140000), or body map area (hip & leg).
<u>answerNo</u>	Integer	3		The actual answer number selected by the user
<u>userComment</u>	Text	specify comment here	Max 300 characters	Any extra information added by the user for this answer.
<u>actionId</u>	Integer	1 to 6		The procedure used by the user to leave the record. Mostly this will be next but there are other ways to exit a record. List of possible values: 1 (Back) 2 (Next) 3 (Change) 4 (Early Exit) 5 (Restart) 6 (End)
blockOrderNo	String	PW1653.500	50	The triagelogic::blockorder no o the question being displayed. This enables the Pathway being used and question at that point to be derived.
ReportText	String	There was no fever present.	250	The report text of the answer selected. In the case of a Tx item being selected this would be TxAnswers::AnswerReportText , in the case of a Cx being selected this would be CareAdviceItems::CxText . This is used in the full report text and on the navigation section of the user interface to be used to go back
IncludeinReport	Integer	0 or 1	1	This indicates whether the record needs to be included in the report or the validation screen. Some questions are not displayed such as rule questions or in such instances where the user went back and chose a different route via the triage

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Layout	String	BodyMapFemaleAdult, Tx1, Tx2, Cx1, bodymap	50	Records the interface style applied for this record i.e. what layout the user can see
DispoRationale	String	Fever, Unwell, Abdominal pain	100	Value from TxAnswers::DispoDisplayText . This is used in the Summary report and on the summary section on screen.
DispoOverride	String	Fixed value list see in figure 15 below	200	Broad reason for disposition being overridden by the clinician
CxKeywords	String	Medication, pain and or fever	255	Value taken from TxAnswers::CxSpecific on selection of the answer stem, or upon a Pathway jump Pathways:: CxKW

2.3.2.3 ServicesReturnedRecord

The table below indicates the data to be collected if the DoS is contacted as a result of the triage. There will be multiple records in this reporting table for the returns provided by the DoS.

Field	Type	Example	Size	Notes
<u>serviceId</u>	Integer <i>Required</i>	1355999568		DoS service ID.
<u>serviceOrder</u>	Integer <i>Required</i>	2		The order in which the service was displayed to the user by your system (i.e. 1 = first result).
<u>serviceStatusId</u>	Integer <i>Required</i>	1 to 4		Integer used to describe status of what happened with the service (default of 1). 1 (Service Returned) 2 (Service Rejected) 3 (Service Selected) 4 (Bulk Reject)
<u>rejectReasonId</u>	Integer	22		If the service was rejected, please include the reason why the service was rejected. Please refer to section 9.1.1 for more information on reject reasons.

2.3.3 Global Fields

In addition to the fields described so far, NHS Pathways needs some global variables setting for the duration of the case. In addition, NHS Pathways is coded to set some flags which may be of use to the host system.

2.3.3.1 XXX, YYY and ZZZ, Substitution

Some **TxQuestions::QuestionText** items contain the text string xxx, yyy or zzz e.g. “xxx felt ill with any of the following symptoms?”. These are used to make the question text more easily communicated to the caller by substituting gender/party specific text. For this to work the host system needs to inform pathways as to the gender and party status of the patient. The following table indicates the rules that determine the substitutions to be made:

Letter	Gender	Party	substitute
xxx	Male	1	Have you
xxx	Male	3	Has he
xxx	Female	1	Have you
xxx	Female	3	Has she
yyy	Male	1	Are you
yyy	Male	3	Is he
yyy	Female	1	Are you
yyy	Female	3	Is she
zzz	Male	1	Do you
zzz	Male	3	Does he
zzz	Female	1	Do you
zzz	Female	3	Does she

2.3.3.2 Pathway Selection

Pathway selection relies on presenting the user with a pre-filtered list of pathways. The filtering relates to values in the **Pathways** table, namely:

 *PW_Gender*

 *PW_Age*

 *PW_Trauma*

 *PW_ClinicalArea*

Patient age and gender have already been captured. Clinical area is user selectable in the pathways selection interface. **Pathways::PW_Trauma** has values of trauma or non trauma. Certain Tx answers have a **TxAnswers::SystemFlag** value of 0 or 1. The default setting of this variable is “0” indicating that the trauma flag is set to “non trauma” and therefore when a user gets to the pathways selection screen the default will be to display non trauma pathways. However, if during triage an answer is selected that has a SystemFlag value of 1 then the trauma flag should be set to “trauma” and trauma pathways should be displayed in the pathways selection screen (bodymap).

2.3.3.3 Other Variables/Global Fields

Variable	Description
<i>gBlockOrder</i>	Various procedures need to know which record to retrieve from the TriageLogic table. Stores the value for the next record to be retrieved. Values will typically be PW123.10 showing Pathway and order number. This is calculated from the TriageLogic::PathwayID and the TriageLogic::JumpID

<i>gLoopBlockOrder</i>	If a Pathway is looped out of to do other questions this variable stores the Triagelogic::BlockOrderNo to return to at the end of the loop
------------------------	---

2.4 Navigating the system

2.4.1 Introduction

When a user interacts with the NHS Pathways system what happens next depends on data held in the **Triagelogic** table and its children. This data determines not only what data to display next but also how the data should be presented.

2.4.2 Triagelogic Procedures

N.B. For the sake of clarity, in the relationship diagrams that follow, only those fields relevant to the relationship are shown.

2.4.2.1 Set up **CaseRecord** and 1st **TriageRecord** record

In order to begin triage the Pathways system needs to know where to start.

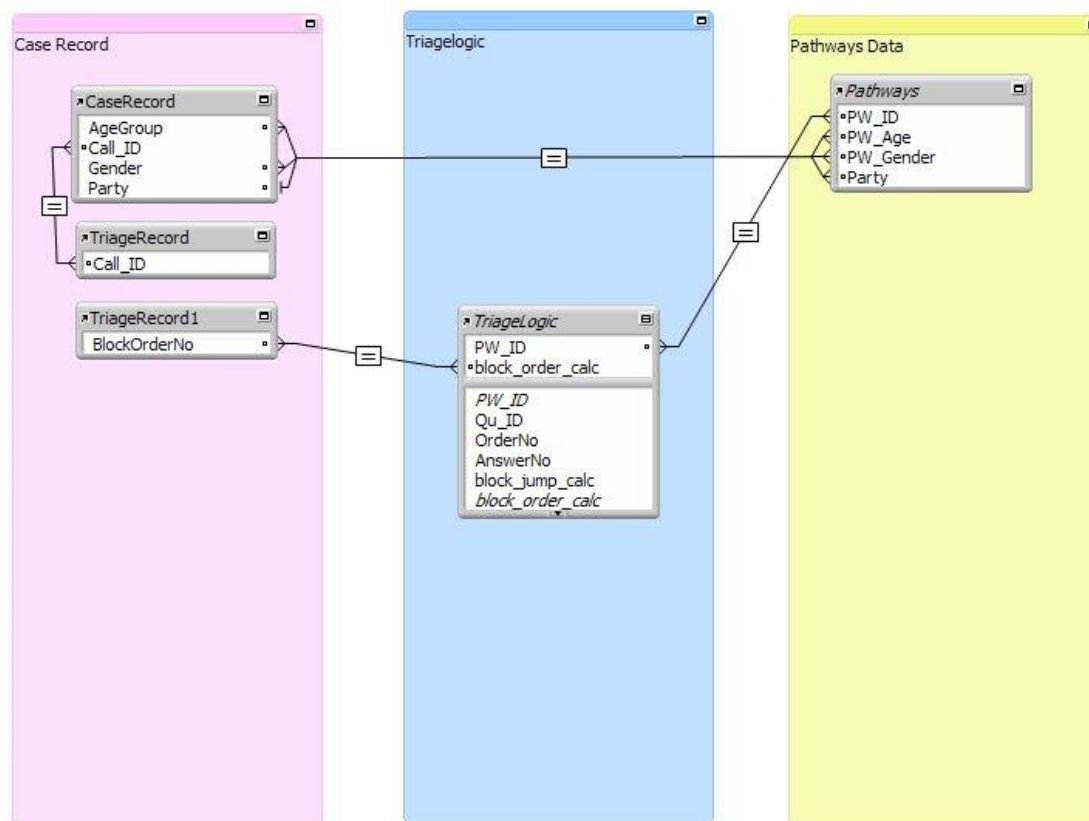


Figure 1 Start Pathways Relationship

For skillsets 1 to 6, there will only be one match in the **Pathways** table for *AgeGroup*, *Gender*, and *Party* of the patient. So the first record presented to the user will be determined by the first row in the **Triagelogic** table for this Pathway. In terms of the starting point order number to use, this is specified in the Skillsets table. This is to enable an introductory question to be asked in OOH/SPA to filter out callers for health information and routine appointments with district nurses etc.

The procedure for starting a new triage would be:

1. Create new record in **CaseRecord**
2. Set the fields *AgeGroup*, *CallID*, *Gender* and *Party* in **CaseRecord** to the host system's values for these fields. (Other fields such as *TriageStartTime*, *SiteID* etc. should also be set at this time)

NHS Pathways Specification

- ❓ Set the temporary variable *gBlockOrder* to:
Skillsets::EntryPathwayId + "." + **Skillsets::EntryOrderNumber**
from the Skillsets table for the current skillset.
- ❓ Where the EntryPathwayId = "AgeGenderParty", find a match in the **Pathways** table for *AgeGroup*, *Gender* and *Party* and replace the "AgeGenderParty" text with the matched **Pathways::PW_ID**
- ❓ Perform New Triage Record procedure (see below) to create a new **TriageRecord** with same *supplierCallRef* as parent in **CaseRecord**
- ❓ Each **TriageRecord** line should have the relevant *clUserCode* or *chUserCode* in the **TriageRecord::UserCode** field
- ❓ Set the **TriageRecord::BlockOrderNo** to *gBlockOrder*
- ❓ Display the data in Interface 1 (Single selection Triage Tx1)

Note: This process differs for HCP and IFT calls, please refer to section 12 for details.

The above applies where the host system has already collected demographics or has patient records and can pass the relevant information to NHS Pathways, e.g. typically OOH. In other settings such as ambulance services, this information may not have been gathered prior to starting triage. In this case, the NHS Pathways system starts with the following screen.

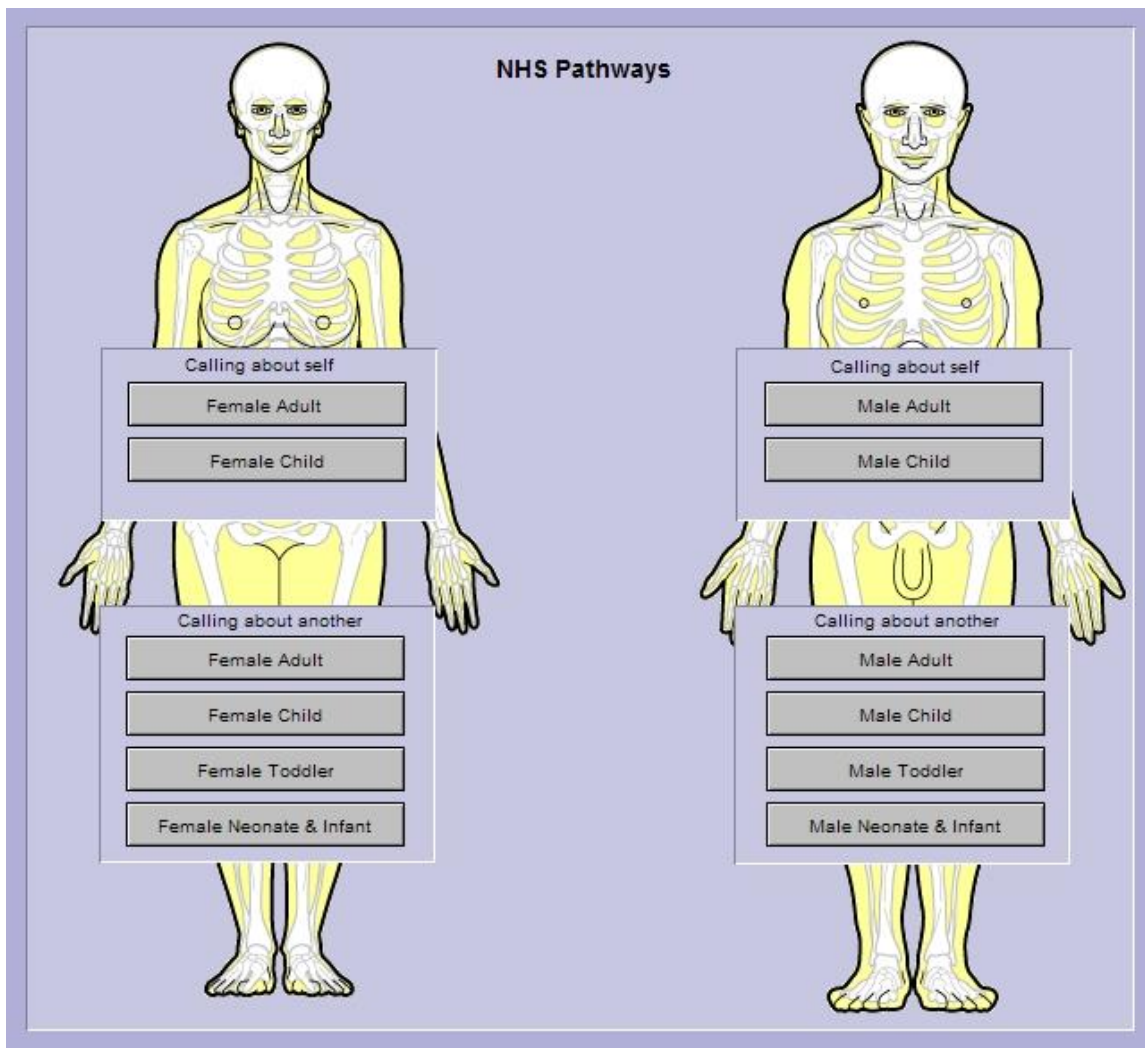


Figure 2 Pathways front screen - Ambulance

Each button sets the relevant party, gender and age group for the patient.

Once **TriageRecord::BlockOrderNo** is set, related data from **Triagelogic** and other tables can be displayed in the interface. The relationships that point to this data are shown below, highlighted in orange.

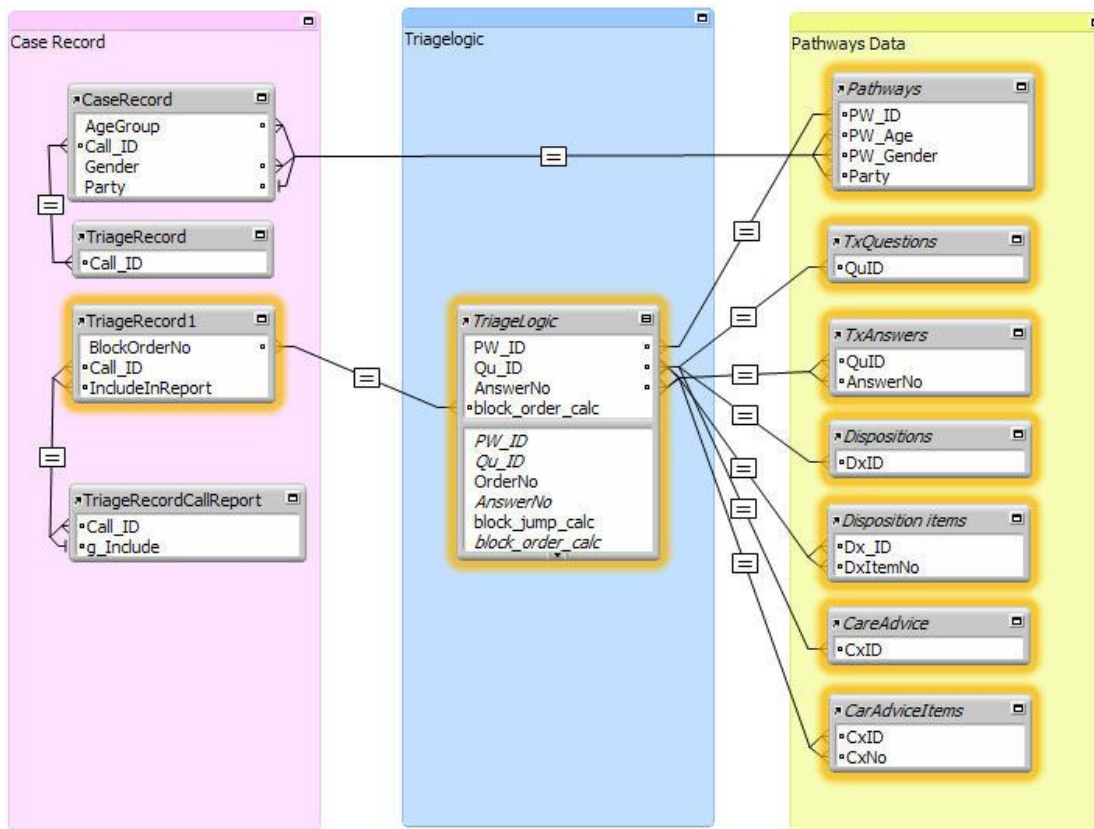


Figure 3 Relationship between TriageRecord and Triagelogic

2.4.2.2 Create new record in TriageRecord

- Create new record in **TriageRecord**
 - Set *supplierCallRef* to **CaseRecord::supplierCallRef**
 - Set *TriageRecordID*
 - Set *TimeIn*
 - Set *UserID*
 - Set *SkillSet*
 - Set *Siteld*
 - Set *IncludeInReport* = 1
 - Set *BlockOrderNo* to *gBlockOrderNo*

2.4.2.3 Write to Triage Record for Triage Questions

When a user selects an answer the data associated with the selection should be written to the record created for this *BlockOrderNo* in the **TriageRecord** table. The procedure given below demonstrates what to do if **Triagelogic::QuID** contains "Tx". Selecting an answer writes:

- Set **TriageRecord::AnswerNo** to **Txanswers::AnswerNo**

- ❓ Set **TriageRecord::ReportText** to **TxAnswers::AnswerReportText**
- ❓ If **Txanswers::specify** = 1 check that **TriageRecord::UserComment** is not empty
 - ❓ If it is empty, show custom dialogue "This answer requires additional information from you. Please enter comment". Dialogue box includes field **TriageRecord::UserComment**. Clicking OK checks for content then proceeds if field is not empty.
- ❓ Set **TriageRecord::CxKeywords** to **TxAnswers::Cxspecific**
- ❓ Set **TriageRecord::SymptomDiscriminator** to **TxAnswers::SymptomDiscriminator**
- ❓ Set **TriageRecord::DispoRationale** to **TxAnswers::DispoDisplayText**
- ❓ Set **TriageRecord::Layout** to "Tx1"
- ❓ Set **TriageRecord::TimeOut** to current date/time

2.4.2.4 Write to **TriageRecord** for Triage in-line Care Advice

For inline Cx items the process is similar to above:

- ❓ Set **TriageRecord::ReportText** to Advice given" & **CareAdvice::CxTitle** & **CareAdvice::CxText**
- ❓ Set **TriageRecord::AnswerNo** to **CareAdvice::CxNo**
- ❓ Set **TriageRecord::Layout** to "Tx2"
- ❓ Set **TriageRecord::TimeOut** to current date/time

2.4.2.5 Write to **TriageRecord** for Triage in-line Disposition

In-line dispositions are possible in ambulance settings i.e. for a skill set 3 using T3 transfer codes. For inline Dx items the process is:

- ❓ set **CaseRecord::chTriageDispoCode** OR **CaseRecord::clTriageDispoCode** OR to **DispositionItem::DispoAdviceCode**
- ❓ set **CaseRecord::chDispoTime** or **CaseRecord::clDispoTime** to current Date/Time
- ❓ Set **TriageRecord::ReportText** to **DispositionItem::DxItemText**
- ❓ Set **TriageRecord::Layout** to "Tx1"
- ❓ Send disposition data to dispatch desk
- ❓ Set **TriageRecord::TimeOut** to current date/time

Having written the data to the **TriageRecord** table, the next step is to create a new record in **TriageRecord** and populate it with the appropriate information from the related **TriageLogic** data. What happens in this "Next" procedure depends on the value of **TriageLogic::JumpID** of the related record.

2.4.2.6 "Next" when **TriageLogic::JumpID** =>0 or -1

By example, if the **TriageLogic::JumpID** is a positive integer, when a user clicks on an answer, a temporary variable (*gBlockOrder*) stores the value in **TriageLogic::block_jump_calc**. If **TriageLogic::JumpID** = "-1" then it is set to T1, T2 ,T3 or T4 of the currently matched record depending on the rules set out below.

A new record is created in the **Triage record** table and the **TriageRecord::BlockOrderNo** is set to *gBlockOrder*. In this way related data from the **TriageLogic** and child tables can be displayed. The display style is dependent on the type of Question and/or its control type. The sequence would be as follows:

-
- ❓ If **TriageLogic::JumpID** is =>0

Set *gBlockOrder* to **TriageLogic::PathwayID** & " " & **TriageLogic::JumpID**

- ❓ If **TriageLogic::JumpID** = -1
- ❓ Set *gBlockOrder* to **TriageLogic::(SkillSets[SystemUser::Skillset]::Tbranch)** – The SystemUsers skillset is used to lookup which Tbranch to use from the TriageLogic table.
- ❓ Set **TriageRecord::Action** to "Next"

❓ Create new record in **TriageRecord**

- ❓ Decide how to display data
- ❓ If *BlockOrderNo* contains "Dx" and **SkillSets[SystemUser::Skillset]::SkillsetUserType** = CH use Disposition Interface Dx1
- ❓ If *BlockOrderNo* contains "Dx" and **SkillSets[SystemUser::Skillset]::SkillsetUserType** = CL use Disposition Interface Dx2
- ❓ If **TriageLogic::QuID** contains "Tx" get related data from **TxQuestions** and **TxAnswers** tables
 - ❓ If **TxQuestions::ControlType** =1 display data using Interface Tx1
 - ❓ If **TxQuestions::ControlType** =2 display data using Interface Tx2 (see below on how to handle other ControlTypes.)
- ❓ If **TriageLogic::QuID** contains "Dx" get related data from **Dispositions** and **DispositionItems** tables
 - ❓ Display data in interface Tx1
- ❓ If **TriageLogic::QuID** contains "Cx" get related data from **CareAdvice** and **CareAdvice** tables:
 - ❓ Display data using Interface Tx2

In the screenshots provided the display style is achieved using different layouts or user interfaces but developers are at liberty to achieve the same visual effect in the most efficient way for their system/development environment (See 2.4). Note that for Cx and Dx items inline, the standard interfaces Tx1 and Tx2 are used but if TriageLogic transfers to them via a "-1" then specific interfaces are used.

2.4.2.7 Control Types 3 and 4.

Some questions have **TxQuestions::ControlType** set to "3" or "4". These questions are not for display to the user, rather they are in the triagelogic to allow rules based branching of the pathway. As such these types should also not be displayed in the call report or call report summary even though they will form part of the triage record. Therefore their **TriageRecord::IncludeInReport** value should be set to "0". This branching is based on the value of a specific variable which can one of three types, system, patient or user set by answer selection. ControlType 3 questions are to set variables while ControlType 4 questions are to evaluate the variable.

2.4.2.7.1 User set variables.

Supposing a call taker is presented with this question:

Is she in severe pain?

1. Yes
2. No

The caller answers “yes” and the call taker selects answer 1. In triagelogic this leads to a ControlType 3 question **which is not displayed to the user** but sets a variable. This question will take the form:

Question Text: STRING:PAIN

1. "severe"

Note the question text is a declaration of the variable type and name. Other types will be Integer (e.g. INT:Height) and Boolean (e.g. BOOL:over 65) So the question above will set a variable PAIN to “severe”.

Later in the pathway the user may select an answer which leads to a Type 4 question.

Question Text: PAIN

1. =="severe"

- 2 *

Answer 1 shows the format for a string variable, =="". Answer 2 in this case is the last answer and will always be a wild card character to handle any values for the variable not explicitly matching the values in other answers above the last one.

For INT answers normal operators can be used. For instance, if INT:DurationOfPain had been set to 9 then answer 2 below would evaluate to true but if it had been set to 25 then only answer 4 would evaluate to true and that would be the answer selected by the system and followed in the triagelogic.

1. < 5
2. < 10
3. < 20
4. *

=, =>, >, <= are also be used.

Another example could be BOOL:Over65 which would be set to 0 or 1 by a type 3 question and evaluated later by a type 4 question with 0 or 1 as answers. In this case there is no need for a wild card value as on or other of the answers will always evaluate to true.

2.4.2.7.2 Patient Variables

Some information about the patient is already known so does not have to be set by a ControlType 3 question. These values can be evaluated. The syntax is as follows:

PATIENT_AGEGROUP – values set by user front screen or calculated from date of birth or PATIENT_AGE. Values “adult” (16yrs or over), “child” (5 yrs to under 16yrs), “toddler” (1yr to under 5), “infant” (under 1 yr)

PATIENT_AGE – this will be the patient’s age (**rounded down**) represented as an INT. **Under 1 year is rounded to ‘0’.**

PATIENT_AGEinDAYS – patient age in days

PATIENT_GENDER – this will be the patient’s gender represented as a STRING – either ‘M’ or ‘F’ or ‘U’

PATIENT_PARTY – this will be an INT (currently either 1 or 3)

So a ControlType 4 question would have, for example, question text and answer text as follows:

Question Text: PATIENT_Gender

1 == "M"

2 *

This would mean that any male patients would go in one direction in the triagelogic and all other patients would go in another.

2.4.2.7.3 System Variables:

Sometimes it will be necessary to turn on or off certain parts of pathways depending on a system variable set by a local administrator. For instance Pathways have developed some specific routes through certain pathways depending on the value of a flu pandemic variable. In cases where the flu pandemic variable is set to "pandemic" then a different route through triagelogic is taken in order to manage this specific situation.

Other system variables may be developed in time and it stands to reason that system suppliers need to provide an interface where site administrators can set these variables. A full list of Pathways system variables can be found in section 2.2.6.2

Examples of system variables in use in the telephone configuration now are:

"SYSTEM_FLUEPIDEMIC" and

"SYSTEM_FLUPANDEMIC"

With values of "epidemic" and "pandemic" respectively.

"SYSTEM_MERS" with a value of "MERS"

In triagelogic there will be a ControlType 4 question that evaluates a system variable.

e.g.

Question text: SYSTEM_FLUPANDEMIC

1. == "pandemic"

2. *

So that depending on the state of the variable the triagelogic will go one way or another. Examples of system variables in use for the Reception point configuration are:

"SYSTEM_RPCContentED" and

"SYSTEM_RPCContentUCC"

With values of "RPED" and "RPUCC" respectively. An integrated service with both an Emergency Department and an Urgent Care Centre will have both values set to be on.

In triagelogic there will be a ControlType 4 question that evaluates a system variable. e.g.:

Question text: SYSTEM_RPCContentED

1. == "RPED"

2. *

2.4.2.8 "Next" when **Triagelogic::JumpID** = -2

If a user completes Module 0 without reaching a disposition they will inevitably select an answer with a **Triagelogic::JumpID** value of "-2". This *JumpID* indicates that the system should transfer the user to the Pathway selection interface. In the screenshots provided there are different layouts for the age/gender combinations but the only difference in these layouts is the body map graphic and the same effect could be achieved by calling the appropriate body map graphic depending on the age/gender of the patient.

The procedure is as follows:

- ❓ If **Triagelogic::JumpID** = -2
- ❓ Set **TriageRecord::Action Id** to “Next to Pathway Selection”

-
- ❓ Create new record in **TriageRecord** (as above)
-

- ❓ Decide how to display data
 - ❓ Set field **TriageRecord::layout** to “BodyMap” & **CaseRecord::GenderId** & **CaseRecord::AgeGroupId**
 - ❓ Go to interface style given by **TriageRecord::layout** (e.g. BodyMapFemaleAdult)
 - ❓ Populate the clinical area groups according to the data in the **BodyMap** table

The Pathways table contains values which determine which pathways are presented to the user for selection depending on:

- ❓ Patient age group
- ❓ Patient gender
- ❓ BMIItem selected by user
- ❓ Trauma/ non trauma value set by SystemFlag as described above.

So for a male adult non trauma case where the user selects BMIItem head from the body map the pathways list for selection will be those pathways that match these values.

Pathways::PW_Age = Adult

Pathways::PW_Gender = male

Pathways::PW_Trauma = non trauma (as set by SystemFlag answer)

Pathways::PW_ClinicalArea = BMIItem as selected by user.

(see interface below)

2.4.2.9 “Next” when **Triagelogic::JumpId** = -3

In an ambulance setting, a set of safety questions have to be asked to make sure the situation is safe for both the caller and the ambulance crew. There are two safety pathways, one for 1st and one for 3rd party callers. A JumpID of “-3” triggers the loop to the 3rd party safety pathway SP3.0 and a “-6” triggers the loop to SP1.0

The sequence is:

- ❓ If **Triagelogic::JumpID** = -3
- ❓ Set **gBlockOrder** to SP3.0
- ❓ Set **gLoopBlockOrder** to **Triagelogic::(SkillSets[SystemUser::Skillset]::Tbranch)** – The SystemUsers skillset is used to lookup which Tbranch to use from the TriageLogic table.

-
- ❓ Set **TriageRecord::Action** to “Next”
-

- ❓ Create New **TriageRecord**
-

- ❓ Decide how to display data
-

A similar sequence is followed for a `JumpID` of “-6” but setting the `gBlockOrder` to SP1.0

2.4.2.10 “Next” when `Triagelogic::JumpID` = -4

Having jumped out to the relevant safety pathway, answers selected follow the normal pattern until the system encounters a `Triagelogic::JumpID` of “-4”. This is used to loop back into the main pathway at the next `Triagelogic` record as defined by the value of the `gLoopBlockOrder` field set above.

The sequence is as follows:

- ❓ If `Triagelogic::JumpID` = -4
- ❓ Set `gBlockOrder` to `gLoopBlockOrder`
- ❓ End if
- ❓ Set `TriageRecord::Action ID` to “Next”

❓ Create New Record (2.3.2.3)

❓ Decide how to display data

2.4.2.11 “Next” when `Triagelogic::JumpID` = -5

Go to interim care advice screen as described below from disposition screen.

2.4.2.12 “Next” from the disposition screen.

Once a final disposition has been recorded what happens next is determined by the logic in the `DispositionItems::TransferCode` field. The values in this field can be:

- ❓ -9 – save and close call (same as when `Triagelogic::jump code` = -9, see below)
- ❓ PT3.0 or similar – transfer to this pathway in triagelogic
- ❓ -5 go to care advice screen Cx1
- ❓ -7 go to care advice screen Cx2

Care advice screen Cx1 should only display type ‘interim’ care advice sets and Cx2 type ‘home care’ advice sets. Additionally, the items displayed are setting, patient and final disposition dependent.

- ❓ `CareAdvice::CxGender`, `CareAdvice::CxAge`, `CareAdvice::Cx_Dx` determine which sets to use
- ❓ `CareAdviceItems::ActiveOOH`, `CareAdviceItems::ActiveSPA`, `CareAdviceItems::ActiveAmbulance`, `CareAdviceItems::FOHINT` determine which items in the set to display

Once at the Care Advice screen (Cx1), the radio button set for the field `TriageRecord::CAValueList` has to be populated with values. Some pathways and answers have specific keywords for care advice sets and it is those that have been collected during triage that will be represented in the `CAValueList` field. Additionally, the “Worsening” keyword should be added if it has not already been collected.

Before populating the radio button list of keywords, a search should be performed so that only relevant care advice is displayed. This should be filtered according to the `AgeGroup` and `Gender` of the patient. The other field to filter on is `CareAdvice::Type`, only sets with a value of “interim” should be displayed. The final filter of `TriageRecord::chFinalDispoCode` OR `TriageRecord::clFinalDispoCode` compared to `CareAdvice::Cx_Dx` is also applied but not displayed on the figure 4 below. If no match is found in the filtered search, the keyword should not be displayed as a radio button (as doing so will only produce zero care advice items when the radio button is selected).

Clicking on the radio button will render the individual care advice items.

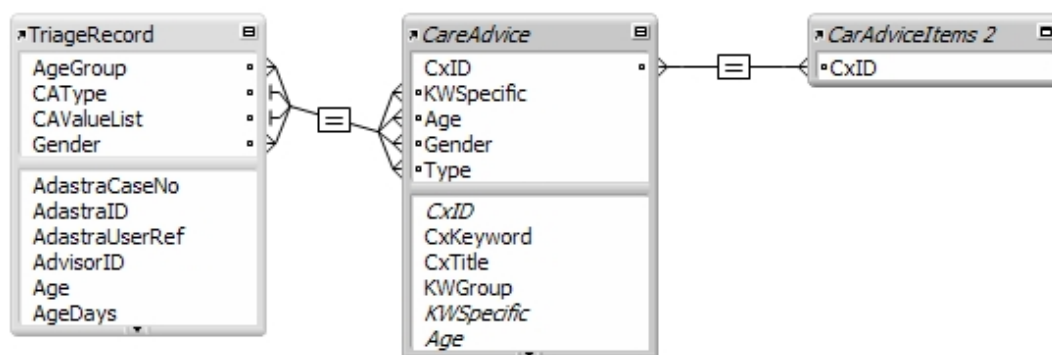


Figure 4 Relationship between TriageRecord and Interim Care Advice

A selection of one of the keywords from the radio button set completes the relationship and selects the **CareAdviceItems::CxItems** for display, sorted by **CareAdviceItems::CxNo**.

There are two exceptions to this general procedure for displaying Care advice sets. In some cases two CxKeywords which may be collected and stored in the TriageRecord provide duplicate or contradictory advice. For instance, there are many instances where the general advice “medication, pain” is collected. However, in other parts of the same pathway more specific advice may be collected, for instance “burns and scalds”. Where this happens, the advice for “burns and scalds” should be displayed but the advice for “medication pain” should be excluded. The exclusion list is held in the table **CareAdviceExclude**.

The second exception is when a triage assessment is either restarted or a user goes back one or more questions to retriage the call. In these situations the **TriageRecord::IncludeInReport** field will be set to 0 (see below) even though the records still exist. Only **TriageRecord::CAValueList** values of records where IncludeInReport = “1” should be included.

Where the transfer code is “-7” the Cx2 style screen is used. However, in this case all care advice sets with *type* = “home care” will be returned for the patient’s gender and age group so the relationship to filter the relevant sets looks like this:

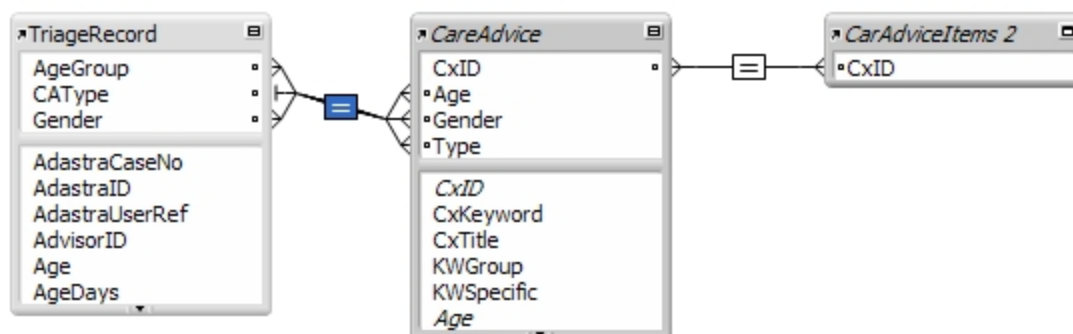


Figure 5 Home Care Advice for clinicians

The field *KWGroup* is used to arrange the care advice sets into groups for easy navigation in the display (see below).

2.4.2.13 Write to **TriageRecord** for Interim Care Advice or Home Care Advice

Before the call moves on from the Cx1 or Cx2 interface the care advice statements that have been selected need to be added to the **TriageRecord**. This can be performed on each keystroke or mouse click or alternatively added en masse on selecting next or save and close button. Each care advice stem needs an individual **TriageRecord** entry

 Create New **TriageRecord**

- ❓ set **TriageRecord::QulID** to **CareAdvice::CxNo**
- ❓ Set **TriageRecord::answerNo** to **CareAdviceItems::CxNo**
- ❓ Set **TriageRecord::Layout** to “Cx1” or “Cx2”
- ❓ Set **TriageRecord::ReportText** to **CareAdviceItems::CxText**
- ❓ Set **TriageRecord::TimeOut** to current date/time

2.4.2.14 “Next” when **TriageLogic::JumpId** or **DispositionItems::TransferCode** = -9

“-9” is the *JumpID* used by Pathways to tell the host system that triage is complete. In an ambulance setting it is envisaged that this would send a message to the dispatch desk to indicate that triage has finished. In any setting the sequence of events would be:

- ❓ Write final data to **CaseRecord** and **TriageRecord** (see 2.4.5)
- ❓ Write summary information to host system
- ❓ Run host routines to return to host

2.4.3 Restart, Jump Back and Early Exit Procedures

2.4.3.1 Restart Triage

Occasionally a call handler may need to restart a triage. For instance, a patient may deteriorate suddenly, and it is quicker for the call handler to go back to the beginning and re-triage rather than stepping back through the triage. The events when a user selects “restart triage” are as follows:

- ❓ Find all **TriageRecords** for the current **CaseRecord::supplierCallRef**
- ❓ Replace the value **TriageRecord::IncludeInReport** with “0” for all records in the found set
- ❓ Replace the value in **TriageRecord::ActionID** with “back” for all records in the found set
- ❓ Set the temporary variable *gBlockOrder* to:
Skillsets::EntryPathwayId + "." + **Skillsets::EntryOrderNumber**
from the Skillsets table for the current skillset.
- ❓ Where the *EntryPathwayId* = “AgeGenderParty”, find a match in the **Pathways** table for *AgeGroup*, *Gender* and *Party* and replace the “AgeGenderParty” text with the matched **Pathways::PW_ID**
- ❓ Perform New Triage Record procedure to create a new **TriageRecord** with same *supplierCallRef* as parent **CaseRecord**
- ❓ Set the **TriageRecord::BlockOrderNo** to *gBlockOrder*
- ❓ Display the data in Interface Tx1 Single Selection Answer Interface

Note: This process differs for HCP and IFT calls, please refer to section 12 for details.

The effect of this will be to create a new triage record with the same *supplierCallRef* as the records already created for the caller and clear the Call Report window of previous triage records. In addition, any variables set or values for *CAValueList* will be effectively hidden because **TriageRecord::IncludeInReport** has been set to 0 for these records indicating their values should not be used.

2.4.3.2 Jump Back Using Call Report

The call report, visible on most interfaces, lists the report text of all Triage records for the current *supplierCallRef* whose **TriageRecord::IncludeInReport** value is set to “1”. Rather than a back button, users are given the ability to jump back to any previous question in the call report in order to facilitate more rapid re-triaging from particular points other than the beginning. The icon shown adjacent to the text initiates the procedure to jump back to that question as follows:

- 🔍 Set the current record’s **TriageRecord::TimeOut** to current date/time
- 🔍 Go to the selected related record as identified by its value in **TriageRecordCallReport::TriageRecordID** self-join
- 🔍 Display the record in the “read only” version of the same interface as it was created in as given by the value in **TriageRecord::layout** that is, the user cannot change the answer in this interface.
- 🔍 The read only version should show the answer(s) previously recorded
- 🔍 Wait for user’s next action

The next actions can be:

- 🔍 Initiate “early exit” (see below)
- 🔍 Initiate “restart triage” (see above)
- 🔍 Initiate “jump back” (see above)
- 🔍 Initiate “change answer” (see below)
- 🔍 Return without change

2.4.3.3 Change Answer

The normal reason for a user to jump back is to change an answer in response to new information or realisation of an error in triage. The “change” button presented on the interface when the user “jumps back” initiates the following procedure:

- 🔍 Find all records for current **TriageRecord::supplierCallRef** whose **TriageRecord::TriageRecordID** => than the current record (i.e. the one jumped back to)
- 🔍 Sort ascending the records by **TriageRecords::TriageRecordID** – the record at the top will now be the record which was jumped back to
- 🔍 Replace **TriageRecords::IncludeInReport** with “0” for all records in found set
- 🔍 Replace **TriageRecords::Action** with “back” for all records in found set
- 🔍 Go to the first record in the found set (the record identified to change)
- 🔍 Replace **TriageRecords::Action** with “change” for this record
- 🔍 Duplicate this record
- 🔍 Display the record in the interface indicated by the value in **TriageRecords::layout**
- 🔍 Set the record’s value for **TriageRecords::IncludeInReport** to “1”

2.4.3.4 Return without change

The read only version of the interface should also contain a button that allows the user to return to the last point in the triage without change. The events when the user elects to do this are as follows:

- 🔍 Find all records for current **TriageRecord::supplierCallRef**
- 🔍 Sort descending the records by **TriageRecords::TriageRecordID** – the record at the top will now be the record which was jumped from
- 🔍 Display the record in the interface as given by the value in **TriageRecords::layout**
- 🔍 Wait for user’s next action

2.4.3.5 Early Exit

For various reasons the user may want to abort the triage process. An early exit Pathway is used to record the reason for the early exit. The Pathway and Order Number used for early exit are specified in the Skillsets table.

- ❓ Set current record's **TriageRecord::TimeOut** to current date/time
- ❓ Set **TriageRecord::ReportText** to "early exit selected"
- ❓ Set **TriageRecords::Action** with "Early Exit"
- ❓ Set **g_blockOrder** to the values as determined by setting

- ❓ Create new record

- ❓ Decide how to display record

2.4.4 Pathway Selection Procedures

Section 2.3.2.4 described how to get to this screen, select the appropriate age/gender specific body map and populate the clinical area pick lists. This section describes how to present the appropriate Pathways for user selection and how to start the selected Pathway. The Pathway selection interface is the gateway between the Module 0 Pathway selected by the system to the Module 1 Pathways, selectable by the user.

2.4.4.1 Display Pathways

In order to know which Pathways to present to the user for selection, the system needs to know the values of the fields shown in the relationship below.

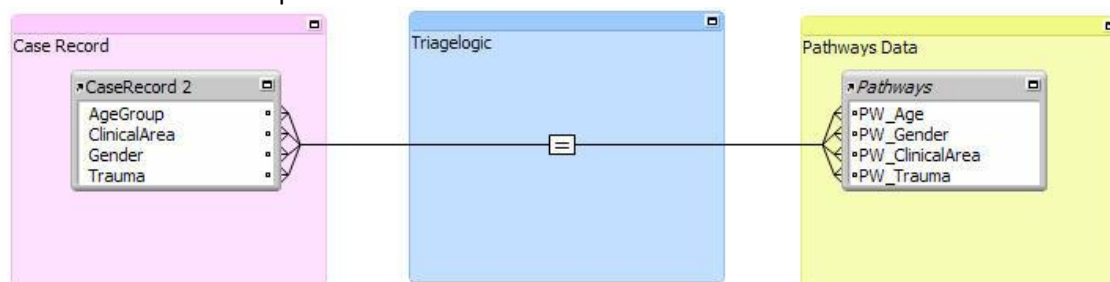


Figure 6 populating the pathways list for selection

CaseRecord::AgegroupId, **genderId**, and **trauma** have already been set on entry to this interface and **ClinicalAreaId** is selected by the user. Once the user has set the value for **CaseRecord::ClinicalAreaId**, the relationship is complete and the Pathway list can be displayed. (see figure 9)

2.4.4.2 Selecting and Starting a Module 1 Pathway

When a user selects a Pathway title this has the action of presenting the keypoints of the selected Pathway and setting the **gBlockOrder** for the first Triage question of the Pathway as follows:

- ❓ Set **gBlockOrder** to **Pathways::PW_ID** & ".0" which effectively says go to the identified pathway at **OrderNo "0"**
- ❓ Display keypoints

A user then needs to select the start Pathway button. This has the effect of:

NHS Pathways Specification

-  Set **TriageRecord::TimeOut** to current date/time
 -  Set **TriageRecord::ReportText** to **Pathways::PathwayTitle**
 -  Set **TriageRecord::CxKeyWords** to **Pathways::CxKW** if there are any
-

-  Create New Triage Record
 -  Set **TriageRecord::BlockOrder** to *gBlockOrder*
-

-  Decide how to display record
-

These are the general navigation and question selection procedures. Other procedures are described with the relevant interfaces in the section below.

2.5.1 Overview

All the relationships, and behaviours described below are, in the main, how an internal NHS Pathways demo tool works. Developers are free to employ their own techniques for producing the same display and behaviours but should bear in mind that data will be supplied in tables as described above. The parent table for all interfaces is **TriageRecord**. Data from related tables is displayed as shown and user actions recorded in the current **TriageRecord** record.

Note: most interface styles have features in common with Interface 1 (Tx1) shown below. Only differences are described in other layouts.

Also note that all Pathway screens should display the text “NHS Pathways, all content Crown Copyright”. This is probably best achieved as footer to the screens.

2.5.1.1 Interface size, fonts and colour schemes

-  All interfaces shown in this section are based on a size of 1024*768 pixels.
-  Pathway title is Arial 12pt bold caps
-  Question Text is Arial 12pt bold
-  Question rationale is Arial 10pt plain
-  Answer text is Arial 11pt bold
-  Answer Rationale is Arial 10pt plain
-  Caller text is Arial 12pt plain
-  Summary Text is Arial 12pt plain
-  Call report text is Arial 10pt plain
-  User comment text is Arial 12pt plain red
-  Button and interface labels are Arial 10pt plain

N.B. An alternative font to Arial is permissible in order to match host system style but this should be a sans serif style for screen legibility.

The colours shown, in restful shades of blue, have been well received by users in pilot sites. However, house style/system restrictions may mean developers will want to use a different palette. This is permissible, but the key feature is to keep the contrasts between the active and information areas. As such, the text objects that can be tabbed to – i.e. the answers - should be in white, important information such as question text should be emboldened, while supporting information such as rationales should be subdued slightly. The aim is to emphasise the decision elements of the screen.

2.5.1.2 Mouse and Keyboard input

Users should be able to interact with the interface elements using either mouse and/or keyboard input. Standard “Windows” keyboard actions should be employed.

2.5.1.3 Tab Orders

As a general rule, only buttons or text areas with actions attached should be in the tab order. This should start with and proceed through the answers. Free text areas such as the user comment box should be excluded from the tab order.

2.5.2 Single Selection Triage (Tx1)

This interface is used to display Triage Questions with a *ControlType* of 1 and Dispositions if they are presented “in-line”. A maximum of 7 *TxAnswers* and their rationales need to be displayed.

1 CALLING ABOUT SOMEONE ELSE, FEMALE, ADULT
 2 What is the problem?
 3 To find out if there may be an specific threats to life or a risk of serious complications.

4 the individual has stopped breathing
 This means there is little or no attempt by the individual to breathe. They will not be alert and usually be sweaty and deathly pale or blue.

5 the individual is collapsed, cold and clammy
 This means the individual is unable to stand. They will not be alert and usually be sweaty and deathly pale or blue.

the individual has a massive blood loss
 This means the amount of blood lost is life threatening. The individual may be collapsed, and usually be sweaty and deathly pale. Bleeding may be from a wound or from one of the body openings.

none of the above

Caller
 Caller: Jane Smith
 Tel: 0121 872543
 Patient: June Smith
 Age: Adult
 Location: 3 City Rd, BS2 2EW

Summary
 The consultation has identified:
 Injury

Call Report
 8 A potentially violent situation was described.
 An injury, as the cause of the problem, was described.
 The individual was described as having stopped breathing.

Restart Triage Early Exit
 Tx : PA1.0

Figure 7 Interface 1: Single selection items

N.B. red numbers are for documentation only – they do not appear on the interface.

For a triage question, the expected data for this interface is described below.

No	Data Source	Notes
1	Pathways::PathwayTitle	
2	TxQuestions::QuestionText	
3	TxQuestions::Qu_Rationale	
4	TxAnswers::AnswerText	All related answers – sorted by TxAnswers::AnswerNo
5	TxAnswers::AnswerRationale	
6	Host system	Patient Demographics – this area could allow for further user input. It is optional if this information is already on screen and visible in the host interface.
7	TriageRecordCallReport::DispoDisplayText	The consultation has identified: Followed by all records matching the relationship but each term displayed once if duplicate values. Values separated by return character
8	TriageRecordCallReport:ReportText	All records matching the relationship (e.g. same <i>supplierCallRef</i> and <i>IncludeInReport</i> = 1). Sorted descending by

		TriageRecordCallReport:TriageRecord_ID
13	TriageRecord::UserComment	User comment field – free text input

For an in-line disposition the data to be displayed would be as follows:

No	Data Source	Notes
1	Pathways::PathwayTitle	
2	Disposition::DispoText	
3	No value	
4	DxItems::DxItemText	Only answer stem 1 is displayed in line
5	DxItems::DxItemRationale	
6	Host system	Patient Demographics – this area could allow for further user input. It is optional if this information is already on screen and visible in the host interface.
7	TriageRecordCallReport::DispoDisplayText	The consultation has identified: Followed by all records matching the relationship but each term displayed once if duplicate values. Values separated by return character
8	TriageRecordCallReport:ReportText	All records matching the relationship (e.g. same <i>supplierCallRef</i> and <i>IncludeInReport</i> = 1). Sorted descending by TriageRecordCallReport:TriageRecord_ID
13	TriageRecord::UserComment	User comment field – free text input

2.5.3 Multi-select Triage (Tx2)

This interface is used to display Triage Questions with a *ControlType* of 2 and Care Advice if it is to be presented “in-line”. Up to 15 items of care advice may need to be displayed in this interface.

STOPPED BREATHING, TRAUMA, CPR
CPR 1st Party (1)

4 X Turn the individual onto their back

X Place your hand on their forehead and gently tilt their head back

X Place your fingertips under the point of their chin and lift the chin to open the airway

X Look for their chest rising and falling

X Listen at their mouth for breathing sounds

X Place your cheek against their mouth and feel for any air on your cheek

X Do not do this for more than 10 seconds, then return to the phone

X Return to the phone

Next

Caller
Caller: Jane Smith
Tel: 0121 872543
Patient: June Smith
Age: Adult
Location: 3 City Rd, BS2 2EW

Summary
The consultation has identified:
Injury
Assistance not available
Willing and able to give CPR

Call Report

Assistance was not available.

Choking or hanging by the neck was not described

It appeared safe and appropriate to give emergency care.

Don't put yourself at risk

A major hazard to safety was not described.

A potentially violent situation or fire hazard was not described.

An injury, as the cause of the problem, was described.

Restart Triage Early Exit

Pathway Txmc : PD1T.820

Figure 8 Interface 2 Multi-select Items

2.5.3.1 Care Advice when in line care advice is end of triage.

In some cases user feedback has indicated that it is confusing to have a “next” button on the screen if this is the end of triage. If the value for the JumpID is -9 this indicates that there are no more triage steps for this pathway. Therefore the “next” button should be replaced with a “save and close” button as is the case for other pages which are last in the triage.

For Triage Items:

No	Data Source	Notes/additional behaviours
4	TxAnswers::AnswerText	More than one selection can be made. Each selection needs to be recorded in TriageRecord as described above, this can be done on each key/mouse stroke or on selection of next/save and close button. The user cannot proceed if no answer is selected. The next/save and close should be unavailable until a selection is made.

For Care Advice Items:

No	Data Source	Notes/additional behaviours
4	CareAdviceItems::Cxltem	More than one selection can be made. Each selection needs to be recorded in TriageRecord as described above, this can be done on each key/mouse stroke or on selection of next/save and close button. The user cannot proceed if no answer is selected. The next/save and close should be unavailable until a selection is made.

2.5.4 Pathway selection by user (BodyMap)

The behaviours for presenting this screen are provided earlier in 2.4.2.8 'Next when TriageLogic::JumpID=-2' Additional behaviours are described below.

The interface displays two human figures (front and back views) with pick lists for body regions. The front view includes pick lists for eye, ear, nose, mouth, throat, breast, chest, abdomen, genital, groin, urinary, and skin and hair. The back view includes pick lists for head, face, neck, shoulder, arm, back, flank, rectum/anus, and hip/leg. A 'Start Pathway' button is at the bottom. On the right, there is a 'Selected Pathways' section, a 'Consultation Report' section with text boxes, and 'Restart Triage' and 'Early Exit' buttons.

Figure 9 Pathways selection screen

The interface as presented above is as a result of the user selecting a prior answer with a **TriageLogic::JumpID** of “-2”. There are 8 pick lists arranged by body region. The values for these lists come from the table **BodyMap**. The order in which the individual Body Map Items are displayed inside the group should reflect the ordering displayed in figure 9, and is also reflected in the “Ordering” column from the data files. Typically, they reflect the order in which they appear on the body (i.e. “shoulder” above “arm”). The figure below identifies the pick list headings and other functional areas.

1 Head

- eye
- ear
- nose
- mouth
- throat

2 NON TRAUMA PATHWAYS FOR HEAD

3

- Alcohol Intoxication
- Aura or Anticipated Fit/Seizure (Declared)
- Behaviour Change
- Dental Problems
- Dizziness or Vertigo
- Earache
- Face, Neck Pain or Swelling
- Hair Loss
- Headache

4

Abdomen

- abdomen
- genital
- groin
- urinary

Chest

- breast
- chest

Face

- head
- face
- neck

Upper Limb

- shoulder
- arm

Lower Limb

- hip_Leg

Back

- back
- flank
- rectum/anus

Not area specific

- skin and hair
- not area specific

Start Pathway

Consultation Report:

The skin on the torso felt normal, warm or hot.

The main reason for contact was not new confusion, declared diabetic hypo/hyperglycaemia, a probable allergic reaction or ICD shock.

The main reason for the assessment was not a heart attack, chest/upper back pain, probable stroke, recent fit/seizure or suicide attempt.

The individual was not fighting for breath.

An illness or health problem was the main problem. User comment - headache for 2 days

Restart Triage

Early Exit

Figure 10 Body map showing functional areas

In the screen shot above, the user has selected “Head” from the pick lists, so completing the relationship required to display matching pathways (3).

No	Data Source	Notes/Additional Behaviours
1	BodyMap::items	There are 8 pick lists shown and these are common across all body maps (gender, age) Only one value from any of the pick lists can be selected at a time. The pick sets the field is CaseRecord::ClinicalAreaID
2	CaseRecord::Trauma & “Pathways for” & CaseRecord::ClinicalAreaID	Lets the user know what criteria are used to present the current list of pathways
3	Pathways::PathwayTitle	All values as a list that match the relationship described in 2.4.4.2 Selecting and starting a Module 1 Pathway Selecting a Pathway results in the changes below
4	none	The body maps for this age group and gender.

When a user selects a pathway, the screen will look like the one below (Dental Problems selected)

1 Head

- eye
- ear
- nose
- mouth
- throat

2 NON TRAUMA PATHWAYS FOR HEAD

3

- Alcohol Intoxication
- Aura or Anticipated Fit/Seizure (Declared)
- Behaviour Change
- Dental Problems**
- Dizziness or Vertigo
- Earache
- Face, Neck Pain or Swelling
- Hair Loss
- Headache

4

5 Dental Problems

Use this pathway when there is:

- Bleeding from the mouth, teeth or gums
- Pain or discomfort in the teeth or gums
- A crown, bridge, fillings, denture or brace problem
- A broken tooth, or teeth not caused by any injury

Do not use this pathway when there is:

- Sudden locking of the jaw (Locked Jaw)

Toothache following a dental injury, triage the injury

A sore throat (Sore Throat or Hoarse Voice) or earache (Earache)

Cold or flu as the main problem (Cold or Flu (Declared))

A probability that something has been swallowed or inhaled (Object, Ingested or Inhaled)

6 Start - Dental Problems

Restart Triage **Early Exit**

Consultation Report:

- The skin on the torso felt normal, warm or hot.
- The main reason for contact was not new confusion, declared diabetic hypo/hyperglycaemia, a probable allergic reaction or ICD shock.
- The main reason for the assessment was not a heart attack, chest/upper back pain, probable stroke, recent fit/seizure or suicide attempt.
- The individual was not fighting for breath.
- An illness or health problem was the main problem. User comment - headache for 2 days

Figure 11 Pathways Selection Screen after selection

No	Data Source	Notes/Additional Behaviours
5	Pathways::PathwayTitle & "Use this pathway when there is" & Pathways::KP_Use & "Do not use this pathway when there is" & Pathways::KP_NotUse	Values for the Pathway selected. Note that the field is brought forward in front of the bodymaps.
6	"Start" & Pathways::PathwayTitle	Values for the pathway selected. Informs the user know which pathway they are about to start. Selecting this button starts the pathway.

2.5.5 Dispositions Interfaces

A transfer code (T1...T4) containing "Dx" informs NHS Pathways to go to the dispositions interface. There are 2 versions of this interface, depending upon the skill set of the user. Unlike the other interfaces described so far, which are populated with related data from the **Triagelogic** table and its children, the disposition screens get their data directly from the **Dispositions** table and its child tables via the following relationship.

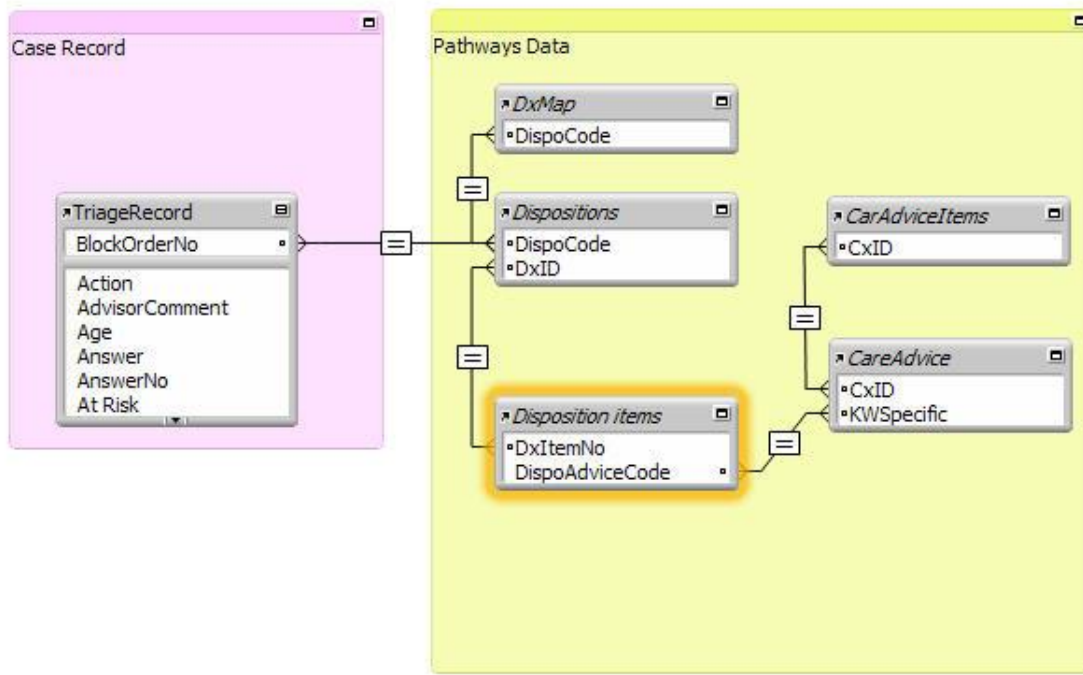


Figure 12 Relationship between TriageRecord and Dispositions table

Clinical users (those with **Skillset::SkillsetUserType** of CL) are able to override the disposition if, in their judgement, the disposition reached is inappropriate. Non-clinicians have fixed override options depending on setting as determined by their skill set.

Inline dispositions are displayed using the Single Selection Triage (Tx1) interface. Please see section 2.5.2 for details.

2.5.5.1 Disposition Interface – For Skillsets with SkillsetUserType set to CH (Dx1)

The screen shot below shows the interface on entry with the recommended disposition pre-selected. As a result, the Disposition Instructions and a DoS return are displayed. If the user changes the disposition then the related disposition instructions (4) and DoS return (where applicable) will change. As the selected disposition has a **Dispomap::CMSGroup** then the DoS has been called (see Section 9: Connecting to the DoS) and the first service returned included between the disposition items and the Disposition Instructions. For information about how the service should be displayed see Section 2.5.6: DoS Interfaces.

To arrive at the interface in this state the procedure is as follows (after writing previous record to **TriageRecord** table)

- ❓ If **Triagelogic::T1** or If **Triagelogic::T3** contains “Dx”
 - ❓ Go to disposition interface 1
 - ❓ Select **DispositionItems::DxNo** = 1 (the recommended disposition is always the first item)
 - ❓ Set **CaseRecord::chTriageDispoCode** OR **CaseRecord::clTriageDispoCode** to **Dispositions::DispoCode**
 - ❓ If the selected disposition item’s **Dispomap::CMSGroup** is not null, call the DoS and cache the result. Display the first service returned
 - ❓ If the user changes the selected disposition item, then the DoS result should be hidden and the DoS recalled if the item’s **Dispomap::CMSGroup** is not null, with the new first return showing in the same place. Again the results should be cached.

This should result in the interface below.

1 Attend Emergency Treatment Centre within 1 hour

2 The individual needs to attend an emergency treatment centre within 1 hour.

3 The DoS will return a list of services that are able to manage the problem.

UTC: Pallion Urgent Treatment CentreDistance (miles): 10.2

Pallion Health Centre, Hylton Road, Sunderland, SR4 7XF

Referral Information: This service offers direct booking appointments.

Accept

Reject

If there are no appointments available please review CMS for alternatives, if there are none then you can advise the patient to walk in.

From 18:00 weekdays or anytime on a weekend/bank holiday: If a patient requires a home visit please pass using the "Home Visit Required" button

You can go straight to this service. You do not need to telephone beforehand.

Contact Details:

0191 607 8516

NOT Open 24hrs

Capacity: High

Monday 10:00-23:59,

Tuesday 10:00-23:59,

Wednesday 10:00-23:59,

Thursday 10:00-23:59,

Friday 10:00-23:59,

Saturday 08:00-23:59,

Sunday 08:00-23:59,

Bankholiday 08:00-23:59

4 Someone else should drive the individual.

Take any medicines and medical records to the hospital.

Take a mobile phone if possible, in case symptoms get worse and an ambulance is needed.

SAFEGUARDING: From what I have heard there may be safeguarding concerns.

Before you go, I will just check whether I need to give you any further instructions or advice.

5

Next

Summary SG:1011 SD:4052Exit

Dispo:Dx02

Injury User comment - a

Warm to touch

Able to weight-bear

Loss of movement, ankle

Call Report

There was loss of movement in the ankle following the injury.

There was no constriction of the limb by an object that could not be removed.

An assault had not occurred.

There was no new limb deformity or swelling.

The foot had not become cold, pale or blue.

Additional Text

Restart Triage

Early Exit

Figure 13 Call handler interface with DoS call – OOH

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Attend Emergency Treatment Centre within 1 hour

☒ The individual needs to attend an emergency treatment centre within 1 hour.

The DoS will return a list of services that are able to manage the problem.

☐ A non-emergency ambulance is being arranged.

Use this option to override the original disposition when it is inappropriate due to social factors, pre-existing disability or local policy only.

☐ An emergency ambulance is being arranged.

This means a "Category 3" ambulance response.

UTC: Pallion Urgent Treatment Centre
Distance (miles): 10.2

Pallion Health Centre, Hylton Road, Sunderland, SR4 7XF

Referral Information: This service offers direct booking appointments.

Accept
Reject

If there are no appointments available please review CMS for alternatives, if there are none then you can advise the patient to walk in.

From 18:00 weekdays or anytime on a weekend/bank holiday: If a patient requires a home visit please pass using the "Home Visit Required" button

You can go straight to this service. You do not need to telephone beforehand.

Contact Details:
0191 607 8516
NOT Open 24hrs
Capacity: High
Monday 10:00-23:59,
Tuesday 10:00-23:59,
Wednesday 10:00-23:59,
Thursday 10:00-23:59,
Friday 10:00-23:59,
Saturday 08:00-23:59,
Sunday 08:00-23:59,
Bankholiday 08:00-23:59

☐ Someone else should drive the individual.
☐ Take any medicines and medical records to the hospital.
☐ Take a mobile phone if possible, in case symptoms get worse and an ambulance is needed.
☐ SAFEGUARDING: From what I have heard there may be safeguarding concerns.
☐ Before you go, I will just check whether I need to give you any further instructions or advice.

Next

Summary SG:1011 SD:4052
Dispo:Dx02

Exit

Injury User comment - a
Warm to touch
Able to weight-bear
Loss of movement, ankle

Call Report

There was loss of movement in the ankle following the injury.
There was no constriction of the limb by an object that could not be removed.
An assault had not occurred.
There was no new limb deformity or swelling.
The foot had not become cold, pale or blue.
The individual was able to walk.

Additional Text

Restart Triage
Early Exit

Figure 14 Call Handler interface with DoS call - Ambulance

The above interfaces are identical but different disposition items are presented according to the setting. The fields **DispositionItems::ActiveAmbulance** and **DispositionItems::ActiveOOH** determine which items to display. SPA and RP is catered for in the same way. A DoS call is also included where the selected disposition item has a **Dispomap:CMSGroup**

No	Data Source	Notes/additional behaviours
1	DxMap::LocalDisposition	
2	DispositionItems::DispoItemText	The items related to the <i>dispoItemText</i> are presented in <i>DxNo</i> order
3	DispositionItems::DxItemRationale	
4	CareAdviceItems::CxCxText	Disposition Advice CxCxText items displayed is filtered by setting e.g. <i>ActiveOOH</i> and sorted by <i>CxNo</i>

5	Next button	The disposition item selected will have an associated DispositionItems::TransferCode. Some dispositions have no care advice therefore this will be the last screen. Others transfer to Care Advice. The label should read “Save and Close if DispositionItems::TransferCode = -9 and the user is not progressing to the full DoS screen. Otherwise it will be “Next”. The Next button should be inactive until a disposition and at least 1 disposition advice item has been selected. If a successful DoS call has taken place then the user must also accept or reject the service shown for the Next button to be activated
---	-------------	--

Selecting the “next” button

- ❓ **TriageRecords** will need to be created for all the Disposition instructions selected in section 4. These will need to be collected with CxID and CxNo collected setting **TriageRecord::Layout** to Dx1.

Then create a new TriageRecord and proceed as follows:

- ❓ set **CaseRecord::chFinalDispoCode** OR **CaseRecord::clFinalDispoCode** to **DispositionItem::DxCode**
- ❓ set **CaseRecord::chDispoTime** OR **CaseRecord::clDispoTime** to current Date/Time
- ❓ Set **TriageRecord::ReportText** to **DispositionItem::DxItemText**
- ❓ Set **TriageRecord::Quld** to **DispositionItem::DxCode**
- ❓ Set **TriageRecord::Layout** to “Dx1”
- ❓ Send disposition data to dispatch desk
- ❓ Set **TriageRecord::TimeOut** to current date/time
- ❓ If **DispositionItems::TransferCode** = -9
 - ❓ Set **CaseRecord::TriageEndTime** to current date/time
 - ❓ Set **TriageRecord::Action** to “end”
 - ❓ Write summary information to host system
 - ❓ Run host routines to return to host
- ❓ Else If **DispositionItems::TransferCode** = -5
 - ❓ Set **TriageRecord::Action** to “next”
 - ❓ New Triage Record procedure
 - ❓ Go to Care Advice Cx1 Interface
- ❓ Else If **DispositionItems::TransferCode** = -7
 - ❓ Set **TriageRecord::Action** to “next”
 - ❓ New Triage Record procedure
 - ❓ Go to Care Advice Cx2 Interface

If the Clinician is using the PaCCS skillset, the Disposition screen should not be displayed. Instead, the **CaseRecord::clTriageDispoCode** should be set to the reached dispositions DispoCode (**Dispositions::DispoCode**), then the Dx2 exit processing (detailed in section 2.5.5.1 above) involving setting more CaseRecord attributes and determining the next step, should be completed as normal, however no Triage Records should be created.

For non-PaCCS clinicians, the screen shot shows the screen on entry with only the recommended disposition on display and “Disposition is appropriate” pre-selected. As a result, the Disposition Instructions are displayed. No DoS returns are shown on Dx2.

Attend Emergency Treatment Centre within 1 hour

☒ The individual needs to attend an emergency treatment centre within 1 hour.
The DoS will return a list of services that are able to manage the problem.

☒ Disposition is appropriate
☐ Disposition considered appropriate - caller unlikely to follow disposition
☐ Disposition considered inappropriate - alternative disposition required
☐ Disposition considered inappropriate - repeat caller (3 calls within 4 days)

☐ Someone else should drive the individual.

☐ Take any medicines and medical records to the hospital.

☐ Take a mobile phone if possible, in case symptoms get worse and an ambulance is needed.

☐ SAFEGUARDING: From what I have heard there may be safeguarding concerns.

☐ Before you go, I will just check whether I need to give you any further instructions or advice.

Next

Summary SG:1011 SD:4052 Exit

Injury User comment - a
Bite or sting User comment - a
Able to weight-bear
Loss of movement, ankle

Call Report

There was loss of movement in the ankle following the injury.

There was no constriction of the limb by an object that could not be removed.

An assault had not occurred.

There was no new limb deformity or swelling.

The foot had not become cold, pale or blue.

Additional Text

Figure 15 Clinician Disposition Screen

As stated above, for Clinicians there is an option to override the disposition. The following screens show this functionality. If the Clinician selects one of the alternatives to “disposition is appropriate” then the alternative disposition appears as below and a specify box should appear to collect the reason for the override. This free text entry should be stored in the **TriageRecord::UserComment**

1 Attend a Treatment Centre within 1 hour

2 The individual needs to attend a treatment centre within 1 hour.
The DoS will return a suitable service that can manage the problem.

3

- Disposition is appropriate
- ☒ Disposition considered appropriate - caller unlikely to follow disposition
- Disposition considered inappropriate - alternative disposition required 4
- Disposition considered inappropriate - repeat caller (3 calls within 4 days)

5

- ☒ Emergency Ambulance due to Clinical Reasons
 - Emergency Ambulance Response for Potential COVID19
 - Ambulance for Clinical Reasons
 - Emergency Ambulance Response for Potential COVID19 Category 3
 - Emergency Ambulance Response for Risk of Suicide (Category 3)
 - 999 for police
 - Ambulance for Transport Reasons
 - Assistance needed at home due to inability to get off the floor
 - Attend a Treatment Centre within 1 hour
 - Attend a Treatment Centre within 1 hour possible Viral Haemorrhagic Fever
 - Refer to Mental Health/Crisis Service within 4 hours
 - Speak to Clinical Assessment service 1 hour COVID risk
 - Speak to Clinical Assessment service 12 hours COVID risk
 - Speak to Clinical Assessment service 2 hours COVID risk
 - Speak to Clinical Assessment service 4 hours COVID risk
 - Speak to Clinical Assessment service 6 hours COVID risk
- Contact Pharmacist within 24 hours
- Home Management

☐ I need to ask you some further questions in order to arrange a service response.

Next 6

Summary SG: 1014 SD:4500 Exit

Injury User comment - arm
Warm to touch
Symptoms caused by injury
Injury to shoulder, elbow or upper

Call Report

- Blunt limb injury, own transport
- There was somebody available to offer safe transport to hospital.
- There was new limb deformity.
- The individual was able to get up from the floor.
- The limb had not become cold, pale or blue.
- There was no new/worsening breathlessness since the injury.

Additional Text

Figure 16 Clinician Disposition Screen

Once the override reason has been collected, a scrollable list of alternative dispositions is presented. This list should be ordered by the `disposmap::DxRank` column (lowest to highest), so that higher (more urgent) dispositions appear at the top of the list. Multiple dispositions exist with the same `DxRank`, and therefore an alphabetical sort of the local disposition text should be used as the sub sort. Selecting an alternative disposition:

 resets the `TriageRecord::BlockOrderNo` to the `Dispositions::DispoCode` of the chosen alternative.

No	Data Source	Notes/additional behaviours
1	<code>Dispositions::DispoText</code>	
2	<code>DispositionItem::DispoItemText</code>	Only the first related item is displayed as nurses do not queue calls for themselves
3	<code>TriageRecord::DispoOverride</code>	Value list as shown
4	Text	Label appears if disposition is not appropriate
5	List of alternative dispositions	All the values in the <code>DxMap</code> table where <code>Include = 1</code>
6	Next button	The disposition item selected (either the original disposition presented or the alternative disposition if chosen) will have an associated <code>DispositionItems::TransferCode</code> . Some dispositions have no care advice therefore this

		will be the last screen. Others transfer to Care Advice. The label of the button should read “Save and Close” if DispositionItems::TransferCode = -9 and the disposition item does not require a DoS call. Otherwise it will be “Next”. The Next button should be inactive until a disposition and at least 1 disposition advice item has been selected.
--	--	--

The procedure for leaving the clinician disposition screen is the same as for the call handlers except that *Layout* would be set to “Dx2”.

2.5.6 DoS Interfaces

Information on how to call the DoS is found in Section 9: Connecting to the DoS

If a service is using the DoS then the programme flow differs between Call Handlers and Clinicians

For Call Handlers:

- On arrival at the disposition screen (Dx1), if a disposition item is selected which has a value in **Dispomap::CMSGroup**, the DoS should be called and the results cached
- The first DoS return should be shown on the disposition screen (Dx1). This service may be accepted or rejected
- On the disposition screen (Dx1), if the user changes the disposition selection to or from a disposition that uses the DoS, the DoS should be called again or the DoS section hidden accordingly
- The user may change the disposition selection after accepting or rejecting a service, in which case that service response should be reverted
- To progress to the next screen from Dx1, if a successful DoS call has been made for the selected disposition item, the user must accept or reject the service and select at least one piece of disposition advice
- If the DoS call fails, the user must select at least one piece of disposition advice to progress
- If the single service on Dx1 is accepted (and disposition advice has been selected), follow the normal routines after the disposition screen (e.g. Care Advice or Call Closed) and record the services in the **ServicesReturnedRecord** table (see section 2.3.2.3)
- If the single service on Dx1 is rejected (and disposition advice has been selected), collect the rejection reason and progress to the DoS screen
- When progressing from Dx1 to the DoS screen, the DoS should not be re-called, the cached services must be used
- On the DoS screen show all services. Include the one which has been rejected, which should be shown as rejected
- On the DoS screen the call handler should be able to change their rejection of the first service to an accept if needed. The service should then be recorded as accepted and not rejected

For clinicians:

- No DoS returns should be shown on the disposition screen (Dx2)
- If there is a value in **Dispomap::CMSGroup**, all DoS returns should also be shown on the DoS screen, which follows the disposition screen

While the DoS is being called, whether on Dx1 or the full DoS screen, the user should be shown an indication that a call is in progress.

Whether as a single service on Dx1, or within a list of returns on the full DoS screen, a service is displayed as follows:

GP: Unregistered/Temporary Resident Patients (North Durham)	Distance (miles): 1.7
Patients in North Durham, DH1 5TS	
Referral Information: If the patient is unregistered, visiting the area or wishes to be a GP Choice patient, advise them that they will need to register as a new patient/temporary resident with a local GP practice. Information on local GP practices can be found at www.nhs.uk	Accept Reject Contact Details: NOT Open 24hrs Capacity: High Monday 08:00-18:30, Tuesday 08:00-18:30, Wednesday 08:00-18:30, Thursday 08:00-18:30, Friday 08:00-18:30

Figure 17 Sample DoS return

Whether on Dx1 or on the full DoS Screen, a service should have the option to be rejected or accepted.

If a service is rejected, then a pop-up (or equivalent) must be presented to allow the user to indicate why the service has been rejected (see section 9.1.1 for more information on reject reasons). If a service is rejected on Dx1 then that service should be included in the list of full returns on the DoS screen, but pre-rejected, with the option to change this to an accept.

When a service is accepted on Dx1 or the full DoS Screen then the normal procedures following the disposition screen are followed – i.e. call closed or go to Care Advice screen but with the services also being recorded in the ServicesReturnedRecord table for reporting purposes (see Section 2.3.2.3). If a service is accepted on Dx1, then the full DoS Screen is not shown.

However, if all services are rejected then the screen should revert to the standard disposition screen relevant to the skill set of the call handler. At this point the call handler is likely to override the original disposition.

Some local customization of how services are presented is allowed. Some implementations include distance to service data and this is obviously very useful. DoS currently returns distances as the crow flies as opposed to actual distance by road, but also includes northings and eastings to enable more sophisticated road distances should the supplier wish to add value.

2.5.7 Care Advice Interfaces

The care advice interface is very similar to the multi-select triage interface except that the care advice items displayed in the list are determined by the user's input using the Selected Topics radio buttons. The values that populate these menus depend on the relationships described above.

The figure below shows the care advice screen Cx1 for interim instructions which are common to both clinicians and call handlers. Interim instructions are those provided to the patient who has been advised to seek help elsewhere (e.g. referred to primary care). The topics presented are determined by the pathways system as most appropriate for this scenario. This is the last screen prior to the closure of the call and therefore should have save and close on the button text.

This screen will also be used for non-clinical call takers (skillsets with SkillsetUserType set to CH) who reach a home care disposition – i.e. no onward referral. At the present time there are no pathways that lead to this disposition for these

skill sets but it may be incorporated in the future. In this case the screen title should be “Home Care Advice” and not “Interim Care Advice”.

Figure 18 Interim Care Advice

No	Data Source	Notes/additional behaviours
1	TriageRecord::CAValueList	Radio button menu populated with values from TriageRecord::CxKeywords for records where TriageRecord::IncludeInReport = 1 , plus the keyword “Worsening”. The list should be filtered before displaying to the user. The filter includes taking into account any CxExclude words, as well as the standard filters for Age, Gender, Interim and Cx_Dx (section 2.4.2.12)
2	CareAdviceItems::CxItem	More than one selection can be made. Each selection needs to have a record created in TriageRecord. The CxID needs to be recorded in TriageRecord::QuID and the CxNo should be recorded in the TriageRecord::answerNo . Deselecting an answer unmarks the check box. Deselecting all items removes the CxItem from the ReportText but a user cannot proceed if no item is selected. The TriageRecord::ReportText should start with “Advice given” then the CxItem data as above. It is up to the supplier whether the data is collected on keystroke or on selecting next but all selected cxitems should have their own triagelogic record.
3	CareAdviceItems::CxItem	Separator with no selection box – i.e. the field CareAdviceItems::Include = 0

4	Save and Close button	The save and close button should perform the same procedure as if a “-9” transfer code had been reached.
---	-----------------------	--

The figure below shows the clinician version of the care advice screen for a home care disposition. The clinician is presented with two drop down lists used to select the care advice set for display. This is the last screen prior to the closure of the call and therefore should have save and close on the button text.

Figure 19 Home Care Advice – clinician

The “Groups” drop down (1) is populated by all unique values in the **CareAdvice::KWGroup** field. Setting this value will filter the “Sets” list (2), presenting unique values from **CareAdvice::KWSpecific** field that have a matching **KWGroup**.

By default, the Group “Not area specific”, and the Set “Worsening” should be pre-selected, resulting in the Worsening Care Advice Items being displayed on screen for the appropriate Age Group/Gender of the patient. At present this will always display Cx170028 as it is configured for all Genders and Age Groups, however that could change in future Pathways Releases, and the appropriate Care Advice should always be displayed based on the Group and Set above.

The rest of the screen functionality is as above.

2.5.7.1 Save and close Procedure following care advice

- ❓ Set **TriageRecord::Layout** to Cx1 or Cx2 depending on style used
- ❓ Set **TriageRecord::TimeOut** to current date/time
- ❓ Set **TriageRecord::Action** to “end”
- ❓ Set **CaseRecord::chTriageEndTime** OR **CaseRecord::clTriageEndTime** to current date/time
- ❓ Write summary information to host system
- ❓ Run host routines to return to host

2.5.8 Call Validation

When a call is queued for a clinician, they have the opportunity to review the call before continuing with the triage. To facilitate this, the call validation screen summarises the information gathered in the **CaseRecord** and the **TriageRecord** for the triage to date. Positive answers to triage questions are emphasised (here in blue) to aid validation as these are the most likely reports of symptoms.

Please note that users of the Senior Clinician Module skillset should be excluded from using this screen. The supplier system needs to bypass this.

2.5.8.1 Entering Validation

On entering validation a new TriageRecord should be created for this case. This is slightly different from the standard new record procedure in that there is no *BlockOrder* to set.

- ❓ Create new record in **TriageRecord**
 - ❓ Set *supplierCallRef* to **CaseRecord::supplierCallRef**
 - ❓ Set *TriageRecordID*
 - ❓ Set *TimeIn*
 - ❓ Set *UserID*
 - ❓ Set *SkillSet*
 - ❓ Set *Siteld*
 - ❓ Set *IncludeInReport* = 0
 - ❓ Set *Layout* to “Validation”

2.5.8.2 Validation Interface

The following screenshot and notes describe the contents and operation of this interface.

Figure 20 Call Validation Interface

The parent table for this screen is **TriageRecord**.

No	Data Source	Notes/additional behaviours
1	Host system	Demographics information
2	"Pathway Selected" & TriageRecord::ReportText	Where <i>layout</i> contains "BodyMap"
3	TriageRecord:ReportText	All triage records for this <i>supplierCallRef</i> where <i>IncludeInReport</i> = 1 and Quld begins with Tx. The list is scrollable. Selecting an item from the list displays the full text of the questions and answers (7). The aim is to display only the triage questions in this section.
4	TriageRecord:ReportText	Where TriageRecord:Quld begins with Dx. The aim is display only the triage disposition in this section.
5	TriageRecord:ReportText	Where TriageRecord:Quld and TriageRecord:Layout = Cx1. The aim is display only the triage care advice in this section.
6	TriageRecord::DispoDisplayText	The consultation has identified: Followed by all records matching the relationship but each term displayed once if duplicate values. Values separated by return character.
7	Question data of selected question from 3	Each TriageRecord has a recorded <i>BlockOrder</i> therefore this data can be retrieved from Triagelogic and child tables using the match between TriageRecordCallReport::BlockOrder and Triagelogic::BlockOrderCalc as per triage interfaces

2.5.8.3 Leaving Validation

There are five ways in which a nurse can leave the validation screen as follows. All would begin by:

? Set **TriageRecord::TimeOut** to current date/time

? Then for

Early Exit (9)

? As per standard Early Exit procedure

? Then for

Restart Triage (10)

? As there is the possibility that the person being spoken to has changed from patient to 3rd person or vice versa, one of the following dialogues will be displayed depending on the *Party* of the **TriageRecord**

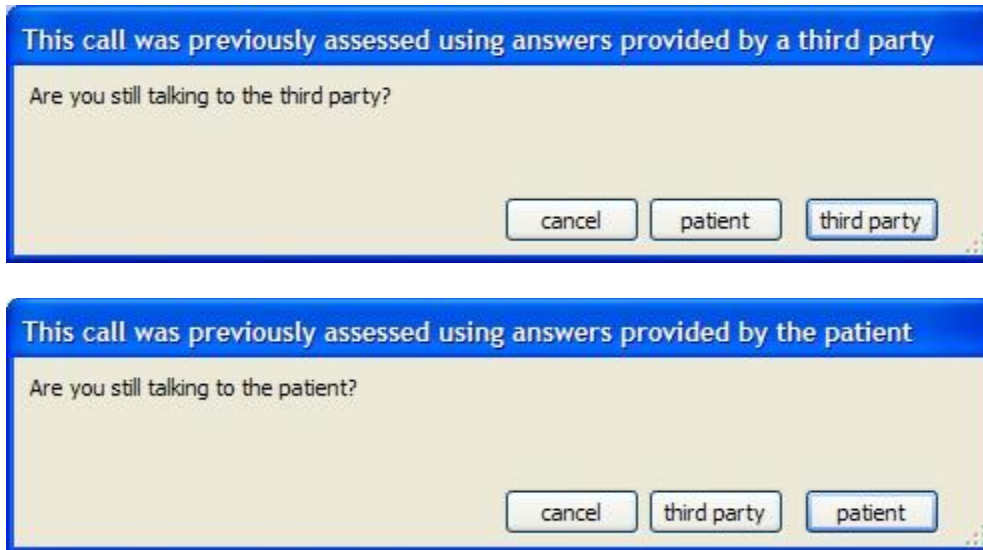


Figure 21 Validation party dialogues

? In either case, cancel will return the user to the validation screen, and selecting patient or third party will set the *party* field for the ongoing triage.

? Then proceed as per standard Restart Triage procedure as detailed in section 2.4.3.1

? Then for

Continue (12): the nurse validates the triage so far and continues from last point in triage. If his/her skill set is 2, any transfers will be taken from **TriageLogic::T2** (skill sets 4 and 6 use **TriageLogic::T4**). However, there is the possibility that the person being spoken to has changed from patient to 3rd person or vice versa. Therefore, one of the dialogues shown in Figure 21 above will be displayed depending on the *Party* of the **TriageRecord**.

In either case, cancel will return the user to the validation screen, and selecting patient or third party will set the *party* field for the ongoing triage.

? Look up the last record in the **TriageRecord** table for this *supplierCallRef*

? Set *gblockOrder* to **TriageLogic::blockjumpcalc**

? Create new triage record

? Display new record

? Then for

Change Pathway (11): the nurse validates M0 but wishes to re-triage using a different pathway. Remove all records post Pathway selection from call report and start again from pathway selection screen. The above party dialogues are first displayed.

All fields are from **TriageRecord**

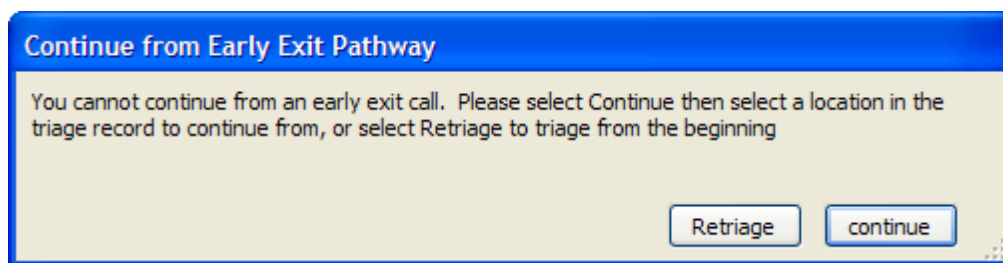
-  Show the party dialogue box
-  Find all records with this *supplierCallRef* and *TriageRecordID* => *TriageRecordID* of the record where *supplierCallRef* is this *supplierCallRef* and *layout* contains "BodyMap"
-  Replace *IncludeInReport* with "0" for all found records
-  Find all the records with this *supplierCallRef* and *IncludeInReport* = 1
-  Sort records ascending by *TriageRecordID*
-  Go to last record in found set
-  Perform new record procedure (this will take user to Pathway Selection screen)

Jump to Question (8) If a clinician considers that an answer has been given in error or the condition of the patient has changed then he/she can re-triage from any point by using this button. Selecting the topmost button will, for instance, effectively start the triage from the beginning.

-  Show the party dialogue box
-  Find all records with this *supplierCallRef* and *TriageRecordID* => *TriageRecordID* of this record
-  Replace *IncludeInReport* with "0" for all found records
-  Find all the records with this *supplierCallRef* and *IncludeInReport* = 1
-  Sort records ascending by *TriageRecordID*
-  Go to last but one record in found set
-  Perform new record procedure (this will take user to Pathway Selection screen)
-  Reset rules variables for any records beyond triage restart point

2.5.8.4 Validation when the call is transferred to clinician via early exit

There is an exception to the above description. A call can be transferred to a clinician via early exit from triage by call handler. Clearly in this case it makes no sense to continue from the last question in the triage record as the clinician will be dumped into the early exit pathway. In this circumstance we need a new routine. It is possible to identify these cases from the last pathway ID in the triage record. PEE1, PEE3 and PEE4 are the early exit pathways.



The above dialogue should appear in these circumstances. Selecting "Retriage" should go to the beginning of Pathways i.e. go to the top most question in the record and reset all existing records below to hide from report as described above but the triage record should still be continuous. Selecting "continue" should return the clinician to the validation screen and proceed as above.

2.5.8.5 Further clarification

It has come to our notice that some clinicians are selecting report stems in the validation screen that are generated from the early exit pathway. In this case they are sent back to the early exit screen and they override the disposition reached. They then expect the DoS to return services which of course it cannot as no complete triage has taken place and therefore no suitable SG/SD combination generated. To prevent this, we would request that system suppliers prevent continuation of triage from any report stem that has been generated by any of the early exit pathways. This can be achieved with the above message or, if preferred by greying out and making non-functional any report stem from the early exit pathways.

2.5.9 Pathways Clinical Consultation Support (PaCCS) Interface

Upon starting a call with a PaCCS skillset, the user will be taken to the PaCCS Interface.

2.5.9.1 PaCCS Template Selection

Pathways Clinical Consultation Support

The screenshot shows the PaCCS Template Selection interface. At the top, there is a search bar with the text 'leg' and a red '1' next to it. Below the search bar, there are two buttons: 'Ambulance / Immediate Threat to Life' and 'Home Care / Close Call'. A red '7' is positioned above the 'Home Care / Close Call' button. Below the search bar, there is a list of symptoms: 'Leg Injury, Blunt', 'Leg Injury, Penetrating', and 'Lower Limb Pain or Swelling'. A red '2' is next to 'Leg Injury, Penetrating'. Below the list, there are two tabs: 'Leg Injury, Blunt' and 'Leg Injury, Penetrating'. A red '3' is next to the 'Leg Injury, Penetrating' tab. Below the tabs, there is a table with two columns: 'Considered' and 'Suspected'. The table lists various conditions/symptoms to consider, including 'Major Haemorrhage', 'Major Trauma Injury', 'Respiratory Distress', 'Faint or Blackout (TLOC -Trauma)', 'Mechanism of Injury', 'Multiple Injuries', 'Signs of Fracture', 'Amputation', 'Limb Ischaemia', 'Features of Nerve Damage', 'Bleeding Tendency/Anticoagulants', and 'Foreign Body Retained'. A red '8' is positioned above the table. The table has checkboxes for 'Considered' and 'Suspected'. 'Limb Ischaemia' is checked in the 'Considered' column, and 'Features of Nerve Damage' is checked in the 'Suspected' column. Below the table, there are four buttons: 'Ambulance', 'Emergency Treatment Centres', 'Service Search', and 'Home Care'. A red '6' is positioned above the 'Home Care' button. To the right of the table, there are two 'Specify...' input fields. On the far right, there is a summary panel with a 'Summary' section showing 'SG:1120' and 'Leg Injury, Blunt', 'Leg Injury, Penetrating'. A red '4' is next to 'Leg Injury, Penetrating'. Below the summary, there is a 'Call Report' section with a list of report stems: 'Leg Injury, Penetrating/Features of Nerve Damage, Suspected', 'Leg Injury, Penetrating/Limb Ischaemia, Considered', 'Leg Injury, Blunt/Multiple Injuries, Considered', 'Leg Injury, Blunt/Mechanism of Injury, Considered', and 'Clinical Consultation'. A red '5' is positioned above the 'Call Report' section. Below the 'Call Report', there is an 'Additional Text' section with a text input field. At the bottom right, there are two buttons: 'Restart Triage' and 'Early Exit'. A red '6' is positioned above the 'Restart Triage' button.

Figure 22 PaCCS Template Selection

1, Search functionality must be provided so that clinicians are able to search for the specified Symptom Templates, with the results showing in a selectable list. The search will look for the entered string being contained in either the Template

NHS Pathways Specification

Titles or any Synonyms associated with the Template. Pressing enter after entering a search string should result in the search being performed.

2, Each Template result will be selectable and when selected will result in the appropriate Template being displayed.

3, It must be possible to open multiple Templates concurrently with the ability for the user to switch between Templates as they need. There will be no hard limit on the number of Templates that can be open concurrently.

The Clinician can close a Template where no changes have been made in the Template. A message will be displayed if changes have been made and they will be prevented from closing it without removing any data entered.

4, Upon navigation to the closure Pathways (after pressing one of the Disposition buttons). the existing Call Summary box will be populated with the names of the Templates that have been updated by the user (i.e. have Suspected/Considered checkboxes ticked).

5, The existing Call Report will be populated with the details of any Condition marked as Suspected or Considered by the user. An additional Call Report record will be created to record the fact PaCCS was used, to contain the text entered in via the Additional Text box and enable navigation back to the Template search functionality when no Conditions are marked as Considered or Suspected.

6, For consistency with existing pathway functionality, the Early Exit and Restart Triage buttons will continue to be available.

7, The navigation buttons on the search bar must be available to the clinician whenever the PaCCS Template searching is available. The labels for these buttons, and the Pathway and Order number that they need to jump to will be specified in the CCSJumps.csv file. Both buttons will jump to the specific points without picking up an SG.

8, Two buttons on each template will be required for DoS searching, one for Emergency Treatment Centres (e.g. Dx02 and Dx03 - ED within 1 and 4 hrs), and one for other Services. These buttons will be configured to jump to specific points in Pathways picking up the SG associated with the Template selected.

Leg Injury, Blunt X

Leg Injury, Penetrating X

Emergency Treatment Centres

Service Search

1

Conditions/Symptoms to Consider

Considered

Suspected

Major Trauma Injury

Respiratory Distress

Mechanism of Injury

Multiple Injuries

Signs of Fracture

Specify...

Specify...

5

Ambulance

Emergency Treatment Centres

Service Search

Home Care

Limb Ischaemia

Unable to Weight Bear/Walk More Than 4 Steps

Bleeding Tendency/Anticoagulants

Features of Nerve Damage

Situational Factors/Environmental Exposure

Assault

Non Accidental Injury

New or Worsening Pain and/or Swelling

Sprain or Strain

Tendon Injury

2

Overview

Tendon injury can occur due to any mechanism of injury, but often high impact or repetitive ones.

Symptoms depend on the tendons involved, such as shoulder (rotator cuff tendons), knee tendons, ankle tendons (Achilles), and smaller tendons in hands and feet.

Symptoms

- Depend on the site of injury and whether partial (some functioning) or partial (partial functioning) of affected site
- Reduced muscle function/weakness/giving way
- Pain
- Often associated with soft tissue symptoms or nerve damage.

Risk Factors

N/A

Patella Dislocation

Compartment Syndrome

Self Harm

3

Past Medical/Social History/Lifestyle Factors

Pregnancy

Psychosocial Factors

References/Resources

Safeguarding Concern (Collect details as required by your organisation's policies)

Figure 23 PaCCS Template Details

- 1, Each Template will contain Conditions specified by the authors ordered by the OrderNo field from the CCSTemplateConditions.csv file, and coloured/marked according to severity, red being the highest, followed by yellow. These are followed by the green General Markers, which are ordered alphabetically by ConditionName.
- 2, The Conditions will be expandable and will show Supporting Information in various sections to the clinician, which may include hyperlinks for guidance. This will be marked up using basic HTML tags. Clicking on links must open the page linked to but not prevent the user continuing using PaCCS. At launch there will be up to three sections of Supporting Information per condition (Overview, Symptoms, Risk Factors), however additional sections could be added at any time. All Supporting Information sections specified in the CCSCConditionSupportingInformation.csv release file should be displayed in the order specified by the OrderNo field.
- 3, The General Markers will be highlighted green and be the same across all Templates. These are defined in the release files as Conditions with their Importance set to 3 (General Marker). They will not be explicitly linked to the Templates in the CCSTemplateConditions.csv file, however every template should include all General Markers defined. These do not have checkboxes or navigation buttons but can have Supporting Information.
- 4, Each High or Normal importance Condition will have Suspected and Considered checkboxes, both will be selectable however selecting one will clear the other (as per radio button functionality). A checkbox can be cleared by re-clicking it. When selected, each one will show a specify box to allow the clinician to enter notes. The maximum number of characters allowed in this box is set to 300. Optionally, to support the requirements of providers, the column of Considered checkboxes can be hidden. If required, this should be configurable to enable the provider to show the column should they change their procedures in the future. Hiding the Considered column should have no affect on any other logic, for example the Specify text boxes should still only be available for a Condition if a checkbox for it has been ticked.
- 5, When the Suspected checkbox is checked, will also show up to four Disposition buttons (Ambulance, Emergency Treatment Centres, Service Search, and Home Care) that cause the navigation to jump to different points in standard Pathways. The jump location will be configured against each button for each Condition. If a condition appears in multiple templates, the same jump locations will be used in each template.

Conditions will be displayed for a Template based on Age Group and Gender. Conditions that aren't appropriate for the current patient based on these criteria, should not be shown.

Should a Condition require a different Symptom Group to the template it is in, it can be configured with its own override value.

2.5.9.3 Call Report/Call Summary

The TriageRecords generated from the PaCCS Interface (detailed in section 13.2) should allow the Call Report and Call Summary to display as required.

For calls utilising PaCCS, the Call Summary should list all Templates populated with any data. The Call Report should list all Conditions marked as Considered or Suspected, as well as having a Clinical Consultation item listed that contains any text that was added in the Additional Information field.

The Call Summary Report presented at the end of the call should include details of all conditions identified as Suspected or Considered, including the userComments. The PaCCS/Clinical Consultation Triage Record should also be included and allow for any Additional Text entered to be displayed.

At the end of the call, the details entered into the PaCCS screen must be transferred to the host system to become part of the Clinical notes for the call. Below is the data that should be transferred:

For each condition marked as Suspected or Considered:

Template/Condition, Suspected or Considered, Free Text entered

For example:

Headache/Recent Head Injury, Suspected, "Tripped over a step 3 days ago and banged forehead on concrete. No LOC but has large bruise to area"

Headache/Transient loss of consciousness, Considered.

3 Output Data

Triage data collected in the **CaseRecords**, **TriageRecords** and **ServicesReturnedRecord** tables form the basis of the history of the triage call. This data is required for reporting purposes both by the host organisations and the NHS Pathways team.

Host system developers will need to provide an interface for provider administrators with the appropriate access privileges to export the data for the provider needs. Typically, the export will be of all data from the three tables for a given date range. Consideration must be given to sensitive patient data and such data should only be included in any export if explicitly required by the provider.

All host systems are mandated to provide triage data in real time to NHS Pathways via the IDT web service at the closing of each triage call.

Aside IDT, reports for each interaction are still needed to be sent to receiving clinicians and services and the patient's own GP. All options are described here.

3.1 IDT Webservice

The data collected in the **CaseRecord**, **TriageRecord** and the **ServicesReturnedRecord** form the data that is required to be sent back to NHS Pathways in order to comply with the CQI section of the licence agreement. The IDT web service as it is now known was originally known as the CQI web service as it was built to support the Continuous Quality Improvement process that NHS Pathways uses to help providers and NHS Pathways learn from the data.

3.1.1 Overview

The Pathways IDT webservice is a RESTful service which can be accessed via standard HTTP methods by a variety of programming languages and HTTP clients including browsers and mobile devices.

3.1.2 Endpoint URLs

UAT – <https://cqiuat.pathwaysdos.nhs.uk/api/v1/cqiCase>

LIVE - <https://cqilive.pathwaysdos.nhs.uk/api/v1/cqiCase>

3.1.3 Authentication

Traffic to and from the IDT webservice will be encrypted over an “https” connection using Transport Layer Security (TLS). In addition to this, **Basic authentication** is also to be used in the form of a *username* and Base64 encoded *password* sent in the https request header.

To obtain a user account, please contact a member of the NHS Pathways team at pathwaysoperations@nhs.net.

3.1.4 IDT Webservice Methods

METHOD	URL	DESCRIPTION
cqiCase	/api/v1/cqiCase	Main method for posting case, triage and service data to the IDT webservice.
cqiExceptionCodes	/api/v1/cqiExceptionCodes	List of possible exception codes returned from the WebApi

3.1.4.1 cqiCase method

A Pathways case is made up of a single **case** record, multiple **triage** records, and if a call to the DoS was made – multiple **service return** records.

Case

```
{
  Case details.....
  Case details.....
  Case details.....
  Case details.....
  listOfTriageItems (mandatory)
  {
    TriageItem 1
    TriageItem 2
    TriageItem 3
    etc etc...
  }
  listOfServiceReturns (optional)
  {
    ServiceReturn 1
    ServiceReturn 2
    ServiceReturn 3
    etc etc...
  }
}
```

The following tables provide a summary of the fields accepted by the IDT webservice. They directly relate to the tables in the '**Pathways Case Data**' section earlier.

'Required' Data Explanation:

Providers should supply data to NHS Pathways via the IDT Webservice, as accurately and complete as possible. Requirements for each Record type are as follows.

- ☐ Many Record fields are mandatory and marked with "**Required**" in the table below.
- ☐ Some fields are "**Required if applicable**" based on certain conditions. E.g. some call handler fields are marked as 'Required if applicable' so should be completed if a call handler was involved in the triage. Similarly, if a clinician is involved in the triage the clinician based fields should be completed.
- ☐ Some fields are marked as "**Required if available**", such as postcode. We ask that you supply all fields that can be provided for a given record.

Where no data exists for a field (e.g. required if applicable or available), it is **preferable** if the JSON packet **excludes** these values altogether rather than sending in blank data such as empty strings or "0" data (for integers). As an example in a particular Triage Record, if no value exists for orderNo;

INCORRECT

```
"timeOut": "2019-07-23T04:10:12+0100",
"pwd": "PW656",
"orderNo": 0,
"quID": "Tx221192",
```

CORRECT

```
"timeOut": "2019-07-23T04:10:12+0100",
"pwd": "PW656",
"quID": "Tx221192",
```

3.1.4.1.1 Case Record

Field	Type	Example	Size	Notes
siteId	Integer <i>Required</i>	75		Unique ID provided by the NHS Pathways team to identify the site.
supplierCallRef	String <i>Required</i>	30ac7afe-1e34-4848-a06a-ff605a299f62	36	System suppliers unique reference or CallId
Postcode	String <i>Required if available</i>	GL1 5QD		Full postcode of patient location.
ageGroupId	Integer <i>Required</i>	1		1 (Adult), 2 (Child), 3 (Toddler), 4 (Neonate & Infant)
ageInYears	Integer <i>Required if available</i>	24		Patient age in years (0 – 120)
genderId	Integer <i>Required</i>	2		1 (Male), 2 (Female), 3 (Unknown)
partyId	Integer <i>Required</i>	3		1 (1 st party), 3 (3 rd Party)
chUserCode	Text <i>Required if applicable</i>	NFT123	Max 50 characters	Call Handler user code
chSkillsetId	Integer <i>Required if applicable</i>	5		Call Handler skillset e.g. 5 (111 Call Handler), or any Skillset from the Skillsets table with SkillsetUserType set to CH (1,3,5,7,8,9,10,11)
chCallStartTime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T03:51:30+00:00	24	Call Handler call start time
chTriageStartTime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T03:53:30+00:00	24	Call Handler triage start time
chDispoTime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T03:56:30+00:00	24	Call Handler dispo time
chTriageEndTime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T03:57:30+00:00	24	Call Handler triage end time
chCallEndtime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T03:58:00+00:00	24	Call Handler call end time
chTriageDispoCode	Text <i>Required if applicable</i>	Dx38	Max 10 characters	Call Handler triage dispo code
chFinalDispoCode	Text <i>Required if applicable</i>	Dx38	Max 10 characters	Call Handler final dispo code
chOverrideFlag	Boolean	true	5	Call Handler Disposition override flag

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	<i>Required if applicable</i>			(true or false)
clUserCode	Text <i>Required if applicable</i>	PSF	Max 50 characters	Clinician user code
clSkillsetId	Integer <i>Required if applicable</i>	6		Clinician skillset e.g. 6 (111 Clinician), or any Skillset from the Skillsets table with SkillsetUserType set to CL (2, 4, 6, 12)
clCallStartTime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T04:43:30+00:00	24	Clinician call start time
clTriageStartTime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T04:45:30+00:00	24	Clinician triage start time
clDispoTime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T04:47:30+00:00	24	Clinician dispo time
clTriageEndTime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T04:48:30+00:00	24	Clinician triage end time
clCallEndtime	DatetimeOffset <i>Required if applicable</i>	2019-07-23T04:49:30+00:00	24	Clinician call end time
clTriageDispoCode	Text <i>Required if applicable</i>	Dx05	Max 10 characters	Clinician triage dispo code
clFinalDispoCode	Text <i>Required if applicable</i>	Dx06	Max 10 characters	Clinician final dispo code
clOverrideFlag	Boolean <i>Required if applicable</i>	false	5	Clinician Disposition override flag (true or false)
symptomGroup	Integer <i>Required if available</i>	1112		The group of symptoms to which the pathway belongs.
symptomDiscriminator	Integer <i>Required if available</i>	4003		The discriminator that is produced by the completion of the NHS Pathways triage.
releaseVersion	Text <i>Required</i>	6.2.2	Max 10 characters	NHS Pathways version in use for this case/triage
clinicalAreaid	Integer <i>Required if applicable</i>	1 to 22		ClinicalAreaID from the pathway selected 1 (eye) 2 (head) 3 (breast) 4 (urinary) 5 (arm) 6 (back) 7 (hip & leg) 8 (skin and hair) 9 (not area specific) 10 (flank)

				11 (rectum/anus) 12 (shoulder) 13 (groin) 14 (genital) 15 (abdomen) 16 (chest) 17 (face) 18 (neck) 19 (ear) 20 (nose) 21 (mouth) 22 (throat)
dosTransactionId	Uniqueidentifier <i>Required if applicable</i>	8abb9e61-941f-4666-b1af-809bf95fc5b9	36	If a call to DoS was made, the unique reference returned from the DoS call.
dosStartTime	DatetimeOffset <i>Required if applicable</i>	2015-12-23T04:43:30+00:00	24	DoS start time
dosEndTime	DatetimeOffset <i>Required if applicable</i>	2015-12-23T04:43:30+00:00	24	DoS end time
dosRequestFailId	Integer <i>Required if applicable</i>	104		If call to DoS fails, the reason why it failed. 300 - Unable to login to DoS web services 301 - The web service is currently unavailable 302 - Authentication failed 303 - Authentication failed missing role 100 - System Search failure 101 - Postcode Validation: Postcode not found 102 - Service Search: No services found 103 - Request Response: Service not found 104 - Parameter Validation: No valid discriminators passed/SGSD combo not valid 105 - Parameter Validation: Invalid search distance/Invalid disposition 106 - Parameter Validation: Invalid gender/ageFormat supplied - supported values
callerType	Text <i>Required if available</i>	Some caller type text here	Max 50 characters	Text field to capture caller type (if used). Mainly applicable to 999 sites.
callReason	Text <i>Required if CallerType is populated</i>	Some reason text here	Max 50 characters	Text field to capture caller reason (if used). Mainly applicable to 999 sites.
nhsNumber	Text <i>Required if available</i>	A made-up example would be: 943-476-5919		We accept the NHS number with spaces, hyphens or purely numeric. This needs to be PDS-traced NHS Number

				only. Required if it has been collected at the start of triage.
journeyId	Uniqueidentifier (8-4-4-12 UUID style) <i>Required</i>	8abb9e61-941f-4666-b1af-809bf95fc5b9		This is a unique number created by the first site that begins a patient journey. As per CDA spec: https://uec-tech.atlassian.net/wiki/spaces/IUC/pages/670215/ITK+Messaging
nhsEnglandAreaCode	Text <i>Required</i>	An example for Yorkshire and Humber would be: 111AD9	Max 10 characters	This field is sometimes known as the NHS 111 Area Code. Each contracted 111 provider will have a unique number. More information can be found by contacting the NHS England IUC team at england.nhsdata@nhs.net or by visiting the link below: https://www.england.nhs.uk/statistics/statistical-work-areas/nhs-111-minimum-data-set/
supplierRelease	Text <i>Required</i>	An example might be: 3.23.11	Max 30 characters	This field is the version of the host system itself, so that data can be linked back to whichever version of the host system processed that call.

3.1.4.1.2 Triage Record

The table below indicates the data collected as the user triages a caller. There will be *multiple* records for each patient triage episode. Triage records should be returned in the json packet in the correct order that the triage occurred in.

Field	Type	Example	Size	Notes
userCode	Text <i>Required</i>	NFT123	Max 10 characters	All triage records need a UserCode as more than one user may triage a call if collected from a queue. Clinician or Call Handler user code.
timeIn	DatetimeOffset <i>Required</i>	2019-07-23T04:43:30+00:00	24	Creation date time of TriageRecord.
timeOut	DatetimeOffset <i>Required</i>	2019-07-23T04:43:30+00:00	24	Date time record is exited – e.g. by activating next procedure.
pwld	Text <i>Required</i>	PW123	Max 20 characters	PathwayID.
orderNo	Integer <i>Required</i>	100		OrderNo of the triage record.

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qulD	Text <i>Required if applicable</i>	Tx220074	Max 20 characters	Question (Tx130011), Care Advice (Cx150010), dispo code (Dx140000), or body map area (hip & leg).
answerNo	Integer <i>Required if applicable</i>	3		The actual answer number selected by the user
userComment	Text <i>Required if applicable</i>	specify comment here	Max 300 characters	Any extra information added by the user for this answer.
actionId	Integer <i>Required</i>	1 to 6		The procedure used by the user to leave the record. Mostly this will be next but there are other ways to exit a record. List of possible values: 1 (Back) 2 (Next) 3 (Change) 4 (Early Exit) 5 (Restart) 6 (End)
skillsetId	Integer <i>Required</i>	6	3	Skillset in use at the time the Triage Record was created. e.g. 6 (111 Clinician), however any skillset ID from the skillset table is valid.

3.1.4.1.3 Services Returned record

The Services returned data is used to describe the services returned from the DoS, giving a picture of the order in which, they were presented, which services were rejected and which service are ultimately selected. For example, if 10 services were returned, each will start off with a statusId of 1 (service returned). Those which were rejected will have a statusId of 2 along with a rejection reason. The service which was finally selected (if appropriate) will have a statusId of 3.

There should not be any duplicate services returned. The Last / FINAL set of services returned should be sent in the data and any initial services should be excluded.

Field	Type	Example	Size	Notes
serviceId	Integer <i>Required if applicable</i>	1355999568		DoS organisation service ID.
serviceOrder	Integer <i>Required if applicable</i>	2		The order in which the service was displayed to the user by your system (i.e. 1 = first result).
serviceStatusId	Integer <i>Required if applicable</i>	3		Integer used to describe Status of what happened with the service (default of 1): 1 (Service Returned) 2 (Service Rejected) 3 (Service Selected) 4 (Bulk Reject)

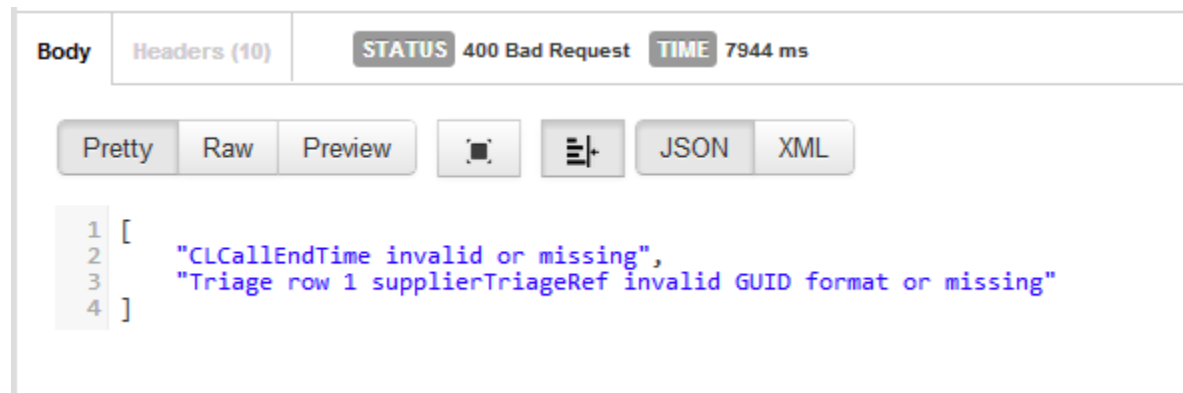
rejectReasonId	Integer <i>Required if applicable</i>	22		If the service was rejected, please include the reason why the service was rejected. Please refer to section 9.1.1 for more information on reject reasons.
----------------	--	----	--	--

3.1.4.2 cqiExceptionCodes method

The cqiExceptionCodes simply returns a list of known exception codes along with a description. Valid basic authentication details still need to be sent in the request header, but no JSON data is required in the request body.

3.1.5 Validation

Upon calling the cqiCase method, validation takes place on the data to ensure that it meets the requirements of the IDT webservice. An http 400 status (Bad Request) is returned along with a JSON list of strings describing where the validation failed.



3.1.5.1 Date/Time validation

The following date/time stamps must be submitted in the **DatetimeOffset** format

“2019-07-23T17:34:07+00:00”

chCallStartTime
 chTriageStartTime
 chDispoTime
 chTriageEndTime
 chCallEndtime
 clCallStartTime
 clTriageStartTime
 clDispoTime
 clTriageEndTime
 clCallEndtime
 dosStartTime
 dosEndTime
 timeIn
 timeOut

In addition, validation checks are also in place to ensure quality. As an example, this means that the “chDispoTime” or the “chCallEndtime” cannot be less than the “chCallStartTime”.

Please note that not all mandatory items are validated, but we ask that your submissions comply with section 3.1.4.1 CQI Case Method to ensure that fields marked ‘required if applicable’ or ‘required if available’ are adhered to.

3.1.6 Sample JSON packet

3.1.6.1 cqiCase method JSON

```
{
  "siteId":2,
  "supplierCallRef":"96d1176d-b954-4c1a-81e6-f68a430067b9",
  "ageGroupId":1,
  "ageInYears":19,
  "genderId":1,
  "partyId":1,
  "postcode": "S11 4TA",
  "chUserCode":"NFT",
  "chSkillsetId":5,
  "chCallStartTime":"2019-07-31T17:26:16+00:00",
  "chTriageStartTime":"2019-07-31T17:28:24+00:00",
  "chDispoTime":"2019-07-31T17:30:06+00:00",
  "chTriageEndTime":"2019-07-31T17:34:01+00:00",
  "chCallEndtime":"2019-07-31T17:34:07+00:00",
  "chTriageDispoCode":"Dx323",
  "chFinalDispoCode":"Dx323",
  "chOverrideFlag":false,
  "clUserCode":"SRT",
  "clSkillsetId":6,
  "clCallStartTime":"2019-07-31T17:37:16+00:00",
  "clTriageStartTime":"2019-07-31T17:38:16+00:00",
  "clDispoTime":"2019-07-31T17:42:16+00:00",
  "clTriageEndTime":"2019-07-31T17:44:11+00:00",
  "clCallEndtime":"2019-07-31T17:45:28+00:00",
  "clTriageDispoCode":"Dx06",
  "clFinalDispoCode":"Dx05",
  "clOverrideFlag":true,
  "symptomGroup":1003,
  "symptomDiscriminator":4003,
  "releaseVersion":"6.6.3",
  "clinicalAreaId":21,
  "callerType":"Police at scene (this field is optional)",
  "callReason":"RTA (this field is optional)",
  "nhsNumber":"000-000-0000",
  "journeyId":"96d1176d-b954-4c1a-81e6-f68a430067b9",
  "nhsEnglandAreaCode":"111AEA",
  "dosTransactionId":"96d1176d-b954-4c1a-81e6-f68a430067b9",
  "dosStartTime": "2019-07-31T17:28:24+00:00",
  "dosEndTime": "2019-07-31T17:30:32+00:00",
  "listOfTriageItems" : [

    {
      "userCode": "NFT",
      "timeIn": " 2019-07-31T17:28:24+00:00",
      "timeOut": " 2019-07-31T17:28:45+00:00",
    }
  ]
}
```

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```
"pwlId": "PFA3",
"orderNo": 5,
"quType": "Tx220074",
"answerNo": 4,
"userComment": "specify any comments here",
"actionId" : 2,
"skillsetId" : 5
},
{
"userCode": "NFT",
"timeIn": " 2019-07-31T17:28:46+00:00",
"timeOut": " 2019-07-31T17:28:55+00:00",
"pwlId": "PFA3",
"orderNo": 900,
"quType": "Tx220715",
"answerNo": 5,
"userComment": "specify any comments here",
"actionId" : 2,
"skillsetId" : 5
},
{
"userCode": "NFT",
"timeIn": "2019-07-31T17:29:08+00:00",
"timeOut": "2019-07-31T17:29:32+00:00",
"pwlId": "PFA3",
"orderNo": 1400,
"quType": "",
"answerNo": 1,
"userComment": "",
"actionId" : 2,
"skillsetId" : 5
},
{
"userCode": "NFT",
"timeIn": "2019-07-31T17:29:35+00:00",
"timeOut": "2019-07-31T17:29:47+00:00",
"pwlId": "PA110",
"orderNo": 11,
"quType": "",
"answerNo": 300,
"userComment": "",
"actionId" : 2,
"skillsetId" : 5
},
{
"userCode": "NFT",
"timeIn": "2019-07-31T17:29:52+00:00",
"timeOut": "2019-07-31T17:30:22+00:00",
"pwlId": "PA110",
"orderNo": 300,
"quType": "Tx221351",
"answerNo": 1,
```

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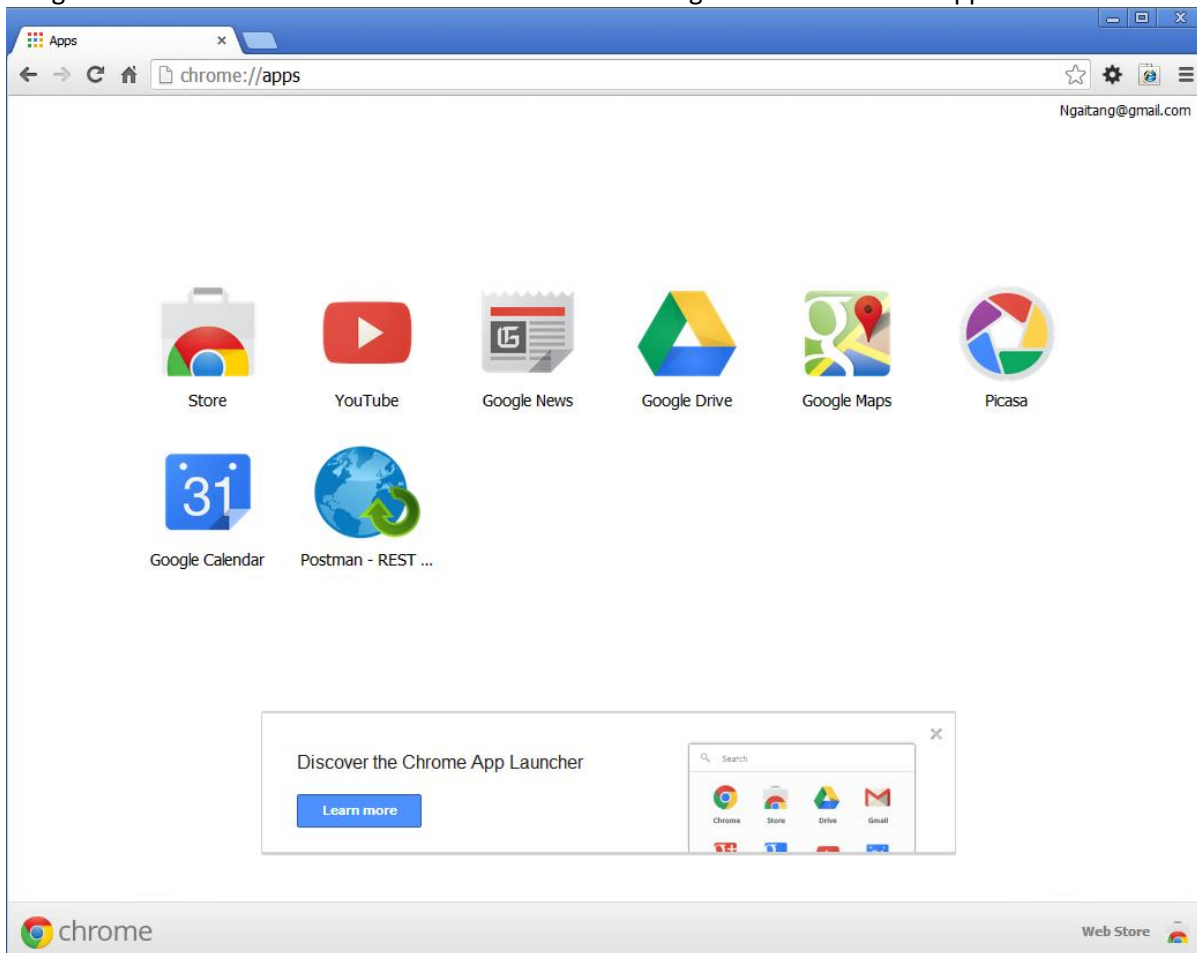
```
"userComment": "",
"actionId" : 2,
"skillsetId" : 5
},
{
"userCode": "NFT",
"timeIn": "2019-07-31T17:30:25+00:00",
"timeOut": "2019-07-31T17:31:02+00:00",
"pwdId": "PA110",
"orderNo": 300,
"quType": "",
"answerNo": 0,
"userComment": "",
"actionId" : 6,
"skillsetId" : 5
}
],
"listOfServiceReturns" : [
{
"serviceId":1323649101,
"serviceOrder": 1,
"serviceStatusId": 2,
"rejectReasonId":22
},
{
"serviceId":132594644,
"serviceOrder": 2,
"serviceStatusId": 2,
"rejectReasonId":24
},
{
"serviceId":13265794487,
"serviceOrder": 3,
"serviceStatusId": 3
},
{
"serviceId":12354567677,
"serviceOrder": 4,
"serviceStatusId": 1
},
{
"serviceId":13264497445,
"serviceOrder": 5,
"serviceStatusId": 1
}
]
}
```

3.1.6.2 [cqiExceptionCodes](#) method JSON

No input JSON is required here. Just provide the Basic Authentication details in the request header, and upon supplying a valid username and password, a list of exception codes will be returned in JSON format.

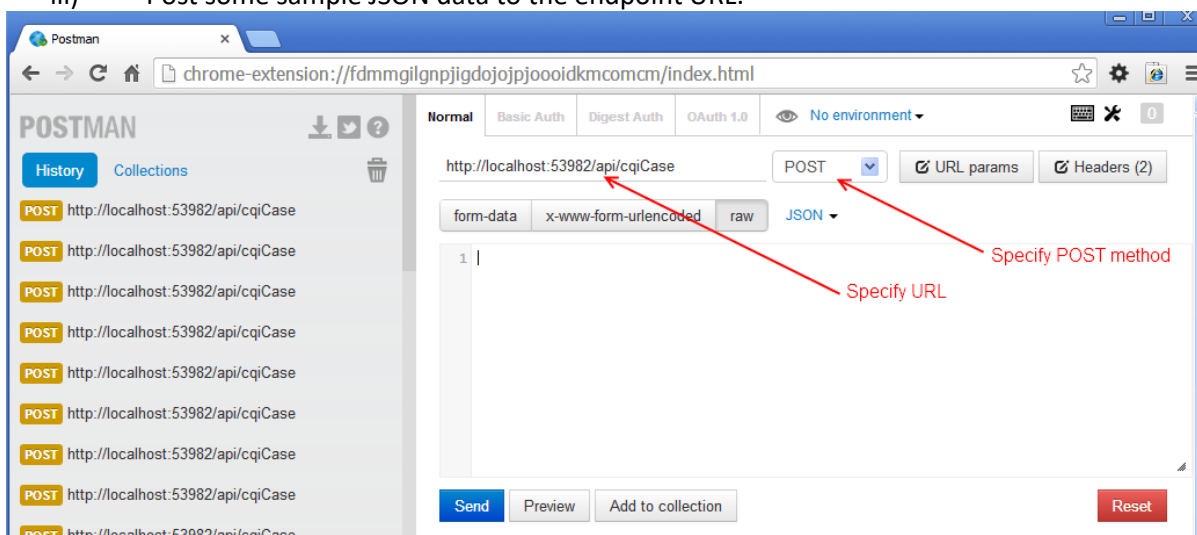
3.1.7 Testing

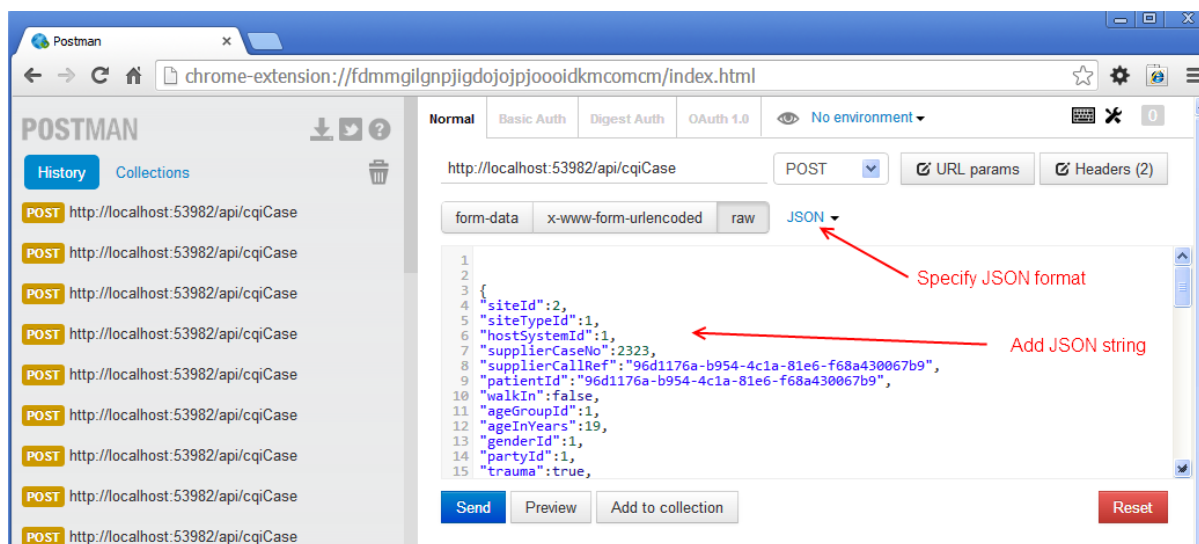
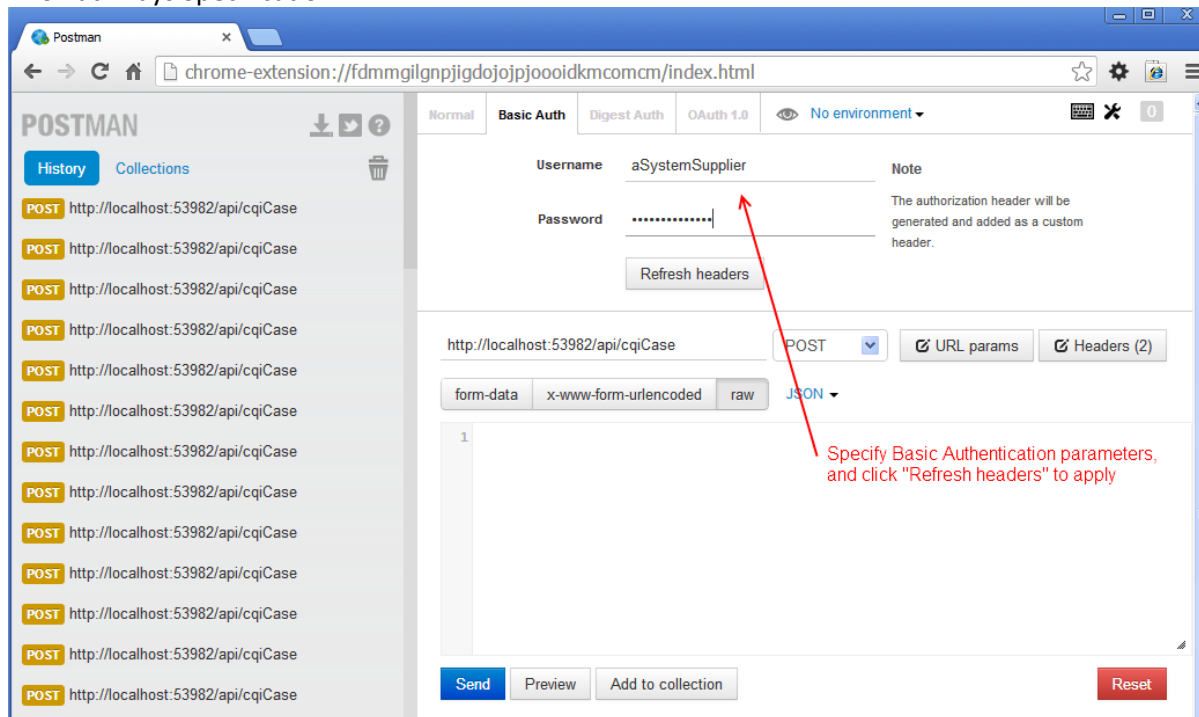
Suppliers are free to create their own test interfaces, but a useful testing resource is the “Postman REST client” app from Google Chrome. This is available to download free of charge from the Chrome App store.



A simple interface is provided which allows the user to:

- Specify an IDT webservice endpoint URL.
- Provide some basic authentication parameters. This automatically Base64 encodes the password sent in the request header.
- Post some sample JSON data to the endpoint URL.





3.1.8 Resending

System suppliers are requested NOT to resend a call once it has been successfully sent, as this will create duplicates in the system.

3.2 GP Reports

In addition to the data export needed additional information needs to be sent to daytime GP practice or Out of Hours services in order to ensure continuity of care is achieved. There are three types of report all of which include data which enables a match to be made to the patients GP record. In **addition** to this the following fields should be utilised:

3.2.1 Short Summary Report

This report should be at the top of any report for daytime GP or Out of Hours

Data Source	Notes/additional behaviours
TriageRecord::userComment	This should be displayed at the point of the call it was taken i.e. start of call or at a particular question response.
TriageRecord::DispoRationale	
Pathways::PathwayTitle	Pathway selected by user.
CaseRecord::chTriageDispoCode or/and CaseRecord::clTriageDispoCode	If the call is passed from call handler to clinician, it is useful to show all the different dispositions achieved during the interaction.
CaseRecord::chFinalDispoCode or/and CaseRecord::clFinalDispoCode	If the call is passed from call handler to clinician, it is useful to show all the different dispositions achieved during the interaction.
TriageRecord::DispoOverride	Include if original disposition overridden.

3.2.2 Full Report

This report should be sent to both Out of Hours and GP in hours

Data Source	Notes/additional behaviours
TriageRecord::userComment	This should be displayed at the point of the call it was taken i.e. start of call or at a particular question response.
TriageRecord::ReportText	Answers with value of positive = 1 should be highlighted in some way (bold, italics, coloured). Both Tx and Cx answer stems should be displayed so question and care advice statements can be included.
Pathways::PathwayTitle	Pathway selected by user.
CaseRecord::chTriageDispoCode or/and CaseRecord::clTriageDispoCode	If the call is passed from call handler to clinician, it is useful to show all the different dispositions achieved during the interaction.
CaseRecord::chFinalDispoCode or/and CaseRecord::clFinalDispoCode	If the call is passed from call handler to clinician, it is useful to show all the different dispositions achieved during the interaction.
TriageRecord::DispoOverride	Include if original disposition overridden.

3.2.3 Detailed Report

This report is usually for investigation and call audit and contains full questions and the possible answers
Full questions and answers

Data Source	Notes/additional behaviours
-------------	-----------------------------

TriageRecord::userComment	This should be displayed at the point of the call it was taken i.e. start of call or at a particular question response.
TxQuestions::QuestionText	Use the QulD number to determine the question.
TxAnswers::AnswerText	Display all the answer stems associated with the QulD indicating which answer stem was selected by the user.
Pathways::PathwayTitle	Pathway selected by user.
CaseRecord::chTriageDispoCode or/and CaseRecord::clTriageDispoCode	If the call is passed from call handler to clinician, it is useful to show all the different dispositions achieved during the interaction.
CaseRecord::chFinalDispoCode or/and CaseRecord::clFinalDispoCode	If the call is passed from call handler to clinician, it is useful to show all the different dispositions achieved during the interaction.
TriageRecord::DispoOverride	Include if original disposition overridden.
CareAdvice::CxKeyword	Use the QulD number to determine the care advice sets chosen by the user.
CareAdviceItems::CxText	Display all the CxText stems offered to the user indicating which stems were chosen.

3.2.4 Call Review

Organisations using NHS Pathways will also want to review individual calls. Therefore, it should be possible for administrators to search for all records for a specific *supplierCallRef*.

It would be a very useful development to “replay” the call using the data from the **CaseRecord** and **TriageRecord** tables in an embedded system. This basically works as follows:

- ❓ The replay system uses the same interfaces as the triage system
- ❓ Records are read only
- ❓ Data from **CaseRecord** and **TriageRecord** populates the interface such that answers provided are highlighted
- ❓ The user can step through the triage using the return key to go to the next record
- ❓ The user can also elect to replay the triage automatically. Each triage record is presented in turn and held on the screen for the duration of time originally spent by the call taker. The duration for each screen can be calculated from the difference between *TimeIn* and *TimeOut*
- ❓ This would enable an audio record of the call to be played at the same time as the computer record
- ❓ There should be a pause facility with the replay system
- ❓ Because interface information is stored with the Triage Record, the replay system knows which to use

3.3 Exchanging Pathways data with other systems

The sending and receiving of messages containing Pathways triage data is now part of the 111 interoperability specification and reference should be made to documents released by the NHS England 111/999 and Integrated Urgent and Emergency Care programme.

4 Training and Testing Database

It is expected that the host system will have a training database and therefore there will be a parallel installation of NHS Pathways data. This is essential for testing and training of new Pathway releases. Similarly, services will require a system

5 Importing NHS Pathways Data

5.1 Data format

Data will be supplied as CSV files. Separate files for each of the Pathways data tables will be supplied. They will always be supplied as complete sets of tables even when only one or some tables are affected by the new release. XML files containing version information and checksum data are also supplied with each release and these should be verified before importing the data.

5.2 Validation of data integrity

Every effort will be made to ensure that the NHS Pathways data supplied by the NHS Pathways team is fully checked for logic. However, host system developers should satisfy themselves that the data is properly formed before importing it into their system.

5.2.1 General Validation Requirements

- ❓ where a value for a field is shown as “required” in Section 2.2 then it should have a value
- ❓ all fields are properly trimmed for leading and trailing spaces

5.2.2 For the **Triagelogic** table

- ❓ *OrderNo* = 0 exists for all pathways
- ❓ *OrderNo*’s for all *JumpID*’s exist
- ❓ *BlockOrderCalc*’s exist for all Transfer codes in fields *T1*, *T2*, *T3* and *T4* except for Transfer codes containing “Dx” where a *Dispocode* should exist in the **Dispositions** table
- ❓ *QuestionID*’s exist in the related **TxQuestions**, **CareAdvice** and **Disposition** tables
- ❓ *AnswerNo*’s exist in the related **Txanswers**, **CareAdviceItems** and **DispositionItems** tables
- ❓ There are no records that have the same *PW_ID*, *OrderNo* and *AnswerNo* combined

5.2.3 For the **TxQuestions** **CareAdvice** and **Dispositions** tables

- ❓ There are no duplicate *QuID*’s in the **TxQuestions**
- ❓ There are no duplicate *CxID*’s in the **CareAdvice**
- ❓ There are no duplicate *DxID*’s in the **Dispositions**

5.2.4 For the **TxAnswers**, **CareAdviceItems** and **DispositionItems** Tables

- ❓ There are no records that have the same *QuID* and *AnswerNo* combined in **TxAnswers**
- ❓ There are no records that have the same *Record ID* in **DispositionItems**
- ❓ There are no records that have the same *CxID* and *CxNo* combined in **CareAdviceItems**

6 Local Customisation of Pathways

6.1 Disposition Items

For a given disposition, e.g. “Contact Primary Care Service within 2-6 hours”, there are alternative dispositions such as “to Clinician”. These override dispositions to the triage disposition will depend on local conditions and therefore the local site administrators will need the ability to turn on or off these options.

It is not the NHS Pathways team’s intention to allow sites to add their own disposition alternatives, rather sites will be able to submit request for change to the NHS Pathways team and approved changes will be added through the normal clinical release process.

To do this, developers will need to provide an interface, accessible only by those who have the necessary permissions, to customise the list of options.

6.1.1 Fields for display in interface

The interface provided can match the host system’s style of administrative tools but should include the following fields.

Field	Example	Edit	Notes
<i>Dispocode</i>	Dx01	no	
<i>DispoText</i>		no	Present this as a drop-down list so that administrators can select the disposition they wish to customise.
<i>DxItemNo</i>	1, 2 etc	no	List of items associated with this <i>dispocode</i> .
<i>DxItemText</i>	The disposition is 999	no	
<i>Active</i>	0 or 1	yes	Radio button set – selecting 0 will exclude an item in user disposition screens. Note: the first Dx item is always the actual disposition, therefore there must always be an active <i>DxItemNo</i> =1 for each <i>dispocode</i> This needs to be validated before exit from the screen.

6.1.2 Dispositions routed to DoS

Only dispositions with a CMS group code are normally sent to DoS. This prevents ambulance dispositions and home care from being sent which would find no matches in DoS. However, there are some cases where there is no national coverage for a service type but there is local coverage and a region may wish a disposition to be sent to DoS. An example of this is Dx79 – Failed Contraception. Provided permission has obtained from the NHS Pathways Clinical Authoring team, a service may request their supplier to add CMS group data to the DxMap table. The pattern is simple, disposition Dx79 will have a CMS Group code of 1079 (i.e. 1000 + the numeric part of the disposition code).

Please note that there are a couple of exceptions to the pattern, notably Dx021 (10021) and Dx1111 (CMS Group code of 11111).

As these changes are held locally they will be overwritten by the import of new content from Pathways.

7 “Attend Incident” calls in Ambulance Settings

7.1 Introduction

Advice on how to construct an Attend incident application using Pathways dispositions has been removed. Attend incident pathways have now been compiled in the same manner as standard NHS Pathways CSV files. It is up to the service and their supplier how these pathways are launched but the most obvious method is to provide a button on the CAD interface to launch the specific attend incident pathway.

8 Ambulance Stand down and Triage cut off point

8.1 Setting Triage cut off point/ambulance stand down

The field **dispomap::DxRank** contains a number which indicates the rank of a disposition (no blanks allowed). The lower the number then the higher the Dispositions rank. For instance, a Category 1 ambulance disposition has a value of 10 whereas home management has a rank of 500. Individual dispositions may share the same rank. For various reasons a service may want to set a cut-off point in triage once a certain disposition is passed. For instance, there is undue operational pressure on the service or the service has a different skill mix to that envisaged by NHS Pathways. In the same way, an ambulance service may want to stand down a response vehicle once all or the most serious ambulance dispositions have been passed.

In order to provide a reliable method of stand down and triage cut off we introduced a “Rank” column into the triagelogic table. This column contains a numeric value where a low rank number indicates a potential ambulance dispatch. For instance, rank 10 indicates a Category 1 response, rank 20 a Category 2 response and rank 30 a Category 3 response.

For each row in triagelogic the rank column contains the rank number of the highest disposition that selecting that answer could lead to. When a call handler selects an answer, the system can check the rank column to see what the highest disposition that route through the triage could lead to. This method also gives the service and system supplier the freedom to determine when they stand down ambulances as there will be sufficient granularity in the rankings to distinguish the categories of ambulance dispatch. i.e. the stand down point could be altered according to REAP levels if required. Similarly, the granularity of primary care dispositions will allow triage cut off point to be determined – the mechanism is the same.

8.1.1 Setting Triage cut off

-  The site can set a value for the cut off disposition rank (e.g. 100) at any time via an administration screen which needs to be supplied by the host system. Who has administration rights is an operational issue. The value groups for the host screen are provided by NHS Pathways in the table DxRank
-  The cut-off point can be set independently for call handlers (skillsets with SkillsetUserType set to CH) and clinicians (skillsets with SkillsetUserType set to CL).
-  If the cut off disposition is set to 0 (or not set at all) then again, this is all ignored, and triage continues as normal.
-  If the rank of the triagelogic row exceeds the set cut-off, a call handler triage should receive a Dx000 disposition, whereas a clinician triage should receive a Dx001.

**NHS Pathways
demonstration version**

Set call handler Triage Cut Off

- ☒ 0 perform full triage
- ☐ 1 cut after emergency ambulance response
- ☐ 2 cut after 1 hour response
- ☐ 3 cut after 4 hour response
- ☐ 4 cut after 6 hour response

Set Clinician Triage Cut Off

- ☒ 0 perform full triage
- ☐ 1 cut after emergency ambulance response
- ☐ 2 cut after 1 hour response
- ☐ 3 cut after 4 hour response
- ☐ 4 cut after 6 hour response

Return

Triage cut off screen

A sample screen is shown above which allows the administrator to set the cut off point for clinicians and call handlers.

9 Connecting to the DoS

The DoS holds the profile of every NHS service in England. It provides search functionality which uses NHS Pathways symptom groups and discriminators to return services such as out of hours providers, walk in centres, minor injury units, etc.

Each route through the pathways system will result in a SymptomGroup and this needs to be collected in the TriageRecord. If the triage transfers to another pathway with a value in this field then the value in the TriageRecord should be overwritten, otherwise it should remain.

Also during triage individual rule questions may have values in the SymptomDiscriminator field. These values should also be collected. They are collected just before the disposition using a Type 3 question. There should only be one symptom discriminator and one symptom group collected at the end of the triage (the most recent encountered of each). These fields, as well as other data, are sent as a search query to the DoS.

Note: For DoS calls from within the PaCCS module, the SymptomGroup should always be set to SG1206 (NHS Pathways In House Clinician) for the DoS call, regardless of the SymptomGroup stored against the case (which should remain unchanged).

9.1.1 Reject Reasons

9.1.1.1 Background

Reject reasons allow the NHS Digital DoS team to understand why the suggested services are not appropriate for patients. Please see https://developer.nhs.uk/apis/dos-api/reject_reasons.html as managed by the NHS Digital DoS team for the current list of rejection reasons.

9.1.1.2 Functional requirements

- ☐ The host system must collect a reject reason for every individual service that is rejected in the DoS results list. Consideration should be given to the user experience of having to select reject reasons, and it is suggested that systems are designed to make this low friction for the user. Ensuring the steps of rejecting and selecting a reason flow well can significantly help here e.g. by tying the choice of a reason into the rejection step to reduce the number of clicks, or amount of mouse movement required.
- ☐ Where a Reject Reason is marked as “Requiring Comment”, the user should be prompted to enter some additional comments about the rejection.
- ☐ Where a Reject Reason is marked as “Clinician Only”, it should not be presented as an option for non-clinical users.
- ☐ The host system must keep records of all rejected services, ensuring that the records include:
 - o The transaction ID of the DoS results (or other identifier allowing a link back to the DoS transaction)
 - o The ID of the service that was rejected from the list
 - o The ID of the reject reason selected
 - o The timestamp at which the service was rejected by the user
 - o Any additional comments made about the rejection
- ☐ It must be possible for the host system to provide rejected service data back to NHS Digital (in the case of host systems using NHS Pathways, this will be via the CQI Web API).
- ☐ The reject reasons list should be maintained in the host system as reference data i.e. data that can change outside of formal specification uplift cycle. Host systems should implement this in a way that reduces the effort and cost of updating the list of reasons for users. It must be possible for these reasons to be updated without requiring a full software release of the host system (i.e. stored in a database table as opposed to hard-coded into application code).

9.1.2 Disposition Override and DoS

A clinician using NHS Pathways has the ability to override the final disposition. If for instance, an original ambulance disposition is downgraded to a visit to ED or Primary Care, or if a home care disposition is upgraded to Primary Care, a mismatch can be created that will result in an illegal request to DoS. Examples: a nurse upgrades a home care disposition to GP 6 hours. The disposition code is marked to trigger a DoS search but no suitable symptom discriminator will have been collected. In this case DxMap contains fields called “SubstituteCode” and “SubstituteGroup”. It contains a value for Dx06 of “4003” which is the SD code for “fully primary care assessment and treatment” and a SG of 1206 which is the SG

NHS Pathways Specification

code for “NHS Pathways In House Clinician”. By substituting any collected SD or SG codes with these values one a legal combination of SG/SD and disposition code will be produced for the DoS search. Note this mechanism is only to be used when a clinician overrides the original disposition.

Screen – clinician disposition

User Action – nurse selects a new disposition to override system disposition

System Action – replace all symptom discriminator codes collected with value in **Dispomap:: SubstituteCode** field for the override disposition selected.

System Action – replace all symptom group codes collected with value in **Dispomap:: SubstituteGroup** field for the override disposition selected.

10 Supporting Materials

System providers will be provided with a link to download additional materials to support the development of NHS Pathways.

10.1 NHS Pathways Data Set

A set of CSV files will be available to download. These are exactly as will be provided for future updates and will be a recent set which has seen active use in an NHS call centre.

10.2 Call Summary Report

The following is an example of the call summary report as displayed to a call handler at the end of triage. Where UserComment data has been entered on the disposition screen, this should be shown at the end of the disposition line (highlighted in bold below). Where user comments are entered for any other point in the triage, this should be included after the relevant line of the report (emphasized in italics below).

Caller information precedes:

DISPOSITION: Dx02 - The individual needs to attend an emergency treatment centre within 1 hour. – **patient getting a lift**

CONSULTATION SUMMARY:

Illness - nose bleed

Warm to touch

Nose bleeding freely

Unilateral nasal obstruction before nosebleed

PATHWAYS ASSESSMENT:

An injury or health problem was the reason for the contact.

Heavy bleeding had not occurred in the previous 2 hours.

An illness or health problem was the main problem. - *nose bleed*

The individual was not fighting for breath.

The main reason for the assessment was not an allergic reaction, heart attack, chest/upper back pain, probable stroke, recent fit/seizure or suicide attempt.

The main reason for contact was not new confusion, declared diabetic hypo/hyperglycaemia, or ICD shock.

The skin on the torso felt normal, warm or hot.

Pathway Selected: PW818, Nosebleeds without Injury - *No injury declared*

The individual was not feeling faint or dizzy.

The nose was continuing to bleed.

There was no nasal foreign body.

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Constant pressure for at least 10 minutes had not been applied.

There was no spontaneous bruising, abnormal bleeding or frequent nosebleeds within the previous 2 weeks.

The individual did not use anticoagulant medication or have a bleeding/clotting disorder.

There had been no recent nasal injury or surgery.

Night sweats, feeling generally unwell, unexplained weight loss, loss of appetite or constant tiredness had not been experienced over the previous 4 weeks.

There were no new lumps or swellings that had been present for more than 2 weeks.

The individual had not received care from a cancer specialist within the previous year.

The individual had unilateral nasal obstruction prior to the nosebleed.

ADVICE GIVEN:

Sit forward and pinch all of the fleshy part of the nose for 10 minutes without releasing the pressure.

Put an ice pack (e.g. a bag of frozen peas) wrapped in a cloth on the bridge of the nose. - *Previously attempted for 2 minutes*

Someone else should drive the individual.

Take any medicines and medical records with you.

Take a mobile phone if possible, in case symptoms get worse and an ambulance is needed.

Before you go, I will just check whether I need to give you any further instructions or advice.

Sit forward and pinch all of the fleshy part of the nose for 10 minutes without releasing the pressure.

Don't pinch the nose if the nosebleed is caused by an object stuck up the nose.

Put an ice pack (e.g. a bag of frozen peas) wrapped in a cloth on the bridge of the nose.

Once the nosebleed has stopped, don't pick or blow the nose and avoid heavy lifting or vigorous exercise. - *Instructions noted by patient*

Please refer to the ITK specification for the format that the summary report should take as part of the CDA interoperability message. Note this is not the full triage record used to present call validation in a receiving Pathways system.

11 Clinical Assessment Service

11.1 Pathways Light

The Integrated Urgent and Emergency Care Programme identified a concept within the CAS project where lightly-trained service advisors could potentially handle some calls of a limited nature and duration.

There were a number of different elements within this concept that were proposed. Two of which were calls regarding repeat prescription and calls from a Health Care Professional. If lightly trained service advisors were able to access those elements without being able to traverse the full NHS Pathways content, then they would not need the full NHS Pathways Call Handler training.

The Pathways Light (PL) modules have reduced training needs and provide a solution which enables easily identified calls to be removed from the 111 call queue via an IVR and which could then be managed by staff who have other duties within a provider organisation, for example.

There are currently 2 NHS Pathways Light modules:

- Health Care Professional Call Back
- Pharmacy/Repeat Prescription

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It is expected that calls will be routed to Pathways Light module users via an IVR within 111 providers. E.g. press 2 for Repeat Prescriptions. Once the caller selects an option, the call will be routed to an appropriate member of staff who can triage the call using the relevant NHS Pathways Light module*.

Staff will check that the caller has selected the correct option on the IVR prior to starting triage. If at any point during the call full triage is required, then the call will be passed to a clinician via outcomes in the pathway or via an early exit option.

The term “light” relates to the NHS Pathways content within each module and the training requirements for users. The NHS Pathways content is specific to the module. An NHS Pathways Light user cannot access the main system and perform full triage; they can only perform triage in the NHS Pathways Light module(s) that they have permission to access.

Based on discussion with providers, the likely users of the NHS Pathways Light modules will be staff with other roles within the organisation, e.g. Service Advisors.

Each module will have its own skill set. Users may be trained to use one module, a combination of modules or all of the modules. System suppliers will need to define how users can be assigned skillsets.

11.1.1 Implementation

To enable the new skillsets required for the NHS Pathways Light modules, the new skillsets table (CSV file) should be used. This includes the current skillsets in use within the pathways system, but also the new skillsets.

SkillsetID	SkillsetDescription	SkillsetUserType	Service	EntryPathwayId	EntryOrderNumber	Tbranch	EarlyExitPathwayId	EarlyExitOrderNumber	Enabled
1	OOH Call Handler	CH	OOH	AgeGenderParty	5	T1	PEE1	5	TRUE
2	OOH Clinician	CL	OOH	AgeGenderParty	5	T2	PEE1	5	TRUE
3	999 Call Handler	CH	AMB	AgeGenderParty	0	T3	PEE1	0	TRUE
4	999 Clinician	CL	AMB	AgeGenderParty	0	T4	PEE1	0	TRUE
5	111 Call Handler	CH	SPA	AgeGenderParty	5	T3	PEE3	0	TRUE
6	111 Clinician	CL	SPA	AgeGenderParty	5	T4	PEE3	0	TRUE
7	Reception Point Call Handler	CH	FOH	PA168	0	T1	PEE1	11	TRUE
8	Attend Incident	CH	AMB	PA166	0	T3	PEE1	0	TRUE
9	Repeat Prescription	CH	SPA	PW1808	0	T3	PEE4	0	TRUE
10	Dental	CH	SPA	PW1809	0	T3	PEE4	0	TRUE
11	HCP Callback	CH	SPA	PW1810	0	T3	PEE4	0	TRUE
12	Senior Clinician Module	CL	SPA	PW1812	0	T4	PEE4	500	TRUE
14	PaCCS	CL	SPA	PaCCS	0	T4	PEE4	9100	FALSE
15	Injury Module	CH	SPA	PW1857	0	T3	PEE4	8200	FALSE
16	ED Streaming and Redirection	CH	FOH	PA168	0	T1	PEE1	11	FALSE

Figure 1 Example Skillsets table (subject to change)

The table above aims to allow new skillsets to be introduced dynamically, without requiring wholesale system development each time. It describes each skillset, and what parameters that skillset should use within the NHS Pathways Triage system (i.e. which entry Pathway, which T branch, which Early Exit pathway etc). In the example above, the Repeat Prescription Pathways Light module starts at PW1808.0, using an Early Exit pathway of PEE4.0, and follows the T3 branch.

11.2 Senior Clinician Module

NHS Pathways Senior Clinician Module is a self-contained module that has been designed to facilitate DoS searching and ambulance dispatch for clinicians (GPs or Advanced Nurse Practitioners) based within a CAS. It is expected that calls will be routed to clinicians within a CAS following triage performed within 111 services or via other commissioned routes.

There are 2 elements to the Senior Clinician module:

- ☐ Ability to request an ambulance or construct a DX/SG/SD combination to search the DoS.
- ☐ Supported clinical assessment using existing clinical content which is structured to provide a checklist/aide memoire

11.2.1 Ambulance dispatch or DoS search

The NHS Pathways content within the Senior Clinician module is specific to that particular module and allows users to access tailored NHS Pathways content following their telephone clinical assessment. The Senior Clinician module will have its own skillset. Users with that skillset will only be able to access content relevant to that module and would not be able to access the main system to perform a full NHS Pathways triage.

Where an ambulance is required, the user is asked to confirm and record the presenting problem, assign an appropriate priority status (and required timeframe) and then request an ambulance.

In less urgent cases where no ambulance is required, the Senior Clinician module can be used to capture brief details of the telephone clinical assessment and the required response.

A body-map is presented to the user which allows them to select the appropriate symptom group. The clinician will then also manually select the level of response and the appropriate timescales required. This results in the creation of an SG:SD:DX combination which is then used to perform a DoS search.

The current version of the Senior Clinician module also contains disposition instructions and worsening advice to support the clinician and produces a written record of the call which can be maintained for auditing purposes.

11.2.2 Supported Clinical Assessment

Development of this is less advanced. This option may offer the functionality in option 1 at the end of assessment but will enable users to select different groups of clinical content which can be used as a checklist during assessment. It will not provide linear questioning such as is used by 111/999 users.

It is anticipated that users would be able to access the system after significantly less training than for full NHS Pathways users training. Users will be subject to the same audit levels as full users. This will be reviewed over time.

11.2.3 Implementation

Note: The first phase of the Senior Clinician Module (SCM) implementation is to enable Ambulance dispatch or searching of the DoS only.

Technically, the same launch mechanism as Pathways Light is used i.e. it will use a set starting pathway defined in the skillsets table. With the current design of the SCM content, the pathway starts with the question “Is an Ambulance Required?”

Ambulance dispatch route

For the ambulance dispatch route, all the logic of the module is achieved using core NHS Pathways functionality, and the pathways content will drive the user through to an end point.

DoS Search route

If an ambulance is not required, the aim is to allow the user to perform a DoS search for the appropriate service required. To facilitate this, the user needs to be able to select an SG, SD and DX. If the call is regarding an injury or an illness, the flow takes the user to the appropriate standard Body Map screen determined by age/gender. From this, the user selects a symptom group by choosing a pathway i.e. headache. The pathway selected by the user should be recorded in the TriageRecord so that it is displayed on the Pathways Selected line of the Call Summary Report. Instead of launching the user to the headache pathway though, the host system should perform a lookup on the new BodyMapOverride table to determine the destination pathway to use. Only the symptom group and pathway title should be used from the selected pathway, the remaining functionality (including picking up Care Advice Keywords) should be done based on the overridden pathway.

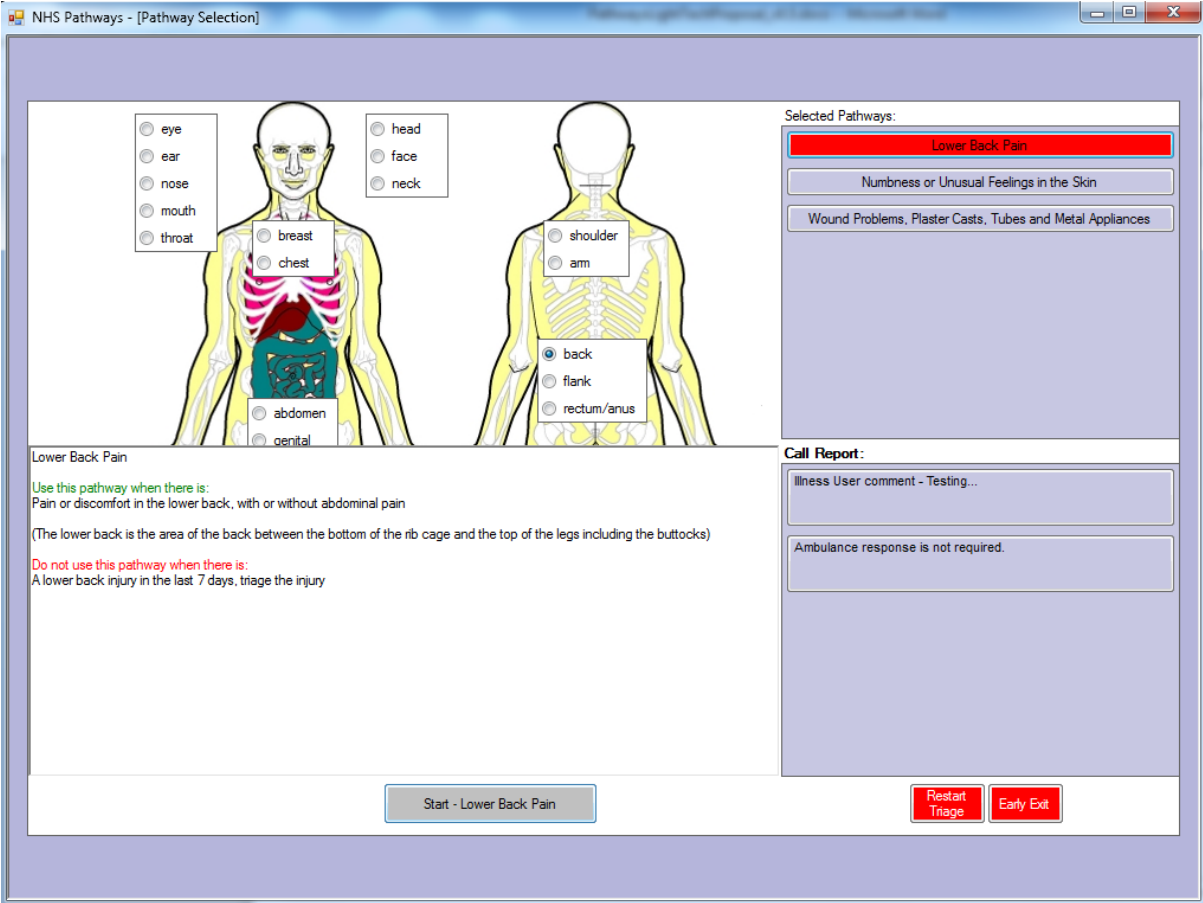


Figure 24 Body Map

SkillsetId	BodyMapPathway	BodyMapPathwayJumpsTo
12	*	PW1811
12	PW1345	PW1821
12	PW1346	PW1821
12	PW1348	PW1821
12	PW1349	PW1821
12	PW1395	PW1820
12	PW1543	PW1821
12	PW1551	PW1821
12	PW1552	PW1821
12	PW1610	PW1820
12	PW1611	PW1820
12	PW1612	PW1820
12	PW1749	PW1821
12	PW1751	PW1821
12	PW620	PW1820
12	PW622	PW1820
12	PW881	PW1821

Figure 25 Sample BodyMapOverrides table (subject to change)

This can be implemented in a number of ways dependent on system supplier, but a possible approach is to re-use the existing body map screen. When the user selects and starts a pathway, perform a lookup on the new BodyMapOverride table to determine the destination pathway to use. If no override exists for the skillset, continue as normal. In the example

above, if the user is a SCM user (skillset 12), and if PW1345 is selected from the Body Map, then the target pathway should be PW1821. The * is used to describe all other pathways.

11.3 Dental Module

The Dental Module is a stand-alone product for triaging dental symptoms. It is designed for dental nurses and call handlers who have completed specific training to use the module. The Dental Module will only triage adults and children over 5 years of age, and does not provide support or access to the main NHS Pathways system to perform a full clinical assessment.

It is envisaged that Dental calls will be routed to Dental module call handlers via telephone IVR (interactive voice recording) where the caller will select a dental option.

11.3.1 Product use

The Dental Module has a new entry pathway which replaces standard Module 0. For safety this pathway identifies potential serious illness or injury that requires full clinical assessment and cannot continue triage via the Dental Module - these patients receive a specific disposition indicating immediate full clinical assessment is required by a fully trained NHS Pathways call handler, NHS Pathways clinician or IUC CAS clinician; Calls that can safely navigate the 'Dental Module' entry pathway continue into the standard NHS Pathways dental suite.

The Dental module will have its own skillset. Users with that skillset will only be able to access the Module 0 content relevant to that module, and would not be able to access the main system to perform a full NHS Pathways triage.

The remainder of the NHS Pathways content within the Dental module is from core Pathways.

11.3.2 Implementation

Technically, the same launch mechanism as Pathways Light is used i.e. it will use a set starting pathway defined in the skillsets table.

At the end of Dental triage the user can perform a DoS search for the appropriate service required. To facilitate this, the user needs to be able to select an SG, SD and DX. As this process follows existing Dental Pathways, this should occur in the same way as main Pathways products.

11.4 Injury Module

The Injury Module allows calls related to injuries to be transferred to Service Advisors to free up Call Handlers to take other calls.

11.4.1 Product use

The main telephony Pathways have been updated to include checks for a new System Variable which indicates whether Service Advisors are available to take Injury Module calls. If they are, the SYSTEM_INJURYMODULE System Variable should be set to "injurymodule". During a standard triage, where the check is performed and the System Variable is appropriately set, the triage will navigate to a Dx500. At this point the call should be transferred to an Injury Module trained Service Advisor.

Upon picking up the call, the Service Advisor should be presented with the Call Report of the call progress to this point. At this point, the Service Advisor should be given the option to continue with the call in the Injury Module or cancel if for any reason the Service Advisor cannot take the call.

The Service Advisor may reach a point where they need to transfer the call to a Clinician, either by reaching an appropriate disposition, or via Early Exit. Calls should be picked up by Clinicians in the normal way, with the standard Call Validation screen (as detailed in section 2.5.8) being used. The Clinician should be able to continue the call at any valid point in the

Injury Module (as determined by the current Validation screen), or Restart the Triage to perform a Triage outside of the Injury Module. Exactly which type of Triage will be performed will be determined by the Skillset of the Clinician. For example they could Restart the triage into a standard Telephony triage, or a Senior Clinician Module assessment.

11.4.2 Implementation

A new Skillset is provided for an Injury Module trained user, which will work in the same way as the other Skillsets, with the Injury Module Entry Point, etc. configured in the Skillsets.csv release file.

The new System Variable will be included in the SystemVariables.csv file.

The System Variable should only be set to the value “injurymodule” if the Injury Module Skillset is enabled and Service Advisers are available to take calls.

The Clinicians Call Validation Screen (section 2.5.8) has been improved with a Restart Triage button added to enable a full triage to be performed without having to first continue the call before being able to Restart the Triage.

12 Healthcare Professional (HCP) Ambulance Admission & Inter-Facility Transfer (IFT) Calls

12.1 Launch Process

In 999 settings, HCP and IFT calls that do not originate via 999 lines (i.e. urgent line) need to follow the frameworks set out by the Association of Ambulance Chief Executives. These frameworks are encapsulated within NHS Pathways behind a new Entry Point.

A facility is therefore needed for users to jump directly into the new entry point (Pathway PHCP, order number 0). The calls should all be 3rd Party for this Pathway and should the PHCP Pathway not exist within the Pathways Release being used, a suitable message should be displayed.

As Hosts Systems differ, the actual implementation of this facility is left to the suppliers to determine. As an example however, Solo implements this as an additional button presented to the user when a Skillset is selected that has the **Skillsets::Service** attribute set to “AMB”. Further validation is performed on clicking the button to ensure the called is set to 3rd Party, and that the PHCP Pathway exists in the Pathways Release selected.

The screenshot displays the 'NHS Pathways - Solo' application window. The top navigation bar includes links for 'Release Manager', 'System Settings', and 'Call Queue'. The main form is divided into several sections:

- Triage Cut-Offs:** Includes fields for 'Call Handler' and 'Clinician', both set to '0 - perform full triage'.
- User Information:** 'User ID' is 'TEST_USER1' and 'User Skillset' is '999 Call Handler'.
- Patient Details:** 'Patient Name' is 'Joe Bloggs', 'Age Group' is 'Adult - 16 years+', 'Specify Age/DOB?' is checked, 'Age' is '0', 'Days' is '15', 'Date of Birth' is '12/02/2019', 'Postcode' is 'DH1 2HP', 'Telephone' is empty, 'Gender' is 'Male', 'Party' is '1st Party', 'Notes' is empty, and 'Home GP ODS code' is empty.
- System Settings:** A list of settings including 'Ebola viral haemorrhagic fever : OFF', 'Flu Epidemic Level : OFF', 'Flu Pandemic Level : OFF', 'MERS virus : ON', 'Reception point ED : OFF', and 'Reception point UCC : OFF'.
- Launch Buttons:** Two green buttons labeled 'Launch' and 'HCP and IFT' are located at the bottom right.

The version number 'v3.6.3' is displayed in the bottom right corner.

Figure 26 Example implementation of HCP and IFT launch facility

Launching the HCP/IFT triage needs to work in the same way as specified in section 2.4.2.1, however, rather than setting the **TriageRecord::BlockOrderNo** using the **Skillsets::EntryPathwayId** and **Skillsets::EntryOrderNumber** fields as detailed, it should always be set to "PHCP.0".

Similarly, the Restart Triage functionality detailed in section 2.4.3.1 needs to restart at "PHCP.0", rather than using the **Skillsets::EntryPathwayId** and **Skillsets::EntryOrderNumber** fields to determine the restart point.

13 Pathways Clinical Consultation Support Module (PaCCS)

PaCCS is based on the existing clinical content of telephony pathways. The module aims to provide a list of symptoms and conditions applicable to common complaints. It is written in a hierarchical order supplying where appropriate additional clinical/symptomatic information to support decision making.

For ease of use, the PaCCS interface specified within this document should be integrated into existing host system consultation functionality, so the PaCCS interface as displayed herein may not exist in its specified form, however the technical aspects of the interface will need to be implemented.

As an example of the technical aspects that must be implemented, the TriageRecords as described will need to be created and sent through the IDT interface regardless of how this is displayed to the user. All PaCCS functionality should also be performed and recorded against the new PaCCS Skillset.

If a more embedded approach to integration is taken, the PaCCS UI will be the default entry point for the new Skillset. This will be indicated by the "PaCCS" fixed value in the EntryPathwayId field of the Skillsets.csv file.

Should the user not require Clinical Consultation Support, they will have the choice to transfer straight to Ambulance or Home Care. If the user wants to perform DoS searches for Emergency Treatment Centres or other Services, they will need to search for a Symptom Template first to pick up a relevant SG, then they will be able to transfer to the DoS screen for searching.

During the PaCCS process, it must be possible for the user to access the rest of the Host System functionality without losing their progress in PaCCS.

Senior Clinician Module (SCM) functionality will be unaffected by the implementation of PaCCS, however providers are able to use PaCCS instead of SCM if they wish.

13.1 Host System Integration

PaCCS should be integrated more tightly with host systems than regular Pathways, however the integration must be done in such a way as to maintain the integrity of the product. For example, it would be possible to integrate the selection of Templates and population of the Suspected/Considered checkboxes from an existing host system clinical notes screen in order to fully capture the consultation before using the navigation provided by the Disposition buttons to dispatch ambulances, call DoS or display Home Care advice, however using the provided data for any other purpose would not be permitted.

While using the PaCCS functionality (both the Template/Condition part and the Pathways navigated to by the Disposition buttons), it must be possible for the user to easily switch between the PaCCS session and the host system screens to enable updates in both areas.

At the end of the call, any consultation text entered in the PaCCS fields should be included with the consultation notes entered on the host system. This will allow a full picture of the consultation to be available in one place. This data could amount to several kilobytes depending on how many templates have been opened, how many conditions have been considered/suspected, and how much free text has been entered for each condition.

13.1.1 Consultation Notes

The notes should be included in one of the following formats:

Where the Specify box has not been populated:

Template Name/Condition, Considered/Suspected.

For example:

Chest Pain/Aortic Aneurysm / Dissection, Considered, “Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris vel finibus arcu. Phasellus eget lorem semper lacus malesuada mollis. Vestibulum id velit egestas, aliquet arcu eget, fermentum nunc. Maecenas at fringilla dolor. Etiam vitae orci velit. Nulla non est nunc.”

Chest Pain/Acute Coronary Syndrome, Considered.

Abdominal pain/Aortic Aneurysm / Dissection, Suspected, “Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris vel finibus arcu. Phasellus eget lorem semper lacus malesuada mollis. Vestibulum id velit egestas, aliquet arcu eget, fermentum nunc. Maecenas at fringilla dolor. Etiam vitae orci velit. Nulla non est nunc.”

13.1.2 Directory of Services Calls

While the Symptom Group for the call is collected from the Template or Condition the disposition button was pressed on, in order to return the services profiled for PaCCS from DoS, the Symptom Group used in the call to CheckCapacitySummary should be substituted with SG1206 (NHS Pathways In House Clinician). This substitution applies to all calls to DoS where the current users Skillset is set to the PaCCS Skillset. The Case record should maintain the collected Symptom Group, with just the DoS call using the alternative code.

The process of substituting the Symptom Group is for the initial release of PaCCS and is likely to change in the future to include the actual Symptom Group collected during the call. Details will be provided in future Tech Spec updates.

13.1.3 Disposition Screens

During a normal Pathways call, a Clinician progressing through a Triage will eventually reach a disposition as determined by the logic built into Pathways by the Clinical Authors. At that point, unless the disposition is an in-line ambulance disposition, the Clinician is given an opportunity to change the disposition and select a replacement.

In PaCCS, once the Clinician presses one of the disposition buttons on the PaCCS screen, they will be presented with a series of questions to determine the disposition required by the Clinician. As the Clinician has therefore just selected the disposition, the ability to override the disposition is not required for PaCCS.

If a user with the PaCCS Skillset reaches the Disposition screen (Dx2), rather than displaying the screen, the **CaseRecord::clTriageDispoCode** should be set to the reached dispositions DispoCode (**Dispositions::DispoCode**) and the Dx2 exit processing (detailed in section 2.5.5.1) involving setting more CaseRecord attributes and determining the next step should be completed as normal, however no Triage Records should be created.

13.2 Pathways Case Data – TriageRecords

On navigating away from the PaCCS interface, a TriageRecord will be created for each Template with data entered on, and another for each condition that has been marked as either Considered or Suspected. A further Triage Record will be created to record the fact that a Clinical Consultation has occurred. This PaCCS TriageRecord should also have the CxKeywords field set if the Pathway being navigated to has a Keyword set (**Pathways::CxKW**). The Keywords collected here should be used when determining which Interim Care Advice should be displayed, as detailed in section 2.5.7.

The TriageRecords should be written in the order of the timeIn fields detailed below. For example, the PaCCS/Clinical Consultation record will be the first to be written as the timeIn field for that record is the time the PaCCS user interface was opened. The order these records are written is important as it affects the order items are displayed in the Call Report (as per section 2.5.2 of the Pathways).

13.2.1 Template TriageRecords

The Template TriageRecord should be set according to the values in the Notes column below:

Field	Type	Example	Size	Notes
TriageRecordID	String	1	50	Unique ID or serial number generated by the host system which identifies this individual triage record
supplierCallRef	String	30ac7afe-1e34-4848-a06a-ff605a299f62	36	This field links the triage record to the case record. Only the caserecord::supplierCallRef is sent back to NHS Pathways
<u>userCode</u>	Text	NFT	Max 50 characters	All triage records need a UserCode as more than one user may triage a call if collected from a queue. This will be the <u>clUserCode</u> or <u>chUserCode</u> already collected for caserecord. Each triagerecord needs to identify which cl or ch was the current user
<u>timeIn</u>	DatetimeOffset	2018-10-25T15:55:13+0100	24	Date time when the Template was opened.
<u>timeOut</u>	DatetimeOffset	2018-10-25T15:55:46+0100	24	Date time record is exited – e.g. by activating next procedure.
<u>pwId</u>	Text	CS0078	Max 20 characters	TemplateID.
<u>orderNo</u>	Integer			Not used.
<u>quID</u>	Text		Max 20 characters	Not used.
<u>answerNo</u>	Integer			Not used.
<u>userComment</u>	Text		Max 300 characters	Not used.
<u>actionId</u>	Integer	1 to 6		The procedure used by the user to leave the record. Mostly this will be next but there are other ways to exit a record. List of possible values: 1 (Back) 2 (Next) 3 (Change) 4 (Early Exit) 5 (Restart) 6 (End)
blockOrderNo	String		50	Not used.
ReportText	String		250	Not used.
IncludeInReport	Integer	1	1	Set to 1
Layout	String	PaCCS	50	PaCCS
DispoRationale	String	Fever, Unwell, Abdominal pain	100	Value from Template::TemplateName
DispoOverride	String		200	Not used.
CxKeywords	String		255	Not used.

13.2.2 Condition TriageRecords

For the Condition TriageRecords, set the fields as per the Notes column:

Field	Type	Example	Size	Notes
TriageRecordID	String	1	50	Unique ID or serial number generated by the host system which identifies this individual triage record
supplierCallRef	String	30ac7afe-1e34-4848-a06a-ff605a299f62	36	This field links the triage record to the case record. Only the caserecord::supplierCallRef is sent back to NHS Pathways
<u>userCode</u>	Text	NFT	Max 50 characters	All triage records need a UserCode as more than one user may triage a call if collected from a queue. This will be the <u>clUserCode</u> or <u>chUserCode</u> already collected for caserecord. Each triagerecord needs to identify which cl or ch was the current user
<u>timeln</u>	DatetimeOffset	2018-10-25T15:55:13+0100	24	Date time when the Condition was ticked.
<u>timeOut</u>	DatetimeOffset	2018-10-25T15:55:46+0100	24	Date time record is exited – e.g. by activating next procedure.
<u>pwld</u>	Text	CS0078	Max 20 characters	TemplateID.
<u>orderNo</u>	Integer			Not used.
<u>gulD</u>	Text	Cn220074	Max 20 characters	ConditionID.
<u>answerNo</u>	Integer	1		Used for option Considered (value 1) or Suspected (value 2)
<u>userComment</u>	Text	specify comment here	Max 300 characters	Value entered into the free text box associated to the condition.
<u>actionId</u>	Integer	1 to 6		The procedure used by the user to leave the record. Mostly this will be next but there are other ways to exit a record. List of possible values: 1 (Back) 2 (Next) 3 (Change) 4 (Early Exit) 5 (Restart) 6 (End)
blockOrderNo	String		50	Not used.
ReportText	String	There was no fever present.	250	The concatenation of Template::TemplateName , Condition::Title , Condition::Suspected or Condition::Considered depending on option selected.
IncludeinReport	Integer	1	1	Set to 1
Layout	String	PaCCS	50	PaCCS
DispoRationale	String		100	Not used.
DispoOverride	String		200	Not used.
CxKeywords	String		255	Not used.

13.2.3 PaCCS TriageRecord

For the PaCCS record, set the fields as per the Notes column:

Field	Type	Example	Size	Notes
TriageRecordID	String	1	50	Unique ID or serial number generated by the host system which identifies this individual triage record
supplierCallRef	String	30ac7afe-1e34-4848-a06a-ff605a299f62	36	This field links the triage record to the case record. Only the caserecord::supplierCallRef is sent back to NHS Pathways
<u>userCode</u>	Text	NFT	Max 50 characters	All triage records need a UserCode as more than one user may triage a call if collected from a queue. This will be the <u>clUserCode</u> or <u>chUserCode</u> already collected for caserecord. Each triagerecord needs to identify which cl or ch was the current user
<u>timeIn</u>	DatetimeOffset	2018-10-25T15:55:13+0100	24	Date time when the PaCCS screen was entered.
<u>timeOut</u>	DatetimeOffset	2018-10-25T15:55:46+0100	24	Date time when the PaCCS screen was exited – e.g. by activating the next procedure.
<u>pwld</u>	Text	PaCCS	Max 20 characters	Fixed value - "PaCCS"
<u>orderNo</u>	Integer			Not used.
<u>gulD</u>	Text	When the always available Home Care button pressed: HC When a Template specific Service Search button pressed: SS CS0078 When a Condition specific Ambulance Dispatch button pressed: AD CS0078 CN10015	Max 20 characters	Code detailing which button was pressed to exit the PaCCS interface. This in the format: "<button type> <TemplateID> <ConditionID>" The <button type> item is mandatory and should be one of the following values: AD – For Ambulance HC – For Home Care SS – For Service Search ET – For Emergency Treatment Centres <TemplateID> is mandatory when the button pressed was part of a Template (either at the top of the Template, or in a Condition on a Template). <ConditionID> is mandatory when the button pressed was part of a Condition.
<u>answerNo</u>	Integer			Not used.
<u>userComment</u>	Text	specify comment here	Max 300 characters	Any extra information added by the user for this answer.
<u>actionId</u>	Integer	1 to 6		The procedure used by the user to leave the record. Mostly this will be next but there are other ways to exit a record. List of possible values: 1 (Back) 2 (Next) 3 (Change) 4 (Early Exit)

				5 (Restart) 6 (End)
blockOrderNo	String		50	Not used.
ReportText	String	Clinical Consultation	250	Fixed value - "Clinical Consultation"
IncludeinReport	Integer	1	1	Set to 1
Layout	String	PaCCS	50	PaCCS
DispoRationale	String		100	Not used.
DispoOverride	String		200	Not used.
CxKeywords	String	Medication, pain and or fever	255	Value taken from Pathways::CxKW where the destination Pathway has a Keyword set.

13.2.4 Jump Back using the Call Report

Clicking any of the PaCCS related Call Report records should navigate the user back to the PaCCS Template selection screen, with the screen being repopulated based on the details stored in TriageRecords.

In an embedded scenario, the screen will need to be in a read-only mode (as per standard Pathways), until the "Change Answer" button is pressed. At this point the standard processing described in section 2.4.3.3 must be performed. Pressing the "Return No Change" button will return the user to where they were before they pressed a Call Summary navigation button (as per section 2.4.3.4).

In a more integrated configuration, it may be undesirable to display the page that allows the selection of the Templates as read-only, as that may limit functionality. In this case, it is essential that the processing that happens with the "Change Answer" and "Return No Change" is followed when appropriate, otherwise the resulting data sent to IDT will be invalid. So, for example, if the user navigates back to the Template selection screen and presses one of the disposition buttons, this is equivalent to pressing the "Change Answer" button, so the same processing is required (section 2.4.3.3).

13.3 Transferring cases to PaCCS Clinicians

Calls may be transferred to PaCCS clinicians by placing the case on the call queue as normal.

There are two possible scenarios for transferring cases between clinicians.

1. Transfer a telephone Triage call to a PaCCS clinician
2. Transfer a PaCCS clinician call to another PaCCS clinician

Transferring a call from a PaCCS clinician to a telephony clinician is not supported, a new Case will be needed should this be required.

When a PaCCS clinician is picking up a call from the call queue the user should be shown the Consultation Summary of the call, before Restarting the Triage and commencing at the PaCCS entry point. The Consultation Summary displayed should be cached to display it again should the clinician require it. The standard Pathways Validation screen should not be provided for the user.

Once the user has started the PaCCS process, they should be given a way to view the cached Consultation Summary previously displayed.

A PaCCS call can be placed on the call queue in the same way that a clinician would add a telephony case to the call queue.

13.4 IDT Changes

CaseRecord (defined in 3.1.4.1.1) contains various fields specifically for Call Handler or Clinician data (e.g. chUserCode and clUserCode).

There is only one set of these fields for Call Handlers, and one set for Clinicians, however a case could potentially have several Call Handlers and/or Clinicians involved, so the fields are ambiguous.

For consistency, where multiple Call Handlers, or multiple Clinicians participate in the call, the ch* or cl* fields should relate to the last Call Handler or Clinician involved with the case.